

Manual

Goods Receipt Inspection Planning WEP-PPW 8.2

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1 Goods Receipt Inspection Planning - Overview

Purpose

This component allows creating inspection plans based on master data defined earlier plus generating and managing inspection requirements which, in combination with the associated inspection steps, form the basis for the collection of inspection data.

Implementation Considerations

Since creating inspection plans is the basis for later collection of inspection data, this component is required to collect measured values and attributive inspection decisions.

Integration

This component mainly serves the components:

- Quality data entry / info functions (AIP)
- Additional inspection planning / inspection steps
- Family inspection planning
- Reports of production control inspects and
- Failure mode analysis / action tracking

Features

The following functions are available:

- Maintenance functions to collect and process relevant master data (article, company, failures, actions, characteristics etc.)
- Maintenance functions to create and modify inspection plans
- Versioning inspection plans including managing historical data
- Inspection plan release and activation using special rights
- Allocating or modifying inspection plan characteristics of various characteristic types (variable, attributive)
- Assigning operations and inspection stations so as to structure the inspection steps on the basis of the inspection plans.
- Defining of tolerance limits, unit, sampling scheme, control charts to be used, activating automatic failure generation etc.
- Allocating any document (drawings, pictures, videos of any format and internal notes) to characteristics, inspection plans, inspection plan characteristics, inspection requirements or inspection step characteristics in a document list

- Generating inspection requirements or steps based on inspection plans created earlier

2 Article

This document describes the “article” application of the Manufacturing Operation Center (MOC). General information on how to use MOC can be found in the document entitled “[moc_cc.pdf](#)”.

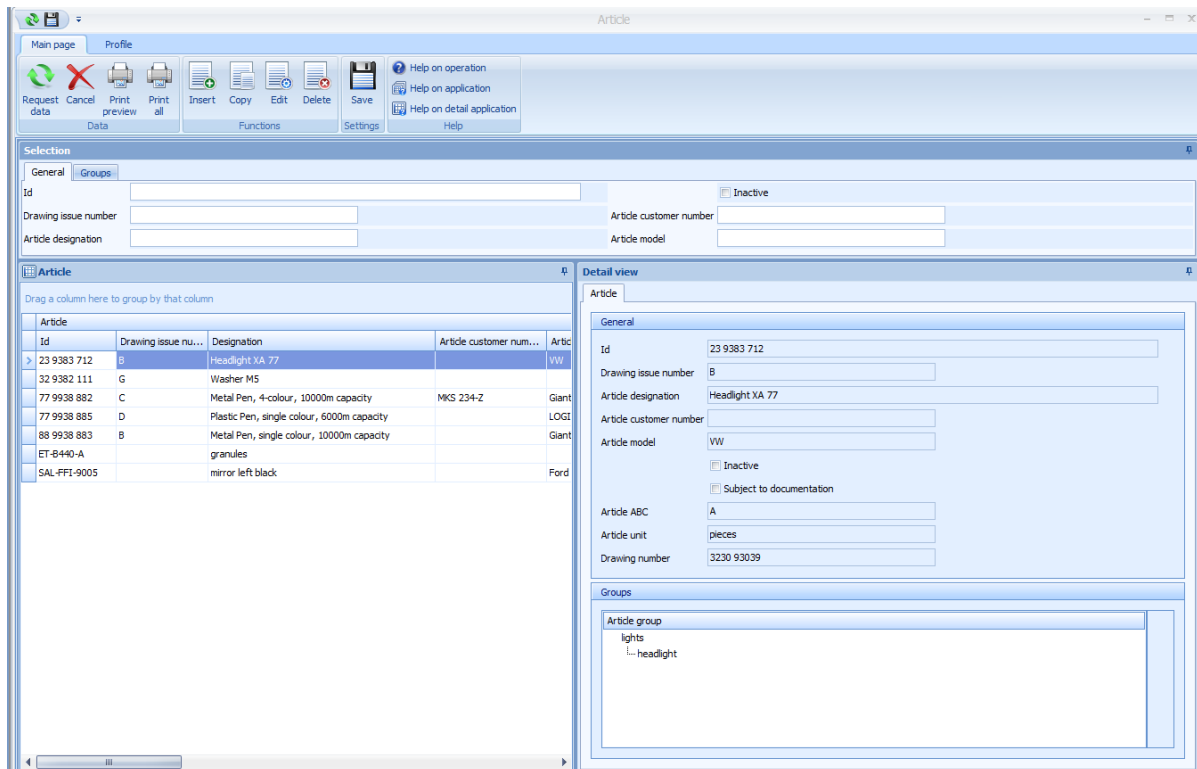
The article catalog has been designed to edit/keep articles. Article data is a global catalog that is used in many CAQ modules and in PDV (Process Data Collection). Provided that there is an interface to a higher-level system (e.g. ERP system), articles may be created automatically via this interface. As soon as a new article is created or changed, e.g. in the ERP system, the article data record is automatically created or changed in the HYDRA-CAQ article catalog based on the defined information.

2.1 Function call

Menu	Master data → Quality management → Article Master data → Process data processing → Article
Transaction code	atc
Function authorization	atc

Available user fields		
Location	Object type/user field key	Source (type)
Table and detail view	ATK/SYSTEM	MF-D
How can I configure user fields? Which user field types are available?		

2.2 Default Application Layout



2.3 Toolbar

The toolbar provides the different functions available for this application and possibly links to other applications. The functions included in the “general” tab of the toolbar are available in all detail applications. In addition to the standard functions, such as help, request data, save application settings, and print preview, the other tabs also include specific functions that are specially tailored to the respective detail application. The following sections describe the individual application functions.

Category Data



Request data

The information to be displayed within the application is requested on the basis of the entered selection criteria. This process might take some time depending on the dataset from which the system filters data and on the selection result to be transferred and displayed.



Cancel

The query sent by clicking the “request data” button can be canceled using this function.

**Print preview**

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

**Save**

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking "Yes".

Category Functions

**Add**

Adds a new article.

**Copy**

Copies the selected article.

**Edit**

Edits an already existing article

**Delete**

Deletes the selected or several selected articles.

Category Help

**Help on operation**

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled "[moc_cc.pdf](#)". It describes how to use MOC in general and applies for all applications.

**Help on application**

This function opens the manual for the respective application from which the help file was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.



Help on detail application

This function opens the application manual at the section where the relevant detailed application is described.

2.4 Selection parameters

The application provides the following selection criteria:

Tab "General"

- Article no.:
Article number
- Drawing issue number:
Drawing issue number of the article, often also referred to as index
- Designation:
Article name
- Inactive:
Inactive, active articles. The checkbox is not enabled by default.
- Customer article no.:
Customer article number
- Article model:
Article model

Tab "Groups"

- Group:
The article group tree can be opened using the  button if an article group is to be filtered. There is a function to accept and cancel the activity.

2.5 Detail application "Article"

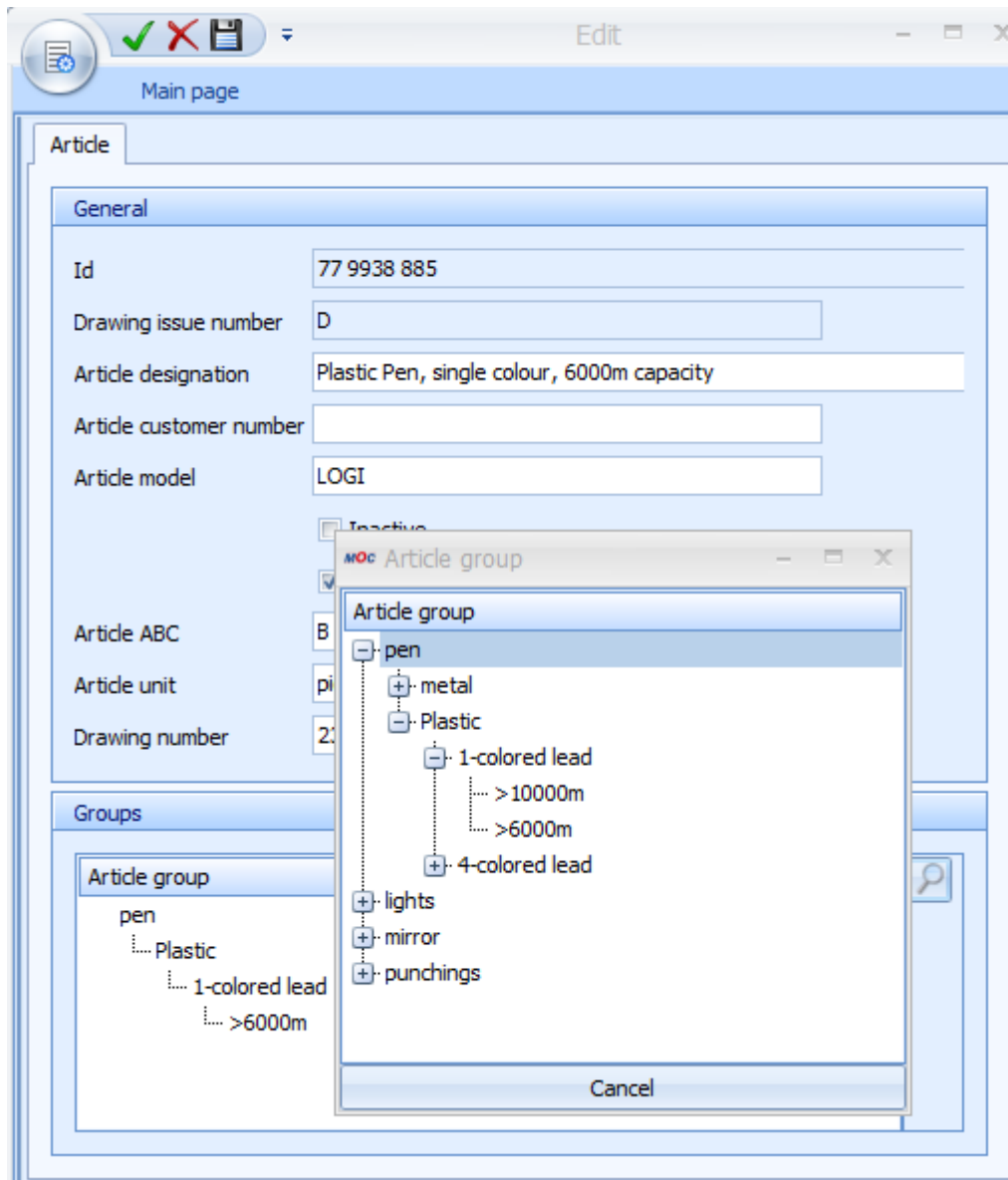
The article number as well as the drawing issue number uniquely identify articles in all areas of HYDRA-CAQ referring to the article catalog. The drawing issue number, also referred to as article index, may be very important, in particular, for inspection planning and when inspection orders are generated. Thus, it is, for example, possible to create an inspection plan for the article 12938 with the drawing issue numbers A and B. Different inspection specifications apply for each drawing issue no. Unless the drawing issue number is indicated and thus may be part of the inspection plan, the system that generates the inspection requirements, must deliver this drawing issue number.

The fields “article no.” and “drawing issue number” fields are key fields, i.e. when a new article is saved, it is first checked whether an article with this key information already exists.

By distinguishing between active and inactive articles, it may be defined whether or not the articles are available in certain selection lists. Thus, no inspection plan can be created for an inactive article. However, inactive articles may be evaluated at any time. Moreover, inactive articles can also be reactivated at any time.

Furthermore, an article can be defined as being subject to documentation. In addition the dialog provides the fields customer article number, article model, article ABC, drawing number as well as the possibility to assign units. To assign units (dimensions), the catalog of units is used.

If you want to make evaluations on article groups or if you use family inspection plans, it is mandatory to assign the respective group. To assign groups, open the group tree using the lens icon. Using the hierarchic tree entries the required group can be selected in the group tree and accepted by double clicking.



The assigned group including the hierarchical group structure then appears in the “groups” field of the editing dialog of articles.

When articles are displayed in a list, the group hierarchy is represented by the columns “group 1 to group 5”.

Groups are maintained in the “article groups” application and is described in the document entitled [“MOC_Groups.pdf”](#).

3 Summary

3.1 General notes on the document

This document describes the “Groups“, e.g. article groups, application of the Manufacturing Operation Center (MOC). For general information on how to use MOC, please refer to the “[moc_cc.pdf](#)“ document.

4 Groups

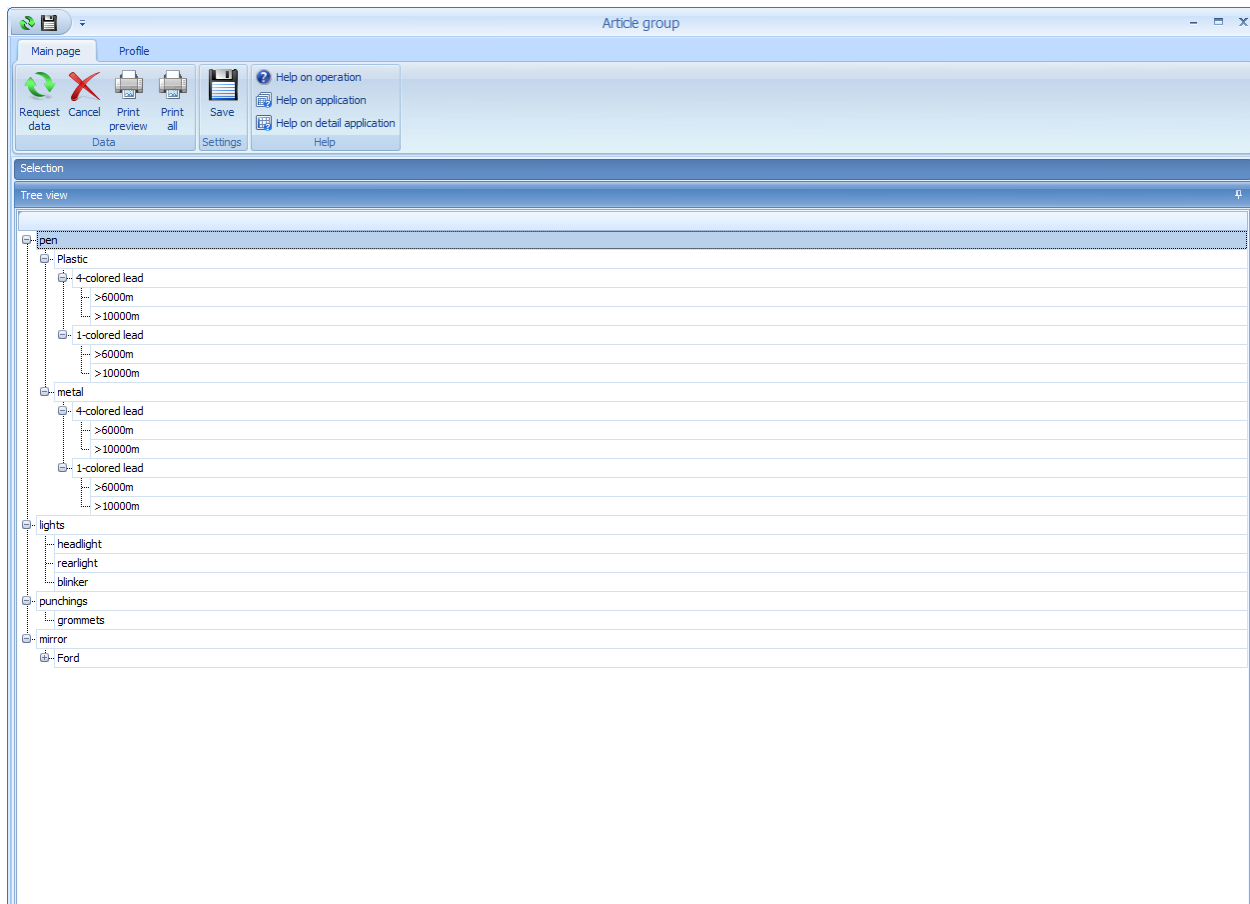
The group catalogs have been designed to create and edit groups for the different applications. The created groups may be assigned to master data of the corresponding application. Consequently, article groups may be created, for example, and assigned to the articles. In this case, it is also possible to create inspection plans on the basis of article groups.

Basically, the creation of groups is also reasonable for failure mode analyses.

4.1 Starting the function

Menu	Master data → Quality management → Article groups Master data → Process data processing → Article groups Master data → Quality management → Measure groups Master data → Quality management → Failure type groups Master data → Quality management → Failure location groups Master data → Quality management → Failure cause groups Master data → Quality management → Causer groups Master data → Quality management → Cost type groups
Transaction code	atcgr → Article groups measgr → Measure groups ftypgr → Failure type groups flocgr → Failure location groups fcaus → Failure cause groups origr → Causer groups costgr → Cost type groups
Function authorization	atcgr -> Article groups measgr.* → Measure groups ftypgr.* → Failure type groups flocgr.* → Failure location groups fcaugr.* → Failure cause groups origr.* → Causer groups costgr.* → Cost type groups

4.2 Default Application Layout



This figure of the article group catalog is exemplary for all groups.

4.3 Toolbar

The toolbar contains the function calls that are available for this application and possibly links to other applications. The functions placed on the "general" tab of the toolbar refer to all detail applications. In addition to the standard functions such as "help", "request data", "save application settings" and "print preview", the other tabs include specific functions that are tailored to the corresponding detail application. The individual functions of the application are listed in the paragraphs that follow.

"Data" category



Request data

The information to be displayed within the application is requested according to the entered selection criteria. This process might take some time depending on the data set from which the system filters data and on the selection result to be transferred and displayed.

**Cancel**

This function cancels the query sent by clicking the "request data" button.

**Print****preview**

The print preview is opened for the selected detail application. The print preview also includes further options to change the resulting printout and functions for exporting the displayed information into other formats, such as PDF, Excel, image files.

**Save**

The application design configured by the user, e.g. columns and categories displayed as well as their respective size and display locations, etc. are only saved if the user requests it. In this case, the user has to affirm the confirmation prompt by clicking "Yes".

"Functions" category

There are no special functions for this detail application. Groups are created, changed and deleted using the context menu of the right mouse button within the tree structure view.

"Help" category

**Help on operation**

Clicking this button opens the help file describing how to operate MOC. The basic document is entitled "moc-cc.pdf". It describes how to use MOC in general and applies for all applications.

**Help on application**

This function opens the manual that describes the application from which the help function was requested. The application manual integrates the application function into the MES context and explains the information to be displayed. The documentation also includes all detailed applications.

**Help on detail application**

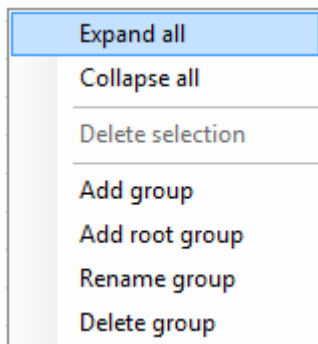
This function opens the application manual at the section where the respective detailed application is described.

4.4 Selection parameters

There are no selection parameters. A specific group can be found by using the “fast selection” function. To use the “fast selection” function just open the group tree structure on the 1st level, select the first entry and enter the first letter of the group in question. Consequently, the first group starting with this letter that is found is selected. The “fast selection” function also integrates subordinate groups that are not opened. If the requested term is included in a group that is not opened, it will be opened automatically.

4.5 "Groups" Detail Application

A group may be created, changed or deleted by opening the context menu of the right mouse button within the display area of the tree structure.



It is altogether possible to define groups up to the fifth hierarchy level. The “add root group” function has to be selected in the context menu of the group tree to create a new main group (1st level). The menu entry “add group” generates a sub-group (level 2 to 5). A designation, which is directly entered in the list view, has to be assigned for this new group. To be able to save the new group, click above or below this new entry within the group tree. Then a confirmation prompt appears asking whether or not the new group is to be saved. Provided that this question is affirmed (“yes”), the entry is saved. The same applies for renaming of groups. Regardless of which hierarchy level is concerned, the entry has to be selected to be able to edit the group designation. Changes can directly be entered in the corresponding line of the tree view. Click above or below the entry to be changed to be able to save the modification. No confirmation prompt appears when it comes to renaming.

An entry is also deleted by selecting a group entry and executing the “delete group” function. Only the group that is at the bottom of the group tree can be deleted.

The “expand all” context menu entry opens all groups up to the lowest hierarchy level. The “collapse all” context menu option closes all entries up to the first level.

The "delete selection" function cannot be used in the maintenance of groups dialog. This function is enabled, for example, in the maintenance of articles application if an article group is selected and this selection is to be removed/deleted.

5 Characteristic Master Data

Overview

Menu	Master data → Quality management → Characteristics
Transaction code	chrq
Function authorization	chrq

Available user fields		
Where?	Object type/user field key	Source (type)
Table and detail view	CMM/SYSTEM	QM
How to configure user fields?		Which user field types are available?

Purpose

The screenshot displays the SAP 'Characteristics' configuration interface. The 'Main page' tab is selected, showing a 'Selection' section with input fields for 'Characteristic no.', 'Characteristic designation', 'Process parameter', and 'Characteristic type'. Below this is a table listing various characteristics with columns for 'Characteristic no.', 'Characteristic designation', 'Input type', and 'Process parameter'. The table lists 18 characteristics, including 'Total length', 'Evenness', 'Length pushbutton', 'Outer diameter', 'Inner diameter', 'Width', 'Angle', 'Hardness HRC', 'length', 'Thickness', and 'Layer thickness'. The 'Details' tab is also visible, showing fields for 'Characteristic no.', 'Characteristic designation', 'Input type', 'Characteristic type', 'Inspection result base', and 'Formula'.

The catalog of characteristics has been designed to define characteristics and, as a result, to predefine characteristic data of inspection plans. For this reason, it aims at people involved in inspection planning.

The catalog of characteristics is one of the most important basic catalogs. You cannot set up any inspection plan without this catalog. As this catalogue is used to predefine characteristics for inspection plans, it includes extensive input options. Basically, the catalog of characteristics should only include such data, which will not have to be modified when the characteristics are assigned to the inspection plan later on. For example, the definition of limit values is usually not reasonable as these values are only known when an inspection plan is set up. Only when you assign data to an inspection plan, a relation between data and article is established. Note this and you will know what kind of information you should predefine. For example, it must be carefully considered whether the characteristic "outer diameter" is only created once and detailed information is stored in the inspection planning later on or whether several "outer diameter characteristics" are created, e. g. with specification of limit values. Usually, it is an advantage to store a restricted number of general characteristics. The required evaluations/reports also play a role in this context. If a new "outer diameter characteristic" is created for almost every tolerance change, this characteristic is "valid" for one article only. In a subsequent failure analysis, a comprehensive evaluation is not possible in this case!

It is important that each detail defined here can be modified in the inspection planning later on or that details, which have not been stated, can still be added.

The configurations made in the characteristics' master data are not final. The characteristics' master data is used as a template for later inspection planning. You can complete and modify all settings of the characteristics' master data during inspection planning.

Integration

The catalog of characteristics is a global catalog that is used in many QM applications. Please find below some possible fields of application that refer to the catalog of characteristics.

- Inspection planning for production, goods receipt, goods issue, initial samples and calibration
- Inspection requirements for production, goods receipt, goods issue, initial samples and calibration
- Failure analysis in complaint management
- Several reports/evaluations

Requirements

There are no special requirements.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
Number of the characteristic
- Characteristic designation:
Designation of the characteristic – Note: You may use wildcards "*"
- Characteristic type:
Inspection type: attributive, inspection chart, variable

Tab Details

- Gage
Select a gage
- Gage designation:
Select a gage designation

Tab User fields

- If user fields are created, they may be selected

If several selection criteria are used, overlapping results are displayed in the characteristics' master data.

In addition, the column filter allows for the content of each individual column to be filtered.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

Tab Characteristics

Characteristic no.

Unique number of the characteristic

Characteristic designation/name

Designation of the characteristic

Input type

Automatic or manual data collection. This field controls the release of HYDRA-PDV fields (in case of automatic collection). If the automatic collection function is selected, the characteristic type is restricted to the "variable" option.

Characteristic type

This option specifies whether the collection of measured values (variable) or the identification of the number of detected failures (attributive) is used for the inspection. If you select the attributive inspection, use the input type to define whether the collection should be based on a catalog or whether the standard collection is performed. Further characteristic types are the inspection chart and the information characteristic. If you select the inspection chart, you can enable the input type *visual defects recording*. The information characteristic is only used to display a document during the inspection process. Subject to the input type, the lower area of the dialog provides the respective sampling schemes.

Visual recording: The characteristic document (not the *inspection requirement document*) is displayed with the position 1. This must be type FILE. The system supports these formats: JPEG, JPG, PNG. To divide a graphic in different areas, you must define the grid for the x-axis and the y-axis (e.g. A,B,C,D,E)

Inspection result base

This setting defines whether all samples or only the sample recorded last is used to identify the inspection result (pass/fail).

Mandatory inspection

If this option is activated, you must enter at least one measured value for this characteristic, before you can complete an inspection order including this characteristic.

Formula:

See chapter Calculation of formulas.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.formula".

Tab Details**Group Gage****Gage**

Defines whether a gage or gage group is to be assigned to the characteristic:

Assignment of the gage (or gage group) to be used.

You can also use resources of resource type "PRM" of the resource management. To display this field in the inspection plan characteristics, you require the authorization "iriscp.gage".

Gage designation (name of test equipment)

Shows the name of the gage

Group Properties

Certificate printing

The selected option defines whether this characteristic is to be printed (display selection or print always) or not (print never) when certificates are printed at a later stage (e.g. acceptance, inspection certificate). If you select the option "display selection", a list of the characteristics with this printing option set is displayed prior to printing. In the list, these characteristics are preselected for the print of a certificate. However, this selection may be removed. Finally, all selected characteristics and the characteristics with the "print always" option are included in the certificate print. Characteristics with the "print always" option do not appear in a selection list, as they are printed in any case. Please note that this option only affects certificate forms.

Failure weighting

If the inspection result for the characteristic is "fail", you can classify the result here for information purposes.

Group Inspect

Analyseauswahlkatalog

Here, you can select an analysis selection catalog. The catalog restricts the selection of possible failures you can enter (failure types, failure location, etc.). (All available failures may still be entered, if you directly enter their number).

Designation of analysis selection

Shows the designations of analysis selection catalogs

Tab Specifications

Once the "specifications" tab has been selected, the sample scheme and constructional measures may be entered. In this context, it has to be considered that (as already mentioned) the definition of tolerance limits in the master data of characteristics is only reasonable if certain conditions are met. The same applies to the definition or calculation of action and warning limits. This section explains the possibilities in detail.

Group Sampling scheme

Sampling scheme

The following sampling schemes are available:

- 100% inspection
- k value inspection
- lot inspection
- n-c inspection
- SPC inspection

The sampling scheme defines the inspection procedure. In case of an n-c inspection and parameters 5-0, 5 pieces are checked and 0 failures may be detected.

Find a more detailed description in section Sampling **schemes**.

Sample size/expected sample size

Specification of the sample size (number of samples) or the expected sample size depending on the sampling scheme, see section Sampling schemes.

Acceptance quantity

Acceptance quantity for the n-c inspection, please also see section Sampling schemes.

Interval type

Input for SPC or n-c inspections: time, pieces, once, none. See chapter Sampling schemes .

Interval value

Specifies the interval subject to the interval unit.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Interval unit

For n-c or SPC inspections, e.g. minutes, hours.

With output batch change

If the output batch changes, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Note:

The option *With output batch change* only triggers the generation of an inspection point, if the respective change of the output batch is included in the dialog "Change of batches" (dialog ID: CA_WL). For example, reel cutting dialogs do not generate inspection points.

With machine status change

If the machine status changes, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Source status

Here, you can specify source statuses (specific non-productive machine statuses) – separated by commas. If the machine then changes from a specified source status into a productive machine status, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

As of SP8, the following configurations are available in addition.

- The field is completely empty: For this characteristic, a machine status change always generates an inspection point, if the machine changes from a non-productive into a productive status.
- "x-y", comma-separated: If the machine changes from source status x to target status y (may be non-productive), an inspection point is generated for this characteristic.
- "x-": If the machine changes from source status x to an arbitrary target status (may be non-productive), an inspection point is generated.
- "-y": If the machine changes from an arbitrary source status to target status "y" (may be non-productive), an inspection point is generated.

With change of shifts

An inspection becomes due on changing shifts.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Inspection due date of last off inspection

For details on the configuration of a last off inspection, refer to the section "[Last off inspection](#)".

Group *Constructional measures*

Unit

Pieces, meter, kg, etc. Unit of the characteristic. Allocate the units by using the unit catalog.

Decimal places

Number of decimal places. Leading zeros before the comma are not displayed in the specification fields. By default, the number of decimal places defined in the system settings is pre-assigned.

Size (measure type)

Plausibility and tolerance limits can be entered as absolute, relative or percentage values. Please note that relative or percentage lower limits (lower tolerance limit, lower process limits) must be specified with a negative algebraic sign.

Standard

Calculation of tolerances based on specific standards (e.g. ISO metric fits). Subject to the selected standard, further information is requested (e.g. engineering fit). The system automatically calculates the tolerance limits on the basis of these specifications.

Fit

Calculation of tolerance limits on the basis of a specific standard and engineering fit. The selected fit depends on the selected standard.

Upper PL

Specifies the upper plausibility limit

Upper TL

Specifies the upper tolerance limit (upper specification limit)

Target value

Specifies the target value

Lower TL

Specifies the lower tolerance limit (lower specification limit)

Lower PL

Specifies the lower plausibility limit

Generate failure (UTL)/(LTL)

If measured values are recorded and the checkbox *Generate failure* is enabled, a violation of the limit value automatically results (in the background) in the failure type "limit value violation" (AUTO:TG> or AUTO:TG<). This option is not available for attributive characteristics, as the specification is only used for information purposes in this case.

User fields tab

If you have defined user fields for characteristics, they are displayed and may be edited here.

Tab Chart 1/Chart 2

In tab chart1/chart2, you can define the control charts to be used. These control charts are later available in the integrated measurement recording and in the measurement recording for terminals (SPCM). You can define a total of two different control charts. Here, you can store for each control chart the action limits, warning limits and the mean value of variable characteristics. There are two different possibilities to define these limit values. You can enter the limit values manually or the limit values are calculated using the specified default values included in tab *Default values chart1/2*. For further information on control charts, refer to sections 5.2 Control charts for variable characteristics and 5.3 Control charts for attributive characteristics.

Chart 1 / Chart 2

Specifies the control chart displayed in the measurement recording dialog on the terminal. You can define action limits on the basis of the control chart type.

Upper AL

Specifies the upper action limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Upper WL

Specifies the upper warning limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

MV (Mean value)

Specifies a mean value, e.g. as basis for the automatic calculation of limits by the system.

Lower WL

Specifies the lower warning limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Lower AL

Specifies the lower action limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Generate trend error

The option "generate trend error" has to be activated to be able to generate an automatic error if a trend exists (e.g. seven values in a row are descending or ascending, the number of values is defined while the system is customized). To identify a trend, the samples of an inspection step characteristic are checked, sorted by their sample number – regardless of the machine where the data has been recorded.

Generate error (UWL) / (LWL)

Enable the checkboxes *Generate error (UWL) / (LWL)* to generate automatically (in the background) the failure type "Limit value violation" (AUTO:WG> or AUTO:WG<), if a limit value is violated during the recording of measured values. Here, the violation of the limit value is identified using the stored control chart. In case an xq chart is stored, the automatic error is only generated if the respective xq value of the control chart, and not the single value, exceeds the warning limits.

Generate error (UAL) / (LAL)

Enable the checkboxes *Generate error (UAL) / (LAL)* to generate automatically (in the background) the failure type "Limit value violation" (AUTO:EG> or AUTO:EG<), if a limit value is violated during the recording of measured values. Here, the violation of the limit value is identified using the stored control chart. In case an xq chart is stored, the automatic error is only generated if the respective xq value of the control chart, and not the single value, exceeds the action limits.

Tab *Default values chart 1* / *Default values chart 2*

For further information on control charts, refer to sections Control charts for variable characteristics and Control charts for attributive characteristics.

Group Default for calculating limit values**Calculation type**

Default values to calculate limit values: Cpk, Sigma, sq/an, Rq/dn, relative deviation from xq, deviation from xq in percent

Cpk

Default value of cpk

Sigma

Default value or calculated sigma value

Rq/sq (RQuer/sQuer)

Default value for Rq/sq (RQuer/sQuer)

Group Non-action probability**Action limits (non-action probability)**

Specifies the action probability (only visible with the calculation type: cpk, Sigma, rQuer/sQuer)

Warning limits (non-action probability)

Specifies the action probability (only visible with the calculation type: cpk, Sigma, rQuer/sQuer)

Group Deviation from xq specification**rel. AL**

Direct entry of the action limits (only visible with calculation types relative/percentage deviation).

rel. WL

Direct entry of the warning limits (only visible with calculation types relative/percentage deviation).

Group Confidence interval**Confidence interval**

One-sided or two-sided. You can select one-sided or two-sided for the control charts R and s.

Group xq**XQ**

Target value, mid-tolerance, mean value of xq chart, input (only visible and can only be selected with an xq control chart)

Editing functions

The below screenshot shows an example of an editing dialog. Design and alignment of fields may deviate.

The screenshot shows the 'Edit' dialog box with the 'Specifications' tab selected. The dialog is titled 'Edit' and has a 'Main page' label. The 'Specifications' tab is active, showing the following fields:

Sampling scheme

- Sampling scheme: n-c-inspection
- Sample size: 5
- Acceptance qty.: 0
- Interval type: ☐ once ☐ none ☐ piece related ☒ time related
- Interval value:
- Interval unit: Minute(s)
- ☐ With output batch change
- ☐ With machine status change
- ☐ With change of shifts

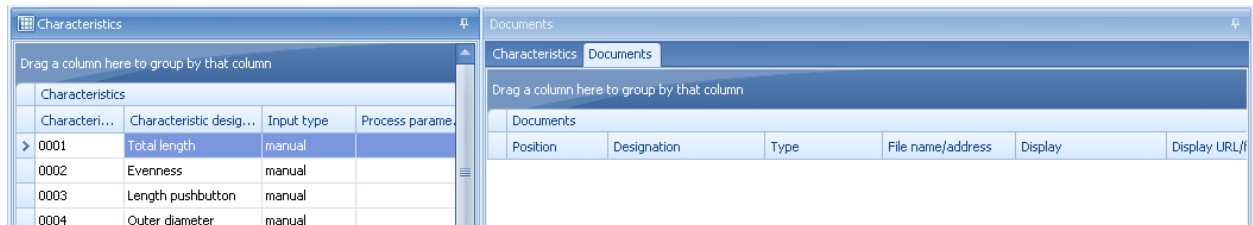
Constructional measures

- Unit: mm
- Decimal places: 5
- Size: ☒ absolute ☐ percentage ☐ relative
- Norm:
- Upper PL: 135.0
- Upper TL: 130.25
- Target value: 130.0
- Lower TL: 129.9
- Lower PL: 125.0
- ☒ Generate error (UTL)
- ☒ Generate error (LTL)

Toolbar

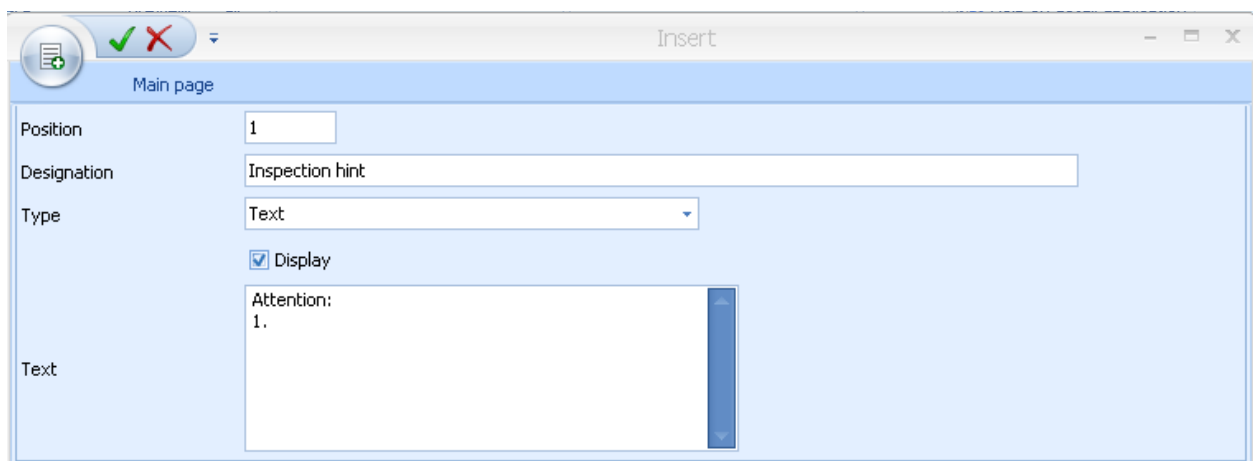
There are no other special function buttons in addition to the standard functions/features.

Detail application *Documents*



The screenshot shows two side-by-side windows. The left window is titled 'Characteristics' and contains a table with columns: 'Characteri...', 'Characteristic desig...', 'Input type', and 'Process parame.'. It lists four characteristics: 0001 (Total length, manual), 0002 (Evenness, manual), 0003 (Length pushbutton, manual), and 0004 (Outer diameter, manual). The right window is titled 'Documents' and contains a table with columns: 'Position', 'Designation', 'Type', 'File name/address', 'Display', and 'Display URL/'.

If you have activated the tab *Documents*, you can assign an arbitrary number of documents to each characteristic. If this tab is activated, the respective buttons in the toolbar to edit the documents are equally activated.



The screenshot shows the 'Main page' form in the 'Documents' tab. It includes a toolbar with a document icon, a green checkmark, a red X, and a dropdown arrow. The form fields are: 'Position' (text box with '1'), 'Designation' (text box with 'Inspection hint'), 'Type' (dropdown menu with 'Text' selected), and a 'Display' checkbox which is checked. Below these is a 'Text' label and a large text area containing 'Attention: 1.'.

All formats registered by Windows are available when assigning documents. You can assign simple documents (e.g. written in Word), drawings of any format and videos. You only have to make sure to install a program that is able to display the used format. The appropriate program linked in Windows opens the documents.

The file types "File", "URL", and "Text" are available. If you select the type "file", you can enter the file name including path manually. Select the file type "URL" to access the internet or intranet. Select the file type "text" to directly enter a text.

Note:

The different types of file format "URL" that the shop floor client supports are listed in the respective manual of the shop floor client. It might happen that "https" URL entries are displayed on the MOC, but not on the AIP shop floor client.

You can assign a designation/name to each document. You can also define the list order of the documents. Use the field "position" to define the order (numeric input). Position numbers must be unique in this list. Enable the checkbox *Display* to define that the document is displayed during inspection process.

Speaking of documents, you also have to decide if a document assignment without precise reference to an article is reasonable. Normally, the document assignment depends on the article.

Taskbar Document

In addition to the standard functions, the application also provides the button to show documents.



Show documents

If a document link is stored, click this button to open and show the linked document. However, a program, which can show the linked file type, must be installed on the PC.

5.1 Sampling schemes

The user can select from five sampling schemes in a specified list. Subject to the selected sampling scheme, some additional information has to be defined. It is subject to the subsequent use in the different inspection plan areas (e.g. production, goods receipt, goods issue), if all or only a smaller selection of sampling schemes is available.

Sampling scheme n-c inspection: The sample size is entered in the "sample size" field (= n) and the maximum number of admissible non-conforming units is entered in the "acceptance quantity" (=c) field. The figure "c" is defined as acceptance number. This means: if $n = 50$ und $c = 1$, the characteristic and thus the piece is only classified as "fail" if two non-conforming units are identified (with sample size = 50).

Sampling scheme 100% inspection: In general, the sampling scheme *100% inspection* is only used in goods receipt and goods issue. The sample size is calculated from the actual quantity of the inspection requirement and corresponds to it.

Sampling scheme SPC inspection: The sampling scheme "SPC inspection" nearly corresponds to the "n-c" inspection plan. The only difference is that the acceptance limit "c" is not used in this case.

Sampling scheme batch inspection: In the standard configuration, the sampling scheme "batch inspection" only applies to the areas "goods receipt" and "goods issue". The percentage specifying how much percent of the batch is to be checked is entered here. Later in the inspection order characteristic the sample size is calculated from the actual quantity of the inspection requirement and multiplied by the specified percentage.

If you must calculate action limits, you must enter the expected sample size here.

Sampling scheme k-value inspection: With the k value inspection the entered k value is checked against the calculated k value and if this value is violated the sample is rated "fail".

5.2 Control charts for variable characteristics

For variable characteristics, the charts xq, s and R are available.

In statistical quality assurance, production dispersion is used for many calculations. One example is the calculation of capability indices and action limits of a quality control chart. Vice versa, if you have specified a process capability index, you can estimate the production dispersion and calculate the action limits on this basis.

The specifications for the calculation of limit values can be found in the tab "default values chart 1" or "default values chart 2", where values to estimate the production dispersion can be entered. The action and warning limits can be calculated on the basis of these specifications. However, it is also possible to enter the production dispersion directly. The system provides three calculation options.

You first describe the specifications using the xq and s chart. The differences with the R chart are explained in more detail in the sections that follow.

There is often a specification for the process capability index cpk. This specification is reasonable. If the process capability index cpk is respected, you can then produce pieces within the range of tolerance. Based on the specified cpk value, the system calculates internally an estimated value for Sigma, which is entered to the right of the option "Sigma" for information purposes. The estimated basic value that has been calculated is used to calculate the limit values of the xq/s chart. The calculation is performed, once further data has been entered using the *Calculate* button. The calculation method "cpk" is set by default. In addition, there are also the calculation methods "sigma" and "sq/an".

The cpk value of 1,33 ensures that 99.725% of the characteristic values are within the tolerance. However, it is often required that 99.994% of the characteristic values are within the tolerance limit, which corresponds to a cpk value of 1.67.

The calculation method sq/a_n means that an estimate of the standard deviation is calculated from the quotient of the medium standard deviation and a correction factor a_n . This correction factor depends on the sample size, which is identified by the index n . The values for a_n are defined in the system and are requested automatically. This estimate of the standard deviation is best in case that there is no specification of the process capability index and the production dispersion is unknown and thus the specification of the sq -value has still to be corrected by a correction factor. It is also the most efficient method under the given conditions. You must specify the sq -value to calculate the limit values later on. Enter the value in the field on the right hand side of the option sq/a_n . If you click the button *Calculate* later on, the estimate sq/a_n is calculated using the specified sq value. The result is entered on the right hand side of the option Sigma for information purposes. This estimate is then the basis for the calculation of action and warning limits of the xq/s chart

The third calculation method requires the specification of a sigma value. In this case it is assumed that sigma is known and consequently the correction factor is not required. Enter the Sigma value to the right of the "sigma" option. In comparison to the previous method, sq/a_n is replaced by sigma. In the majority of cases sigma is not known. Therefore, it is best to use the calculation method using the specified sq value to automatically calculate the estimate sq/a_n for variances in case of doubt.

If you select the "relative deviation from xq " or the "deviation from xq in percent" as "specification to calculate limit values", the input option for "action probability in %" disappears. Instead, you can enter the "deviation from target value". These values and the specified value of xq are then used to calculate the limit values (target value, middle of tolerance, mean value of $\bar{x}$ chart, input).

Further details have to be made in order to identify action and warning limits of the xq chart. You must specify an xq value. The system offers the possibility of setting the xq value equal to the middle of tolerance or the target value or of specifying a value manually. If the process is supposed to be aligned to the mean value, the middle of tolerance should be preferred as xq value.

The action probability must be entered in percent in order to calculate action and warning limits of the xq/s -chart. For this purpose, you must first decide, if you want to use one-sided or two-sided limit values for the calculation. Select one of the two options.

Once you have specified the option 'one-sided' or 'two-sided', enter the action probability in percent. The possible and reasonable action probabilities are defined in the system and only need to be selected from the list. For the xq -chart, the calculation is based on the standard distribution. For example, if 99.725% of the characteristic values must be within the action limits, select the value 99.725. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed with the respective action probability for information purposes.

If warning limits are also to be calculated, an action probability must be entered here. Note: The probability value of the warning limit must be lower than the action limit value.

The sigma area is not displayed in the selection list, since the distribution of χ^2 is used to calculate the limit values of the s-chart. Apart from that, the input is the same as for the xq chart.

If you save the specifications, the limits are calculated – if the *Calculate* checkbox has been enabled before.

As already mentioned, the user can enter the limit values directly without any specified values.

If the R-chart is selected instead of the xq or s-chart, the option σ/a_n is replaced by R_q/d_n in the specifications of the calculation. The estimate of sigma is now calculated using the specified mean range R divided by the correction factor d_n . This correction factor depends on the sample size n. The corresponding values are defined in the system and are selected automatically. Apart from that, the rest is the same as for the xq or s-chart. Note: In case of an R-chart, the calculation of the limit values is not based on a χ^2 distribution, but on a table stored in the system, which is based on standardized ranges.

Notes:

- You can only use the calculation specifications "relative/percentage deviation from xq" if you use an xq control chart. For this reason, the fields of the group "Xq" are only visible, if you have previously selected the xq control chart.
- The user must select the confidence interval (one-sided/two-sided). Usually, you do not define a lower limit for an s-chart. In this case, select "one-sided".
- If only one of the two tolerance limits is available, you cannot use the calculation method "cpk". The calculation formula of the cpk method requires both tolerance limits.

5.3 Control charts for attributive characteristics

The p- and u-charts are available for attributive characteristics.

p identifies the proportion of defective units in the sample and u identifies the failures/defects per unit in the sample. As to the p chart, it is important that each item is either defined as defect-free or defective. If an item has several failures/defects it is only once referred to as defective.

In contrast to the variable characteristics, there are no lower limit values. Furthermore, it is normally not necessary to state the values UTL, LTL and target value.

It is necessary to enter a pq or uq value in percent for the automated calculation of specifications. This can be done in the default values tab.

If you save the specifications, the limits are calculated – if the *Calculate* checkbox has been enabled before.

Calculation is respectively based on normal distribution. The value 99,725 has to be selected if, e.g. 99,725% of the characteristic values are supposed to lie below the upper action limit. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed with the respective action probability for information purposes.

5.4 Calculation of formulas

If you store a formula, you can automatically calculate measured values by way of measured values or statistical values of other characteristics that have been inspected before.

If the extension [QMSingleValue.FormulaArguments](#) is enabled, you have the possibility to use extensive arguments to calculate the single value you want to collect. In addition, you have more possibilities to access specification values and values of inspection results of other characteristics. For more details, refer to the section "".

If this extension is not enabled, you can only calculate characteristics using the inspection results of other characteristics that have already been entered. Find details in the section "Calculation via reference to other inspection results".

5.4.1 Operators, functions and constants

The following operators, functions and constants for calculating measured values are supported:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the natural logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the common logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x to the power of y
Constants	
pi	3.141592654
e	2.718281828

If constant numeric values are used in formulas, you must be careful not to use thousand separators. If these constants are floating point numbers, be careful to use a dot as decimal separator instead of a comma.

5.4.2 Formulas referring to other inspection results

Formulas including a reference to other inspection results are always calculated when an inspection result referenced in the formula is created, changed or deleted.

For these characteristics, you must first specify the level of the formula calculation. The following types are available:

- **V – Calculation on the level of single values (Value).**
For each single value of the characteristics involved, one single value is generated for the calculated characteristic.
- **S - Calculation on the level of samples (Sample).**
For each sample of the characteristics involved, exactly one single value is generated for the calculated characteristic.
- **C - Calculation on the level of characteristics (Criteria).**
Exactly one single value is generated for the calculated characteristic (with respect to the overall statistic of all characteristics involved)

The actual formula follows this identifier (see previous chapter).

The following syntax applies for the variables identifying the single values or statistical values of the order characteristics involved [x:y:z].

The x parameter identifies the statistical value to be used. The available values are listed below. Please bear in mind that the calculation level might cause restrictions.

- **X – Single value**
(is only available for calculations on the level of single values)
- **AVG – Mean value**
(is only available for calculations on the level of samples or characteristics)
- **MIN – Minimum**
(is only available for calculations on the level of samples or characteristics)
- **MAX – Maximum**
(is only available for calculations on the level of samples or characteristics)

- SUMX – Sum of single values
(is only available for calculations on the level of samples or characteristics)
- R – Range
(is only available for calculations on the level of samples or characteristics)
- S – Standard deviation
(is only available for calculations on the level of samples or characteristics)
- N – Sample size
(is only available for calculations on the level of samples or characteristics)
- M – Number of samples
(is only available for calculations on the level of characteristics)

The y parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- SENO – identification via the OP sequence of the characteristic (serial number)
- INCR – Identification via the characteristic number (inspection criteria)
If the characteristic number is not unique within the inspection requirement, it is not predictable which one of the applicable characteristics is used at the time of calculation.

The parameter z identifies the characteristic using the field content defined by parameter y . Either the OP sequence or the characteristic number of the calculation source is entered in this field. If the characteristic number includes a space character, it should be replaced by an underscore within the formula.

Example 1:

A new characteristic is calculated from the single values of the characteristic assigned to the number "LENGTH"/"LAENGE" divided by 2.5. A corresponding single value is supposed to be calculated for each single value of the source characteristic (calculation on the level of single values).

→ Formula: V: [X:INCR:LAENGE] / 2.5

Example 2:

The characteristic "surface" results from the product of the characteristics with the characteristic number "LENGTH"/"LAENGE" and "WIDTH_TOTAL"/"BREITE_GES". A single value of the characteristic "surface" is supposed to be calculated for each single value of both source characteristics (calculation on the level of single values).

→ Formula: V: [X:INCR:LAENGE] * [X:INCR:BREITE_GES]

Example 3:

The characteristic "maximum margin width" results from the subtraction of the minimum of the characteristic "inside diameter" (OP sequence 10) from the maximum of the characteristic "outside diameter" (OP sequence 20). A single value of the characteristic "maximum margin width" is supposed to be calculated for each sample of both source characteristics (calculation on the level of samples).

→ Formula: S: [MAX:SENO:20] - [MIN:SENO:10]

For the calculation of formulas including references to other inspection results, it is allowed to calculate new formula characteristics that are based on calculated formula characteristics. However, this nesting may not have more than 10 references one below the other. Furthermore, double concatenations must not be created (Example: characteristic A is calculated from characteristic B and characteristic C; characteristic C is calculated from characteristic A).

5.4.3 Extended formulas

The extended formulas provide the following advantages compared to the formulas including references to other inspection results:

- You can enter arguments for these characteristics that are used to calculate the measured value. In most cases, you do not need to use other "source characteristics".
- On saving the inspection result, the measured value is calculated and is immediately available. You do not need to refresh the measured values in the AIP to see the measured values.
- For the calculation, you can optionally use single values or sample or characteristic statistics of other characteristics. You may combine these in any way.
- You can use variables for the target value, the upper and the lower tolerance limit of the current characteristic or of other characteristics in the formula.

For this reason, you should primarily use the extended formulas.

The following syntax applies for the variables identifying the single values, statistical or specification values of the order characteristics involved [x : y : z] .

The x parameter identifies the statistical value to be used. The available values are listed below. Note: Depending on the respective shop floor client used, it is possible that not all 10 argument fields are available.

- VAR1 – Argument 1 of the inspection result of the own inspection step characteristic
- VAR2 – Argument 2 of the inspection result of the own inspection step characteristic
- VAR3 – Argument 3 of the inspection result of the own inspection step characteristic
- VAR4 – Argument 4 of the inspection result of the own inspection step characteristic
- VAR5 – Argument 5 of the inspection result of the own inspection step characteristic
- VAR6 – Argument 6 of the inspection result of the own inspection step characteristic
- VAR7 – Argument 7 of the inspection result of the own inspection step characteristic
- VAR8 – Argument 8 of the inspection result of the own inspection step characteristic
- VAR9 – Argument 9 of the inspection result of the own inspection step characteristic
- VAR10 – Argument 10 of the inspection result of the own inspection step characteristic
- X – Single value of another characteristic
- AVG – Mean value of the sample of another characteristic
- MIN – Minimum of the sample of another characteristic
- MIN – Maximum of the sample of another characteristic
- SUMX – Sum of the single values of the sample of another characteristic
- R – Range of the sample of another characteristic

- S – Standard deviation of the sample of another characteristic
- SREL – Relative standard deviation of the sample of another characteristic
- N – Sample size of another characteristic
- AVG_ALL – Mean value of all samples of another inspection step characteristic
- MIN_ALL – Minimum of all samples of another inspection step characteristic
- MAX_ALL – Maximum of all samples of another inspection step characteristic
- SUMX_ALL – Sum of the single values of all samples of another inspection step characteristic
- R_ALL – Range of all samples of another inspection step characteristic
- S_ALL – Standard deviation of all samples of another inspection step characteristic
- N_ALL – Total sample size of all samples of another inspection step characteristic
- M_ALL – Number of samples of another inspection step characteristic
- TV – Target value of an inspection step characteristic
- UTL – Upper tolerance limit of an inspection step characteristic
- LTL – Lower tolerance limit of an inspection step characteristic

The *y* parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- SENO – identification via the OP sequence of the characteristic (serial number)
- INCR – Identification via the characteristic number (inspection criteria)

If the characteristic number is not unique within the inspection requirement, it is not predictable which one of the applicable characteristics is used at the time of calculation.



The characteristic number must not include any special characters. A minus sign "-" is not permitted, for example.

- **SELF** – Identification of the own calculated characteristic

The characteristic that is to be calculated identifies itself. Only in this case, the parameter *z* is not required.

Note: You may only use the identification of the own characteristic for the argument fields and for the target value and the tolerance limits.

The parameter *z* identifies the characteristic using the field content defined by parameter *y*. Either the OP sequence or the characteristic number of the calculation source is entered in this field. If the characteristic number includes a space character, it should be replaced by an underscore within the formula.

Example 1:

The measured value of the current characteristic is calculated from the sum of the argument fields 1 to 4.

→ Formula: `[VAR1:SELF] + [VAR2:SELF] + [VAR3:SELF] + [VAR4:SELF]`

Example 2:

The characteristic is the result of the product of the maximum measurements of the inspection step characteristics with the characteristic numbers 'LAENGE' and 'BREITE_GES' ('LENGTH' and 'WIDTH_TOTAL').

→ Formula: `[MAX_ALL:INCR:LAENGE] * [MAX_ALL:INCR:BREITE_GES]`

Example 3:

The measured value is calculated from the sum of the following three summands:

- Content of argument field 1
- Middle of the tolerance of the current characteristic
- Sample mean value of the characteristic with OP sequence 10

→ Formula: `[VAR1:SELF] + (([UTL:SELF] + [LTL:SELF]) / 2) + [AVG:SENO:10]`

Note the following when using extended formulas:

- Contrary to the formulas including references to other inspection results, the measured values are not calculated when the "source characteristics" are changed. The measured values are only calculated, if the inspection result of the respective calculated characteristic is explicitly collected or changed (e.g. via the argument fields).
- When the inspection result is saved, the system must be able to identify valid values for all variables used in the formula (single values, sample or characteristic statistics, specification values of other characteristics, all used arguments).
Otherwise, an error message occurs and the inspection result is not saved.
- You cannot directly edit the calculated measured value. The measured value is always the result of a calculation.
- If you use the parameter [X:...], the respective single values of other characteristics are searched for using the absolute single value and sample number. For the current characteristic, the parameter [X:...] is not available.
- If you use the statistical parameters [MAX:...], [MIN:...], [AVG:...], [SUMX:...], [R:...], [S:...], [SREL:...], or [N:...], the respective statistical values are searched for using the absolute sample number. Here, you cannot use statistical parameters of the own characteristic.
- If you want to use the statistical parameters of the complete characteristic using the parameters [MAX_ALL:...], [MIN_ALL:...], [AVG_ALL:...], [SUMX_ALL:...], [R_ALL:...], [S_ALL:...], [M_ALL:...], or [N_ALL:...], only the data of other characteristics is available (not the data of the own characteristic).
- Via customization, extensions can be made available to obtain any variables in the syntax [VAR:<Object>:<Identifier>].
- You cannot use characteristics that include extended formulas as sources to calculate formulas including references to other inspection results. But you can use these characteristics for other characteristics with extended formulas.

5.4.4 General notes on calculated characteristics

If unknown variables are used within a formula (faulty parameters x and/or y), the escalation CPAUMW.CALCULATED_CRITERIAS_GET_VARIABLE_VALUE is triggered.

If problems occur on assigning an identified value to a variable of the formula, the escalation CPAUMW.CALCULATED_CRITERIAS_SET_VARIABLE is triggered.

Both actions described require the escalation management license.

Tool numbers, machine numbers, cavity numbers or similar information are not stored for the calculated single values.

To transfer a corresponding number (batch number, sample number, serial number, etc.), all source samples of the calculation must be assigned the same number. If there is no number that is assigned to all source samples, you cannot assign a number to the calculated sample. If several numbers are found that have been assigned to all source samples, only the first number found is assigned to the calculated sample.

This function only applies to numbers, which have been assigned on sample level.

5.5 Last off inspection

As part of the function extension for the in-production inspection, the function of the last off inspection is available. For this function, you must have created the CAQ system option 1222 manually as a precondition. For details, refer to the procedure document "[Configuration_OM_Options.pdf](#)". The documentation of the CAQ system option specifies which characteristic user fields must be created (master data characteristic, inspection plan characteristic and inspection step characteristic), so that you can specify a characteristic for a last off inspection.

To specify a characteristic for a last off inspection, the user field of the last off inspection must have a content.



The function "Last off inspection" is not offline capable. And you cannot use the last off inspection with operations that are specified as *Inspection OP* via processing code.

If the operation is logged off or interrupted in offline mode, the buffered activities/postings are processed one after the other when the online mode is restored. This has the effect that the operation is logged off or interrupted although the last off inspection is missing.



The processing code generally defines if any check for a defined last off inspection is performed at all during logoff/interruption. If the processing code in tab *Quality* is set to *Inspection OP*, no check and no last off inspection is performed.

If an inspection step has been "logged on" with the logon of an operation on the AIP, the system proceeds as follows for the last off inspection when the operation is logged off or interrupted.

1. The system checks if an inspection point with cause for creation *Last off inspection* exists for this inspection step at the workplace in question. It does not matter if the inspection point is completed or not. If the check is also performed with an interruption, the system checks if an inspection point with cause of creation *Last off inspection* has been created since the last logon. If an inspection point is found, the operation is interrupted or logged off.
2. If no inspection point is found in item 1, the system checks if the relevant inspection step logged on includes characteristics with the inspection due date *Last off inspection*. If this is not the case, the operation is logged off or interrupted.

3. If the processes described in item 1 and 2 have the result that an inspection point must be created with the cause of creation *Last off inspection*, the following message is shown:

Last off inspection is missing!

Enforce posting using the option "posting required"?

Using the option *Posting required*, you can perform the logoff/interruption without having made a last off inspection. Use the CAQ system option 1154 to activate the logoff/interruption using the option *Posting required*.

4. If the operation is not logged off or interrupted because of the missing inspection point with the cause of creation *Last off inspection*, you must go to the inspection to create an inspection point with the cause of creation "Last off inspection". You can use the option *Last off inspection* in the inspection list on the level *Inspection step* to create an inspection point with cause of creation *Last off inspection*.

If you try to log off or interrupt the operation that includes an inspection point with cause of creation *Last off inspection* that has not been completed, the standard processes apply. This means that the operation can optionally be logged off or interrupted although the last off inspection has not been performed/completed using the option "Posting required".

6 Specification List

1.1 Summary

Menu	Master data → Quality management → Specification list
Transaction code	sclq
Function authorization	sclq

Utilization

The specification list has been designed to create specifications, e.g. within the framework of family inspection planning.

To be able to use the specifications of the specification list for the generation of an inspection step with inspection step characteristics, the property "from list" has to be set in the corresponding inspection plan characteristics. In this case, only active entries of the specification list are used. A specification list entry is searched on the basis of the following key fields, when generating an inspection step.

- Area
- Article number, drawing issue number
- Resource number
- Characteristic no.
- Operation number and operation designation
- Customer no.
- Supplier number

The order of searching the specification list may be configured while customizing the system.

The specification list does not replace inspection planning. It rather is a supplement to family inspection plans.

Integration

This function is a fundamental components of family/group inspection planning, as the inspection specifications that vary with each item in the article group of a group inspection plan are defined in the specification list.

Prerequisite

An inspection plan for the article group including corresponding characteristics referring to the specification list has to exist to be able to use specification lists (configuration within inspection plan characteristics: "from list").

Selection criteria

The application provides the following selection criteria:

Area

List of available CAQ areas

Specification no.

Unique specification number

Version no.

Unique version within the specification. Only active, provided that the version control function has been activated within system settings.

Active

Active or inactive entries are filtered.

Special case

Special cases/normal cases or both are filtered.

Article number

Article number of the specification list entry; can be selected from the catalog for article master data.

Article designation

Article designation

Customer number

Customer number can be selected from the company catalog

Customer name

Customer name

Supplier number

Supplier number can be selected from the company catalog

Supplier designation

Supplier name

Operation

Operation number

Workplace

Workstation, e.g. machine

Please note: The definition of machine-related specification list entries is not supported by default (standard).

This field can be used for customer-specific implementations.

Characteristic no.

Characteristics number

Characteristic designation

Characteristic designation

Characteristic type

The characteristic type is filtered: variable, attributive, inspection chart

If several selection criteria are used overlapping results are displayed in the specification list entries.

Field descriptions**"Characteristic key" tab*****"Characteristic" group*****Area**

Specifies the area for which the specification list entries are to apply.

Specification no.

Number of the specification

Version no.

Version number of the specification

Active

Checkbox. Shows whether the entry is active or inactive.

Workplace

Indicates the workplace.

Please note: The definition of machine-related specification list entries is not supported by default (standard).

This field can be used for customer-specific implementations in order to define machine-related inspections for the same article. This might be required if different types of machines are used.

Machine designation

Designation of the workplace/machine

Resource

A resource (e.g. tool) may be indicated.

Designation

Name of the resource

Characteristic no.

Unique number of the characteristic

Characteristic designation

Name of the selected characteristic number

Operation

Number of the operation

Operation designation

Designation of the operation

"Properties" group**Characteristic type**

Variable, attributive or inspection chart

Input type

Specifies whether data is collected manually or automatically.

Special case

Indicates whether this characteristic does no longer need to be checked after x "pass" inspections in a row.

Number of pass inspections in a row

The number of "pass" inspections in a row that is required for the characteristic to be no longer required to be checked.

No characteristic

Specifies whether this characteristic is not required to be checked for the combination of the indicated key fields.

"Article" group**Article number**

Article number for this specification list entry

Article designation

Designation of the selected article number

Drawing issue number

Drawing issue number of the selected article number

"Companies" group**Customer number:**

Customer no.

Customer name

Customer name of the selected customer number

Supplier number

Supplier no.

Supplier name

Supplier name of the selected supplier number



The key fields of the "characteristic key" tab can no longer be changed if a list entry is changed (edited).

"Specifications" tab

Go to

The fields included in the "specifications" tab correspond to those of the characteristic master data and are described in the documentation [MOC_CharacteristicsQm.](#)

"Chart 1/2" tab



Go to

The fields included in the "chart 1/2" tab correspond to those of the characteristic master data and are described in the documentation [MOC_CharacteristicsQm.](#)

Default values chart 1/2

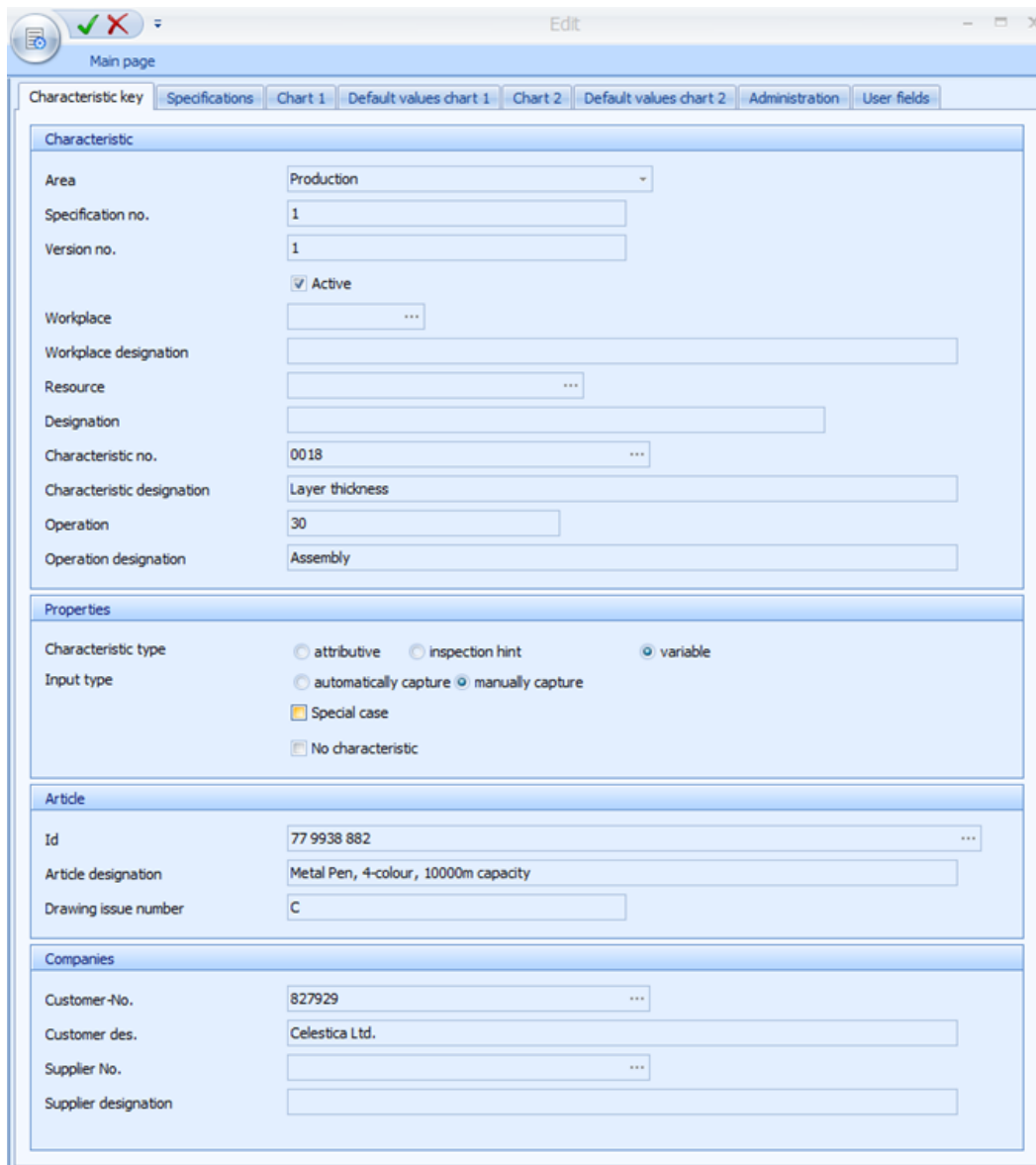


Go to

The fields included in the "default values chart 1/2" tab correspond to those of the characteristic master data and are described in the documentation [MOC_CharacteristicsQm.](#)

Editing functions

The following dialog opens to edit a data record:



Toolbar

The below-mentioned additional functions are available besides the standard functions.



Activate

Function authorization: sclq.active

Activates a specification list entry. A previously released version is automatically deactivated.

**Deactivate**

Function authorization: sclq.release

Deactivates a specification list entry. The specification list entry is no longer used.

7 Workplace and Resource Configuration

Overview

HYDRA menu	Master data → Resources → Resource configuration Master data → Workplaces/machines → Workplace configuration
FEDRA menu	Detailed Scheduling → Master data → Resource configuration
Transaction code	res
Function authorization	mdres mdresgenh for fields in combination with Test Equipment Management

Available user fields		
Where?	Object type/user field key	Source (type)
Tab <i>User fields</i>	<Res.type*>/depending on data record	Resource (MF-D)
Table	RES/SYSTEM	Resource (MF-D)

*) <Res.typ> = resource type

The resource configuration is the central function to manage resources in the MES.

Purpose

This application manages the master data of workplaces/machines and other resources (tools, DNC resources, etc.). The resource type classifies resources. Each resource type is also linked to specific functions and applications, which provide further functionalities of the MES for resources of the specified type.

Integration

Use this application to view the resource information of all resource types available in the system. The resource type also specifies how and if data records can be edited. Depending on the resource type, you cannot edit all fields or create and delete all resources.

Based on the resource type, the MES also includes further applications that are especially tailored to these types. The *machine data collection* application package, for example, is based on resources of the type "machine".

In addition to the resource configuration, the *resource overview* application is available. You cannot use the *resource overview* application to edit data. This application only allows administrative operations for the daily handling of resources such as the stock transfer of resources.

Requirements

Create a year model/shift calendar prior to creating a workplace or machine. If you want to use the various resource types effectively, you also need the advanced licenses for these types.

Selection criteria

The application provides the following selection criteria:

Resource from ... to ...

This selection criterion refers to the resource. You can also use wildcards (placeholders *).

Short name

Short name of the resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Workplaces and machines always have the resource type MNR. But you can assign individual resource types to the other resources by configuration. Predefined resource types include:

DNC	NC/DNC program
DOC	Document
ENE	Energy meter
ENT	Removal device
ENT	Removal device
MNR	Workplace/Machine
PAC	Packaging, transportation container
PRM	Test and measuring equipment
PER	Production staff / general
PRU	Setup staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

We recommend using the predefined resource types.



The displayed detail resource information varies with the resource selected in the table overview.

Name

Name of the resource.

Group

Workplace/machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of the resource.

Short name

Short name of the resource.

Resource family

Family the resource is assigned to.

Responsibility area

Responsibility area the resource is assigned to.

Storage location

Regular storage location of the resource.

MD user fields

MD user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

Field descriptions

This detail application includes four main tabs:

- Resource configuration
- Resource list
- Resource attributes
- DNC versions

Main tab *Resource configuration*

Here, you can define the configurations and master data of resources.

General tab**Resource type**

Resource type of the resource. The system delivery includes some default resource types. Create additional resource types in the application .

Resource

Enter the number of the resource or workplace to be collected in this field.

The resource type also specifies the maximum number of characters that are allowed for the resource number:

- Resources of the type MNR: a maximum of 8 digits
- Resources of a type <> MNR: a maximum of 20 digits

Permitted characters: ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890_+-. Do not use spaces and other special characters. For technical reasons, you can enter * (asterisk) and % (percent), but they are nonetheless not permitted because they are not valid characters. When you exit the input field, the system automatically converts lower case letters into CAPITAL LETTERS.

Please note for workplaces/machines (resource type MNR):

For technical reasons, the system does not check the maximum number of digits allowed for resources of the type MNR. For this reason, make sure that the resource number length (= workplace/machine number) does not exceed 8 digits.

Please note: If you set the resource type MNR before entering the resource ID (machine number), the GUI only allows you to enter eight digits.

If you select the option "numeric machine number" (basic parameter settings) for use with DOS terminals, you must ensure that the resource number (= workplace/machine number) only includes numerical digits and that its length is exactly 8 digits. If necessary, prefix leading zeroes to the number to extend it to eight digits, when creating the workplace/machine.

Short name

Short name of the resource. Only use this field with workplaces/machines (resources of the type MNR).

Name

Use this field to assign a short, unique name to each resource. Reports and overviews as well as terminal dialogs show this name, which is also useful for orientation purposes.

Responsibility area

Use responsibility areas to restrict the data users can view in different evaluations/reports. Users can only view the data they are allowed to according to their responsibility area authorization.

The *responsibility area* field can also remain empty. In this case, the resource is always displayed regardless of the user's assigned responsibility authorizations.



If you leave the *responsibility area* field empty, the system automatically enters the value "--DEFAULT--" in the field. Resources including this value are always displayed regardless of the user's assigned responsibility authorizations.

Cost center

This field includes the cost center the resource is assigned to.

Inventory number, engraving number, drawing number, manufacturer, owner

Additional information in form of comments.

Acquisition date, acquisition costs

Additional information in form of comments.

Configure the currency for the entire system in the basic settings.

Storage location

Location where the resource is stored when it is not being used (original storage location).

In connection with the Material and Production Logistics (MPL) product group, this field specifies a material buffer. If you log on an input batch, the logged on input batch(es) will be transferred from the previous material buffer to the material buffer entered in this field (upstream of the machine).

Delivery date, start-up date, guarantee date

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

External designation, resource type designation, usage, purchase order number

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Supplier and party in charge including detail fields

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Workplace configuration tab

This tab is only available if you select a resource of the type "MNR".

Workplace master data**Workplace category**

N Machine

P Workplace

Defined as machine or workplace. If you exclusively use BDE and/or MDE and PDV, the two categories are identical as regards processing.

J Machining center (BDE-BEA only)

The "Machining center" category and its functionality are described in detail in the BDE-BEA product documentation.

L Line (MDE-SFL only)

A Aggregate (MDE-SFL only)

The categories "Aggregate" and "Line" and their functions are described in detail in the MDE-SFL product documentation.

Q CAQ inspection station

Workplace is defined as mere CAQ inspection station and does not affect BDE or MDE statistics.

R Coil-based manufacturing (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

S Cutting unit (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

D Parallel output batches (only MPL)

You can produce parallel output batches on the machine for an operation that requires batch management.

C Packing station (only MPL)

You can use specific posting functions of the machine to represent a packing station. The functions are described in detail in the AIP-LCS product documentation.

M Melting aggregate

This option defines a machine as melting aggregate in terms of composition.

F Laboratory/in-production inspection

This workplace is configured as inspection station. The inspection points are displayed, which are assigned to this workplace or machine group of this workplace because of the higher-level inspection point. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

```
<MachineSpecifiedLayout>True</MachineSpecifiedLayout>
```

W Goods receipt inspection

This workplace is configured as inspection station. The goods receipt inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

```
<MachineSpecifiedLayout>True</MachineSpecifiedLayout>
```

K Calibration

This workplace is configured as inspection station. The calibration inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

```
<MachineSpecifiedLayout>True</MachineSpecifiedLayout>
```

Workplace type**E** Single workplace (SWP)**G** Group workplace (GWP)



Group workplaces are workplaces without machine data collection or MDE evaluations.

In case of group workplaces, you cannot post to resource performance accounts in an operation-related manner with postings based on the current machine status. Only main production times (RPA 11) are recorded. You must define a status with the control indicator "production" in the .

The system does not generate for group workplaces. Therefore, MDE evaluations that evaluate MDE log records are not possible.

Like single workplaces, you can assign group workplaces to terminals. In this case, you have to make sure that the is set to operation mode "BDE" or the option *Processing* is set to "BDE processing" in the .

External workplace

This field identifies external workplaces. Currently, it only functions as a comment.

Locked

If this option is checked, the machine/workplace has been (logically) deleted. In this case, the system does no longer permit the following changes:

- Order postings on the terminal
- Order postings on the MOC (e.g. using the "order overview" function)
- Changes when editing events

The graphic planning board of the Shop Floor Scheduling and the application *Workplace assignment* do no longer show the machine/workplace.

Blocked machines/workplaces are shown in evaluations and overviews. If blocked machines/workplaces are not shown, this is then described in the relevant documentation of the MOC application.

Tip: In applications where data is selected according to the responsibility area authorization, you can hide machines/workplaces if you remove the responsibility area.

Company

Use this field to differentiate the individual machines/ workplaces. The system can use this field for evaluation purposes.

Group

Use this field to assign the workplace/machine to a logical group. In planning, this is a capacity group. Capacity groups combine primary capacities.

If you create a new workplace, it is automatically assigned to a group of the same name (menu BDE: Master data > Workplaces/machines > Groups), which is defined as a capacity group. If the capacity group does not yet exist, the system automatically creates a capacity group and assigns the workplace.

Category

Enter the category of the machine. By means of this, you can enable a validation check according to the BDE configuration: *Master data > Order configuration > Order types*, tab *validation*, option *Check planned workplace/group/category on OP logon (value category)*.

Year model

Enter a valid year model. The times to be posted are compared with this shift model when they are recorded. If you have not defined a planned year model in the HLS tab, the shift model entered here is also used in the Shop Floor Scheduling.

Standard rate, machine

Enter the arithmetical standard rate of machines for calculations. The Shop Floor Scheduling uses this value for some (evaluated) KPIs.

Standard labor rate

Enter the arithmetical standard labor rate for calculations. The Shop Floor Scheduling uses this value for the KPI "Evaluated labor utilization".

Performance level

You can enter the performance level of the workplace/machine in percent in this field. The Shop Floor Scheduling and the evaluation of material requirements integrate this value when calculating the remaining run time.

Incentive wage indicator

Defines the type of calculation used for incentive wages. This option is mostly used in combination with the incentive wages based on formulas for customer-specific configurations. In addition, use the "incentive wage indicator" as selection criterion for the *wage type determination* to calculate incentive wages.

Leave this field empty, if you do not use the incentive wage module.

The *incentive wages indicator G=group calculation* has a special meaning. If this option is set for a workplace/machine, you have to assign a premium group every time you log on an order. You can do this either via

- the "assignment of premium groups" option of the product group *Incentive wages* or, optionally, via
- an additional field in the terminal dialog for the logon of orders. If no assignment is available, the system rejects the logon of the order by issuing a validation error.

Therefore, you may only assign the incentive wage indicator *G = Group calculation*, if the group premium conditions are met in the incentive wages calculation, as otherwise orders can no longer be logged on!

You can specify the meaning of the other incentive wage indicators according to your requirements while customizing the system.

File

You can assign a graphic to each machine/workplace. The *workplace overview* or the AIP shows this graphic, for example. The following image formats are supported: jpg, gif, tif, bmp, ico, emf, wmf.



In the path configuration, you must have configured the following:

- the path ID "MOCWPIMG" for the MOC or SMA
- the path ID "HYDRA" (also see) for the AIP. The file name length of graphic files is restricted to 12 characters (8.3 notation). Note for Linux installations: only use lower case letters for file names.

Maximum capacity (KG)

If a machine is configured as melting aggregate, define the maximum capacity in KG here.

Accuracy class, unit, etc.

Information fields in order to describe the accuracy. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Data collection

Display 3rd list

Use the options described here to show/enable a third list in the main view of a Windows terminal (CTWIN / AIP). You can switch between the respective terminal lists depending on the options set. The following settings are possible. Please note that the contents displayed in the lists depend on the product group in use:

- Input material (MPL): shows logged on input materials/ batches.
- Resources (WRM): shows logged on resources and tools.
- Staff (BDE): shows logged on staff.
- Output material (MPL): Produced output batches are displayed.

Show material/PRT list when OP is logged on

This option is only relevant in connection with the WRM module and the resources logged on to the Windows terminals (CTWIN / AIP).

If this option is set and you log on an OP, a specific login dialog opens. This dialog includes a list of components/production resources and tools. This list shows resources that meet at least one of the following requirements:

- the option "posting to terminal" is set in the resource type;
- the option "log on with OP" is set to "explicit logon" for the resource.
- the resource is a so-called "required resource" (option is set for the resource).

Please note: If the workplace is relevant for MPL, the list also shows material components.

Sequencing list

This option defines which operations are displayed in the sequencing list of the terminal. The following settings are available:

- S Basic setting. The system takes the value from the option of the same name in the *HYDRA basic settings*.
- M Pool of workplaces. The terminal sequencing list only shows the operations planned for the workplace.
- G Pool of workplaces and groups. The terminal sequencing list shows operations that are:
 - planned for the current workplace or
 - for another workplace of the group or
 - that are still located in the pool of groups.
- K Pool of workplaces and categories. The terminal sequencing list only shows the operations that are planned for workplaces of the selected category.
- H Group control. The terminal sequencing list shows the operations that are
 - planned for the current workplace or
 - for another workplace of the group.

Number of OPs in sequencing list

Enter the maximum number of operations that are to be displayed in the terminal sequencing list. Enter 0 if you want to show all operations.

Compulsory sequence

Use this option to specify if it is mandatory to log on the OPs in the planned sequence. The following parameters are permitted:

- N Disabled
- J Enabled

If the parameter is "enabled" and you log on an OP, the system checks whether the order backlog for this machine/workplace includes an OP that is planned for the same time or previous to this OP, but has not yet been started (i.e. status = V/prepared). If yes, the system rejects the logon of this OP.

Note: If you plan orders in the system using the *Order sequencing* (menu Production control → Production support → Order sequencing) and you configure the sequencing list with any other option than "M" (pool of workplaces) and you enable the *compulsory sequence*, this might lead to a combination that does not make sense.

Please note for the sequencing list:

- If the sequencing list includes operations that are in the status "interrupted", you can log on these OPs at any time, irrespective of the specified compulsory sequence.

Dialog control

To meet this requirement, define a dialog control that deviates from the standard behavior for the workplace in the dynamic dialog configuration of the Windows terminal (CTWIN / AIP). Then refer to the dialog control in the dialog.

Use this configuration only as part of customizing the HYDRA system. Otherwise the configuration is not relevant.

Logon of several OPs

Select this option, if several different operations should be processed on the machine. Otherwise, the system only allows one operation to be logged on to the machine.

Possible values:

Y Log on as many OPs as required at the same time.

Please note: The system allows a maximum of 20 operations to be logged on simultaneously to a machine, if the machine is assigned to a terminal with operation mode *MDE*. If more than 20 operations must be logged on at the same time, MPDV must review the conditions in order to remove the limitation. If MPDV agrees to remove the limitation, you can do so, otherwise search for alternative solutions. MPDV analyzes the conditions as part of a service.

N You can log on one OP only.

1...9 You can log on a maximum of n OPs.

Posting

Quantity posting to staff

Use this function to post the quantity of order interruptions/logoffs to the person who is logged on for the longest period.

Detailed information about quantity posting to staff can be found .

Posting for OPs that are not logged on

Use this option if you want to

- interrupt
- finish
- report part quantities for operations that are not logged on to this workplace.



If you record quantities for an operation that is not logged on, the system posts these quantities onto the operation in the BDE module. The MDE module does not post the quantities.

If you want to use this function with the AIP terminal, the BDE posting dialogs that are installed by default require the following:

- use the simplified BDE posting dialogs (the so-called "") or
- customize the dialogs.

Then you will be able to enter an operation that is not logged on.

Posting of machine time with simultaneously logged on operations

If this option is set and OPs are logged on simultaneously, the system posts the machine time proportionately onto the operations.

- | | |
|---|---|
| Y | Proportionate posting on OP according to the number of OPs |
| N | No proportionate posting. If the option is not set, the complete machine time is posted for each operation. |
| V | According to the default quantity of the OPs. Make sure that the default quantity (target quantity in primary quantity unit) of the operation is > 0. |
| Z | According to the standard time of the OPs. Make sure that the standard time (processing time) of the operation is > 0. |

Please note:

This option is also evaluated for group workplaces and in general you should better not use this option for group workplaces.

Automatic logoff of staff when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

Use this option to configure the personnel-related data collection at MDE workplaces. If you use HYDRA MDE, the terminals can generate fully automatic shift ends. You can configure here if

- the staff logged on to the workplace should be logged off automatically at the end of the shift or
- if they should remain logged on.

- | | |
|---|--|
| Y | Always log off staff when the shift ends. |
| N | Always save staff when the shift ends except for manual logoff. |
| X | Evaluate the person's settings. The system searches for the corresponding settings of the person . |

Automatic OP posting when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

- Y Interrupt and log on again at beginning of shift
- N Interrupt

Shop Floor Scheduling

Find further information about the HLS product group in the relevant HLS documentation.

Planning function

This option specifies whether a workplace or a machine will be displayed and if so, in which planning function.

- P Planning in the graphic planning board of the Shop Floor Scheduling or in the graphic order sequencing (GAV), i.e. you plan the workplace via the Shop Floor Scheduling or the graphic order sequencing; the workplace is then displayed in these applications, but not in the tabular order sequencing (AVG).
- Note: There are also other settings that specify whether a workplace is displayed in the Shop Floor Scheduling or in the graphic order sequencing:
- the workplace must be assigned to a group identified as a "capacity group"
 - you must be authorized for the responsibility area of this workplace
 - planning profile
- H Only relevant, if you use the HYDRA Shop Floor Scheduling module (HLS).
Like P.
- T Reserved
- A Planning in the tabular order sequencing (AVG), i.e. you plan the workplace using the AVG product group.
- N No planning; the tabular order sequencing (AVG), the graphic order sequencing and the HLS module do not show the workplace.

Planned year model

Here, you can enter a special year model only used for planning in the Shop Floor Scheduling. This year model does not affect data collection and posting in the product groups BDE/MDE. If you do not define a planned year model, the system uses the year model (*Master data* tab) for the planning.

Availability

Define the available capacity of a workplace/machine. The default value for the available capacity is 1000 [per mill].

In the Shop Floor Scheduling, the capacity check and automatic assignment assume that each operation has a capacity requirement of 1000 [per mill], i.e. exactly one operation can run on the workplace/machine at a time. In case of a manual multiple assignment, a dialog informs you about the double assignment. If you use the automatic assignment, multiple assignments are generally not feasible.

Use this setting to extend the availability of the workplace such that a multiple assignment is permitted. If the workplace capacity allows, for example, processing of two operations at the same time, set the available capacity to 2000 [per mill] in this field.

If nothing is entered in this field or if you enter the value 0, the system interprets this as the default value of 1000 [per mill].

This functions requires a corresponding license.

Check personnel availability

Choose from the following options:

- Check if at least one person is planned
- Check personnel availability
- Check personnel availability and qualification

When you operations in the , the system checks whether persons are planned in the application for the time of the scheduling You will find further information on the display of personnel capacities in the Graphic Planning .



This option is only available if you enable the extension .

MPL

For further information on the MPL product group, refer to the relevant MPL documentation.

Batch management

Activates the entry of the batch number for this machine within the terminal posting dialogs.

Possible values are:

- N No batch processing
- L Batch tracing (input/ output batches) as part of HYDRA MPL/TRT
- D Throughput batch processing as part of HYDRA MPL/TRT
- J Individual batch tracing (CHV)

The following functions are only available in connection with the product group Material and production logistics and are supported only by Windows terminals (CTWIN / AIP).

Preceding material buffer

Irrelevant.

Subsequent material buffer

If you specify a material buffer in this field, the field *Target buffer* in each of the entry dialogs (e.g. output batch change, log off operation) is automatically populated with this value.

If you do not enter a material buffer in the input dialog (e.g. deleted from the input field), the system automatically posts the output batch to the material buffer specified in the "subsequent material buffer" field.

Automatic generation of batch number

If you set this option, the system automatically generates a batch number for the output batch to be produced. Otherwise, the system expects you to enter the batch number for the new output batch to be produced, when you log on an operation or change the output batch.

Please note: If, in the field *Batch management* you set the option D (= *Throughput batch recording*), the system automatically sets the value for the *Automatic generation of batch number* to "J". In this case, you cannot enter the batch number manually.

Consumption balance

When you log off an OP, the system opens an additional dialog (V_BLZ) displaying the material components and their consumption quantities in relation to the OP that is currently logged on. In this dialog, you can also log off input batches that are still running. This option is only activated, once you have enabled the consumption balance for the material type of the output material.

Generate transport order for output batches

This option creates a transport order relating to batches for a generated output batch. The transport starts from the material buffer where the output batch is included. The configurations of the material type override the corresponding options of the resource configuration.

Generate transport order for input material

This option creates an article-related transport order relating to a material component, when you plan an operation for a machine via the Shop Floor Scheduling module. Transport starts from the output material buffer of the preceding operation. The configurations of the material type override the corresponding options of the resource configuration.

Quantities tab

This tab is only available if you select a resource of the type "MNR".

Conversion factors for base quantity

At the machine or workplace, you can collect the quantities in different quantity types and for different quantity accounts. In general, the system supports the following **quantity accounts**:

Yield

Scrap

Rework (Windows terminal CTWIN/AIP only)

Open quantity (problem quantity; Windows terminal CTWIN/AIP only)

The following **quantity types** are supported with each quantity account:

Primary quantity

Secondary quantity (Windows terminal CTWIN/AIP only)

Tertiary quantity (Windows terminal CTWIN/AIP only)

Basic quantity (Windows terminal CTWIN/AIP only)

The system design specifies the use of several quantity types or accounts. For example: If you want to enter the rework quantity manually, a corresponding input field must be configured in the input dialog (customization).

Use the quantity type "primary quantity" if you want to collect quantities automatically.

Quantity units and conversion factors for base quantity

Define a quantity unit for each quantity type. Use the alternative quantity accounts to enter data/quantities manually. In this case, the system does not convert quantities automatically.

If you do not enter data manually in the alternative quantity accounts, the server converts the quantities into the alternative accounts using:

- the conversion factors or
- the units that are configured in the MOC machine master data.



For further information on the conversion of quantities and examples, refer to the document

Basis for HYDRA-MDE quantity conversion

Define the basis for the quantity conversion.

- A Use the conversion factors of the OP that is logged on. If no operation is logged on, the system uses the quantity conversion stated in the machine/workplace configuration.
- M Use conversion factors from the workplace configuration for the quantity

conversion.

Units and conversion factors for base quantity (P)

Quantity unit (P)

Indicate the quantity unit you want to use for data collection at this machine/ workplace. If you collect quantities automatically, these quantities are generally primary quantities.

If you want to convert quantities automatically into another quantity type, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (S)

Quantity unit (S)

Indicate the secondary quantity unit you want to use for posting the quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (T)

Quantity unit (T)

Indicate the tertiary quantity unit you want to use for posting quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity

Quantity unit (B)

Indicate the base quantity unit you want to use for posting quantities to the workplace/machine.

Manual entry of quantities, yield

Manual entry of yield

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of yield

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting yield as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, scrap**Manual entry of scrap**

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of scrap

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting scrap as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, rework

Manual entry of rework quantity

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of rework

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting the rework quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, open quantity

Manual entry of open quantity

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of open quantity

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting open quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

"MDE configuration" tab

This tab is only available if you select a resource of the type "MNR".

Monitoring

Monitoring type

Choose from the following monitoring types:

Monitoring via operating signal

No monitoring

Cyclic monitoring

If you select **cyclic** or **operating signal** monitoring, you can only enter a malfunction if the terminal prompts you to do so ("Assign malfunction"). If you do **not** use **automatic monitoring**, you can enter a new machine status at any time.

If you use the **cyclic** monitoring option, the machine automatically switches to the "production" status when counting pulses occur. If you select the "operating signal" option, the machine automatically switches to the status "production" as soon as the operating signal is set. If you do not use the "automatic monitoring" option, you must assign the "Production" status manually.

Entry of disturbance reason required with specified delay time in [s]



You can only use this function, if the following requirements are met:

- it is a Windows terminal (CTWIN, AIP)
- The Process Communication Controller (PCC) does not run in stand-alone mode.

If the system identifies a downtime without a reason, the terminal opens the input dialog "Change machine status" after the specified delay time. If the terminal goes back into production, the window still remains open.

If you now enter a machine status (during production), this data input activates a transfer posting event that changes the most recently recorded status from "General disturbance" to the newly entered status. If this change is ok, the window closes; otherwise, it remains open.

However, if the system identifies the next downtime (with or without a reason), you can no longer change to the previously noted status. The window closes automatically.

If the system identifies another downtime without a reason and the delay time has expired, then the input window opens as described above.

If the system identifies a downtime without a reason and the machine switches to production before the delay time expires, then the terminal does not automatically prompt you to enter a malfunction reason.

Important note:

This change only affects the HYDRA Machine Data Collection. The system does not correct the resource performance accounts of the currently running OP online!

Please note for data maintenance:

The tabular event maintenance of the MOC shows all changed machine statuses. However, you cannot edit the transfer posting event as it is locked. In order to perform recalculations correctly with respect to orders and machines, change the original event with the status "NOT ASSIGNED" to the correct status. The transfer posting event does not affect recalculation!

Minimum malfunction time

Specify a time in seconds for the minimum malfunction time. This value defines the time that a malfunction/disturbance must continue before the machine changes from the status "Production" to the status "Not assigned".

If operating signals are monitored, the status is directly changed. You can use the following explicit option in the MDEB2.ini to disable this behavior (deactivation of direct status change). Result: the status is only changed when the **minimum disturbance time** has expired:

MDEB2.INI

```
[INIT]
;Activating the direct status change (globally or for a specific machine)
SetMStatusDirect=1
SetMStatusDirect@<machine number>=1

;Deactivating the direct status change (globally or for a specific machine)
SetMStatusDirect=0
SetMStatusDirect@<machine number>=0
```

Minimum cycle time

If you select the **cyclic** monitoring option, specify a minimum cycle time in seconds in this field.

The terminal uses this minimum cycle time and the target cycle that is stored with the (logged in) operation and that is set off against the cycle extension to calculate the maximum value. The terminal uses this maximum value as the default cycle time.

If both, the minimum cycle time and the target cycle stored for the operation, are 0, the default cycle time is set to 60000 seconds [per 1000 machine clocks].

Cycle extension

If you select the **cyclic** monitoring option, enter the percentage for extending the target cycle time in this field. Enter a value ranging between 0 and 5000.

The system offsets the target cycle stored with the (logged in) operation against this percentage. A value less than 100 is a shortened cycle; a value greater than 100 is an extended cycle.

Number of target cycles

If you select the "**cyclic** monitoring" option, enter the number of cycles (0 to a maximum of 9) after which the terminal automatically switches from a status unequal to "production" into the "production" status within the cycle time (requirement: the status that is unequal to production is not locked for the "production" status).

Some production processes provide machine cycles during the setup phase. Set a value greater than 0 in order to prevent the current machine status from changing immediately. Please note: The quantities you collect until the machine switches to the "production" status are neither posted as yield nor scrap.

Cycles to be evaluated

Reserved Enter 0 in this field.

Management

Posting during production lock

Use this setting to specify how to post the counting pulses that are collected while the status "production" is suspended. This configuration takes effect for all counters configured as "Yield".

Posting as scrap If this option is configured for the counter, the system offsets the counting pulses against the partitioning/ pulse factor and posts these pulses as scrap. Even if you defined another quantity account for offsetting, this one will not be used.

Posting as yield parts the system posts the counting pulses as yield

No posting the system does not post the quantities while the "production" status is suspended.

Pulse factor specific to machines

Use the pulse factor, for example, if you want to collect lengths (e.g. using a wheel).

Set the value to 0 for machines where a discrete or integral number of quantities (e.g. pieces) is collected per pulse. In this case, the pulse factor is not evaluated. That means, the number of cycles posted corresponds to the actual pulses transferred via the MSS (machine interface).

The MSS (machine interface) records the signals transferred from the machine (counting pulses). According to the configured number of pulses, the system calculates and posts the quantities as follows:

Quantity for the machine = pulse * partitioning for the machine/ pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation/ pulse factor for the operation

Please note: The pulse factor will be calculated as a fraction. When the quantity is calculated, the pulse is used as denominator and the partitioning is the numerator.

The system interprets pulses that occur during a malfunction or a production lock (configuration of *Posting during prod. lock > scrap*) as scrap. Also use the above-mentioned formula to calculate the scrap quantities.

Partitioning specific to machines

Enter the partitioning specific to the machine in this field. Multiply the machine-specific partitioning by the partitioning stored with the operation in order to integrate the machine-specific partitioning into quantity calculation. Enter the value 1 in this field, if you do not want this to happen.

Extended weekend automatic

If you select this option and the system is configured accordingly, the system assigns at the beginning of the shift the status that was available before status 999 was activated.

Note:

To use this option, the workplace must already be assigned to a terminal.

Find detailed information about the automatic activation of status 999 in the document .

Waiting period, short-term disturbance

Configure a *short-term disturbance* status for each machine/ workplace to improve the overview, e.g. in the machine history. Use this status as a "repository" for unconfirmed statuses, which only existed for a specific (short) period.

If the terminal automatically identifies a downtime and the machine automatically goes back into production, the system checks if this disturbance is shorter than the time period configured for short-term disturbances.

If this is the case, the still unfounded malfunction is justified with the status that is configured as the "short-term disturbance" status for the machine.

Inputs/ outputs

Machine lock/ Target quantity reached/ Machine downtime/ Free I/O

Enter the logical output where a digital signal should occur when the corresponding status is available.

Machine lock output	The system sets this output, if you enabled the option "set machine lock output" in the current machine status.
Target quantity reached output	The system sets this output, if the collected yield reaches the target quantity of the OP.
Machine downtime output	The system sets this output, if the machine is in a status unequal to <i>Production</i> . When changing to the <i>production</i> status, the system sets the output back to 0.
Free I/O	Free input/ output for customizations.

Use these statuses for connecting a monitoring light or a horn, for example.

Enter the corresponding number in one of the fields in order to assign an output and to specify which relay is interconnected by the terminal when the predefined status occurs. Enter "0" to prevent any action. Note that you cannot assign a terminal output more than once.

Please note

Specify the statuses that trigger the activation of the machine lock in the *Status assignment*.

Generally, enter the value "1" in the input field, when the machine lock is activated via the available relay output of a DS 100. In this case, the system sets the machine lock if

- a correspondingly configured status occurs and
- the status is not assigned.

Output batch change**

Customer-specific assignment of an input with an automatic output batch change (MPL). By default, enter 0 in this field.

PDE (Process Data Collection)

Collect process data

This parameter specifies if the system collects process data for this machine. If this parameter is not set for a machine, you cannot collect process data for this machine.

External connection



The AIP 8.2 and/or the PCC in stand-alone mode (MDE-Blade 2 Version 8.1.0.1) do no longer support the options marked with **. As they use other configurations for the connection.

External connection

If this machine is assigned to a master terminal the following connection options are available:

No external device	External devices are not connected
DS100	DS100 connection
Arburg control system**	Arburg connection
Engel interfacing**	Connection of Engel machines
MT3**	MT3 connection
PDE**	Process data collection

If you activate a DS100 or MT3** connection, you can select the field "device address". If you activate the option "Engel interfacing", you can select the field "serial number". If you activate the option "Arburg server system", you can select the field "class".

Note regarding the combination of connections on a master terminal:

"DS 100" and "No external device": allowed

"MT 3" and "No external device": allowed

"MT3" and "DS 100" not allowed!

Serial number (Engel interfacing)**

Enter the serial number of the connected Engel machine. Set the option "EMS machine interface" in the HYDRA basic parameter settings if you want to use Engel machines.

Device address

You can select this field, if you activate a DS100 or MT3** connection. Enter the device address of the sub-bus participant.

"Resource configuration" tab



For resources of type "MNR", only the fields marked with "*" are available:

- Family (section *resource master data*)
- Cycles (section *target utilization*)
- Runtime (section *target utilization*)

Resource master data

Type

Identifies the type of resource:

Resource: A resource can be uniquely identified, i.e. the resource is actually present. Its quantity is always 1.

Anonymous resource: An anonymous resource cannot be uniquely identified. If the identifier is set, then you can change the value in the field *Number* from 1 to another positive integer value. You cannot post data onto anonymous resources because anonymous resources do not relate to one specific resource.

Required resource: A required resource stands for one or more actual resources that can be identified. Specify in the configuration *WRM: Master data > Required resources* which resources are represented by a required resource. The number results from the number of actual resources assigned to the required resource.

Please note: If this field is empty, the resource is implicitly an ("actual") resource.

Equal type

Reserved for future modifications.

Version

Revision number; store here the program version for resources of the type DNC.

Quantity

You can only edit this field, if it contains an anonymous resource and the option *Anonymous resource* is set (see above). A value > 1 indicates how many of these resources are available.

This field is calculated automatically for required resources.

Family*

Assign a resource family. If you change the resource family subsequently, an information dialog appears as a warning because user fields might possibly be assigned via the resource family.

Target utilization**Cycles***

The field *Cycles* provides additional information. The cycles value defines how long the resource is to be used.

Runtime*

The field *Runtime* provides additional information. It defines how long the resource is to be used.

Input unit

Input unit

Absolute value limit (EMG 8.1, function authorization: resablim)

Enter the absolute value limit of the (meter) resource. The energy monitor shows this limit value in addition to the current meter reading. Use the Escalation Management to generate an escalation message, if the counter value of the resource exceeds the specified absolute value limit. You need the function authorization "resablim" to view this field.

Actual utilization

The periods when a resource was logged on to a workplace are the basis for posting the *cycles (clocks)*, *runtime*, *yield*, and *scrap* as *actual utilization*.

Clocks

The cycles (clocks) posted for the resource up to now.

Runtime

The total time in hours posted for the resource up to now. The total time is the sum total of all times posted onto RPA 1 to 11.

Yield (B)

The yield posted for the resource up to now (base quantity unit).

Yield (P)

The yield posted for the resource up to now (primary quantity unit).

Scrap (B)

The scrap posted for the resource up to now (base quantity unit).

Scrap (P)

The scrap posted for the resource up to now (primary quantity unit).

Configuration

Target cycle

Target duration in seconds for 1000 machine cycles if this tool is used.

Please note: The target cycle stored in the OP is relevant for the planning in the HLS module and for the machine data collection at the terminal.

Original partitioning

Partitioning of the tool (= number of cavities) when using this tool.

Current partitioning

Current partitioning of the tool. This value can deviate from the original partitioning, e.g. if the original quantity can no longer be produced with one cycle/clock due to a tool defect.

Always use the current partitioning to post cycles to the tool.

Please note: The partitioning stored in the OP is relevant for the planning in the HLS module and for the machine data collection via the terminal.

Partitioning due to cavities

If you set the option "partitioning due to cavities", the system (re-)calculates the fields "current partitioning" and "original partitioning" using the values defined in the cavity management. Then, you can no longer change the fields manually.

Log on with OP

Use this option to specify whether or not you want to log on the resource with the OP. To do so, the resource must be included as a component in the operation's list of production resources and tools.

Possible values:

None: The resource is not logged on.

Implicit: The system automatically (implicitly) logs on the resource that is assigned to the operation as a production resource and tool; you can neither log on the resource manually (explicitly) nor change the logon.

Explicit: You can manually (explicitly) log on the resource that is assigned to the operation as a production resource and tool or you can log on another resource instead. If you do not log on the resource or another resource explicitly, the system implicitly (automatically) logs on the current resource; in this way, the current resource serves as a "default".

Please note:

If you log on another resource explicitly (manually), this resource will be logged on for the resource that has the same *resource type* in the operation's list of production resources and tools. For this reason, you can only log on those resources explicitly (manually) that are included as a requirement in the operation's list of production resources and tools. In this way, you cannot log on a resource that is not included as a requirement in the list of production resources and tools (the resource must be entered in the list).

In general, you should not enable this option for the resource type DNC. The DNC product group handles this differently (NC programs are logged on separately).

The system also logs on resources that are defined in the BOM of the machine.

Parallel logon/ planning possible

You can log on/plan the tool simultaneously.

Please note: You can only log on a resource to one machine more than once. Consequently, the option "Parallel logon possible" refers to several different OPs logged on to one machine.

In this case, the system posts data proportionately as follows:

- Post quantities proportionally.
- Post times 100% for each resource. This means that the system posts double the time to the resource, if the resource is logged on twice.

Post to resource

Specifies whether or not the quantities and times are posted to the resource. Due to a high degree of complexity, you should only assign this option to those resources that you actually want to evaluate.

Collective lock

If you lock a lower-level (assigned) resource using the BOM function, the system sets a collective status for the higher-level resource. If this collective status is set, the system treats the higher-level resource as locked when a download request is made.

If you enable this function, the system passes the collective lock to the higher-level resource.

Planning**Setup time**

Duration in hours for setting up the tool.

Please note: The setup time stored in the OP is relevant for the planning in the HLS module.

Teardown/retooling time

Duration in hours for removing the tool.

Please note: The retooling time stored in the OP is relevant for the planning in the HLS module.

Assignment

Not used. The system uses the configuration option of the same name stored in the resource type to integrate the resource allocation in the HYDRA Shop Floor Scheduling.

Evaluation**Integrate in evaluations**

Reserved for future modifications.

File**File exists**

Shows whether or not the file is stored in the specified path. A cyclic process checks the files and sets the options subject to whether or not the file is available.

File name

File name; without file extension for DNC. The system adds the file extension according to the configuration in the resource type. The defined paths specify the storage location.

Comparison resources

Enter two comparison resources for energy consumption resources. They will then be shown in comparative evaluations/reports, e.g. the energy monitor.

Resource 1

Resource number of the resource to be compared.

Resource type 1

Resource type of the resource to be compared.

Resource 2

Resource number of the resource to be compared.

Resource type 2

Resource type of the resource to be compared.

Accuracy

Enter more detailed information on measuring accuracy and measuring range for test equipment resources.

Tab *User fields*

You can use user fields to store additional customer-specific information in the MES. The *user fields* tab includes eight sub-index tabs, which each has eight additional user fields. The so-called user field key specifies the available user fields and their meaning.

The workplace and resource configuration provides data of two basic object types. You can also edit this data in the workplace and resource configuration: on the one hand these are machines and workplaces and on the other these are the resources. Machines and workplaces are also "resources". But resources are not automatically machines and workplaces.

Object type

The system configures the user fields of machines/workplaces in relation to the object type "MNR". The system stores data contents to the machines/workplaces table and the resources table of the database to ensure data consistency.

The system configures user fields for resources in relation to the object type matching the resource type of the resource (example: create resources of the type "PAC" in relation to the object type "PAC"). The system stores data contents to the the resources table of the database.

User field key

Each user field key describes a combination of user fields. The management of the user field key (and therefore the meaning of the fields) is different for each object.

User fields

The following user fields are available after configuration:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16

Field data type	Number of fields
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

By default, no user field keys are defined. Configure the system accordingly to support this kind of user fields.



As the table shows resources of different types, use the user field key "SYSTEM" of the object "RES" to identify the column headings for the user fields.

Comment tab

Store additional resource comments in the "comment" tab.

Main tab Resource attributes

Shows additional resource attributes via the user field definitions of the resource family. Use the "resource attributes" button for editing.

Main tab Resource list

Shows the resource list for the selected resource. Click the "resource list" button to go directly to the BOM application for editing purposes.

Main tab DNC versions (available as of DNC 8.2)

Shows the available versions of a DNC resource including a flag indicating the currently applicable version. HYDRA provides this valid version for machine downloads.

Toolbar

General tab



Insert

Function authorization: mdres.create

Opens the dialog for adding a resource. This dialog provides the fields that match the selected resource type.



Copy

Function authorization: mdres.copy

Opens the dialog for copying an existing resource. Subject to the selected resource and its resource type, the copy function differentiates the following:

- Copy function for resources of **resource type = MNR** (workplaces, machines)
- Copy function for resources that do **not** have the **type MNR**

Copy function for resources of resource type = MNR (workplaces, machines)

From: resource type, resource, short name, name

- Resource type (fixed "MNR")
- Workplace/machine number
- Short name
- Name

of the workplace you want to copy. You cannot change these values. They derive from the selected data record.

To: resource type, resource, short name, name

- Resource type (corresponds to the resource type of the workplace you want to copy; cannot be changed).
- Workplace/machine number
- Short name
- Name

of the target workplace.

Copy machine status

Function authorization: mdmst.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy counter configuration

Function authorization: mdctr.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Note that the counter numbers of the new workplace are identical with the counter numbers of the workplace you copied. If necessary, you have to adjust the counter numbers.

Copy reasons

Function authorization: mdreas.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy function for resources that do not have the resource type MNR

The copy function for all resources that do not have the type MNR opens the "insert" dialog and takes over the details from the previously selected resource. But you can edit and change all fields.



Edit

Function authorization: mdres.edit

Opens the dialog to edit a resource and provides the tabs and fields of the relevant resource type.

As of MES Weaver 4.0_{pe}, you can change master data of several selected resources of the same resource type at the same time. You can select up to 10 fields and assign a value. You require the function authorization **mdresmm** to edit several resources at once.



Delete

Function authorization: mdres.delete

Deletes one or several selected resources.

Resource tab



Configuration – resource status

Opens the application "resource status" to define statuses for all resources that do not have the type MNR.



File - show file

Opens the file view – only available for document resources, which are configured as file-based resources without DNC processing in the *Resource type*. And only available if the relevant license and function authorization are available.



Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Go to – required resources

Opens the "required resources" application. The selected resource is entered as default value for the required resource.

**Go to – cavity assignment**

Opens the "cavity assignment" application. The selected resource is entered as default value.

**Go to - resource attributes**

Opens the application "resource attributes". The selected resource is entered as default value.

**Functions – Measures**

Opens the *Measures* application.

**Functions – Status change**

Opens the dialog to change a resource status. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.

**Functions – Release of resource**

Opens the dialog to release a resource. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.

**Functions – Stock transfer**

Opens the dialog to transfer/relocate a resource.

Workplace tab**Configuration – status assignment**

Opens the application "status assignment". The system enters the selected resource in the corresponding field.

**Configuration – counter configuration**

Opens the application "counter configuration". The system enters the selected resource in the corresponding field.

**Configuration – terminal assignment**

Opens the application "terminal assignment". The system enters the selected resource in the corresponding field.



Entry – reasons

Opens the application "reasons". The system enters the selected resource in the corresponding field.



Entry – Operator positions

Opens the application "operator positions". The system enters the selected resource in the corresponding field.



Entry – premium indicator

Opens the application "premium indicator". The system enters the selected resource in the corresponding field.



Groups - groups

Opens the application "groups". The system enters the group of the selected resource.



Groups – group assignment

Opens the application "group assignment". The system enters the selected resource in the corresponding field.



Miscellaneous – cycle parameter

Opens the application "cycle parameter". The system enters the selected resource in the corresponding field.



Miscellaneous - workforce requirements of workplaces

Opens the application "workforce requirements of workplaces". The system enters the selected resource in the corresponding field.

DNC tab

The tab is only available, if you select a DNC resource. These are resources configured as resources with DNC processing in the resource type.



Configuration – resource status

Opens the "resource status" application.



Configuration - assignment of DNC family to machine

Opens the application "assignment of DNC family to machine".



Copy resource attributes (as of DNC 8.2)

Copies values of resources attributes from one resource to another. Both resources must use the same user field key.



File - comparison editor

Opens the comparison editor for the selected resource or resources. See [below](#) for further information.



File - export

Exports the file specified for the resource. You use the file explorer to specify the target file.



File - import

Imports the file specified for the resource. You use the file explorer to specify the source file.



File - viewer

Opens the file specified for the resource using the defined viewer program.



File - editor

Opens the file specified for the resource for editing using the defined editing program.



Set valid version (as of DNC 8.2)

Only active, if you select a version in the *DNC versions* tab. The selected version is set as the new and valid version.



Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.



Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Functions – Status change

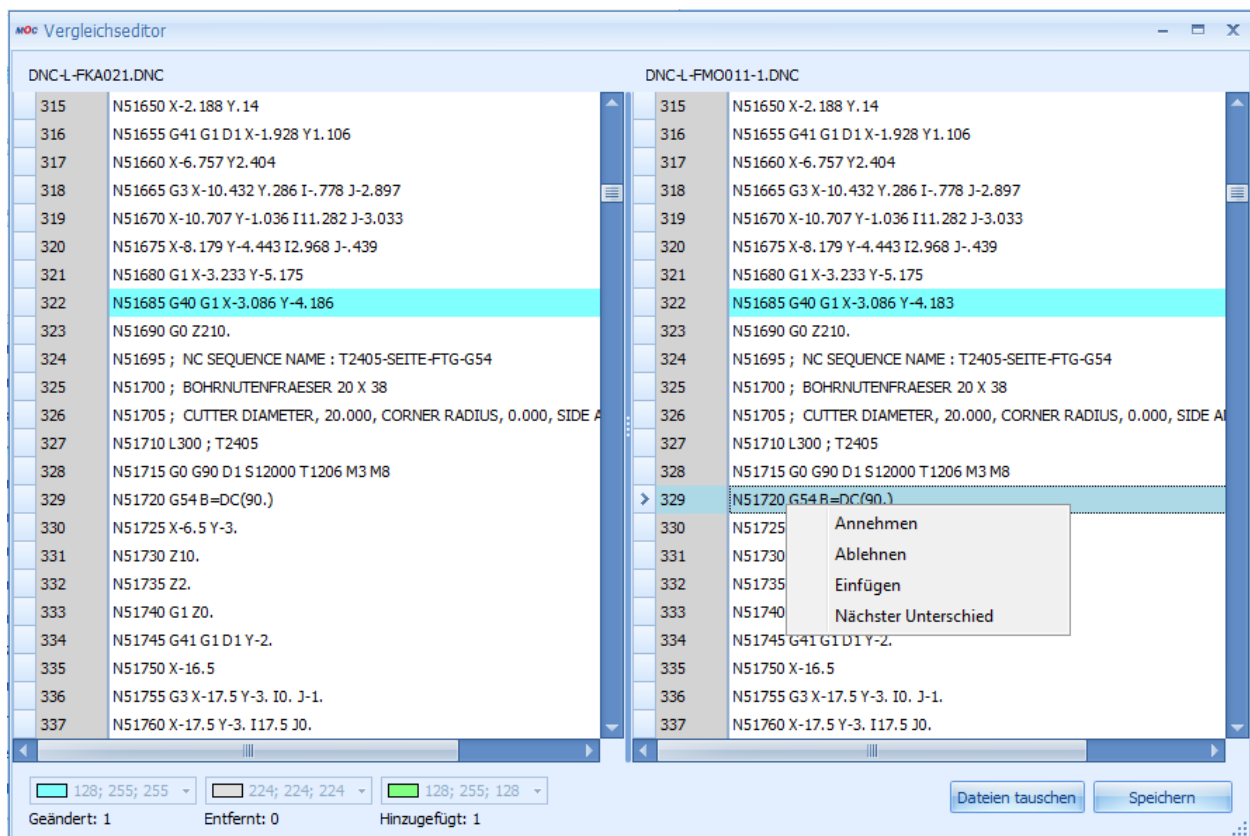
Opens the dialog to change a resource status.



Functions – Release of resource

Opens the dialog to release a resource.

How to use the comparison editor



The comparison editor compares the files attached to the DNC resources. Two operation modes are available:

Selection of one resource:

The editor shows the released resource and the optimized version of the resource for comparison. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor. You can only use this mode for DNC types with the file processing type "optimized".

Selection of two resources:

If you select two resources before you open the comparison editor, the editor compares the two selected resources. You can select the file type. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor.

Click the relevant buttons or use the context menu (right clicking) to start the functions of the comparison editor:

- Reject: Rejects the difference identified (on the right). Accepts the value from the left file. The editor does no longer highlight the difference.
- Keep: Accepts the difference identified (on the right). The editor does no longer highlight the difference.
- Next difference: Goes to the next difference.
- Insert: Inserts a row at the current position.
- You can always change the contents of a row. Click the row and enter a value. Press ESC to quit the row without changes. The editor then highlights the row as "changed".
- Swap windows: Click this button to swap the windows. This function is necessary if you compare two resources. The place where a resource is displayed results from the display order in the table; the system does not know, which resource must be changed. If you only select one resource, this button is not available as in this case you can only change the optimized program version.
- Save: Saves the changes made to the file on the left-hand side.

Processing notes for workplaces and machines

Configuration changes

Restart the terminal which the workplace/machine is assigned to in order for the terminal program to interpret the configurations or modifications made to this workplace/machine.

Deleting a machine/ workplace

In a first step, the system shows a confirmation prompt asking if you really want to delete the machine. If you confirm this prompt, the system makes an attempt to delete the workplace. You can only delete a workplace successfully, if:

- you have not yet collected data for the workplace;
- you have currently not assigned the workplace to a terminal or a line;
- you have currently not logged on operations to the workplace;
- you have not planned operations for the workplace.

If you delete the workplace successfully, the system also deletes all configuration data, e.g. status assignments, for this workplace.

Checking Business Parameter Containers (BSCs)

See for further details on how to check the system against business parameters.

8 Resource Families

Overview

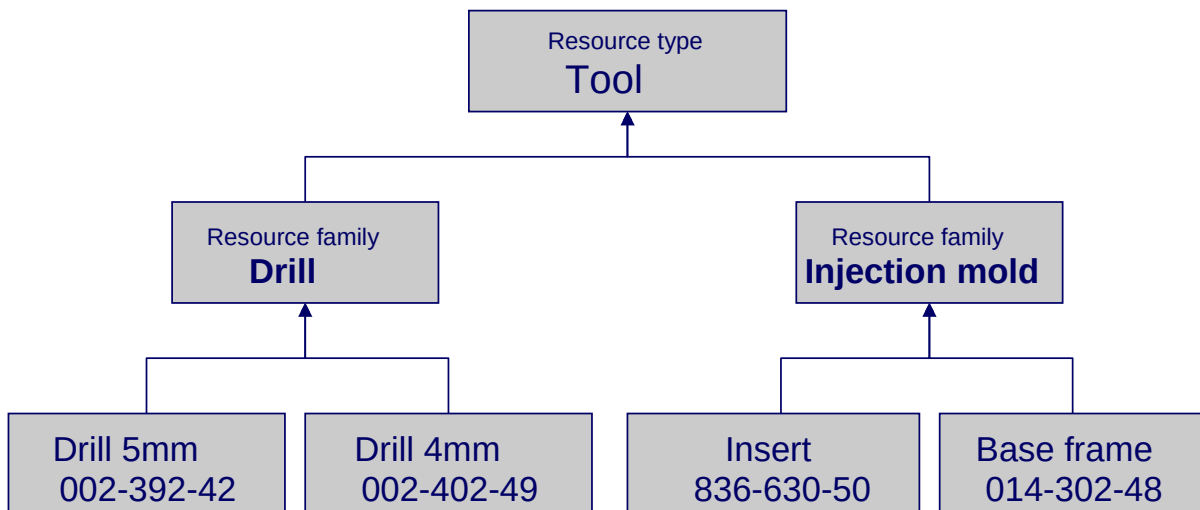
HYDRA menu	Master data → Resources → Resource families
FEDRA menu	Detailed Scheduling → Master data → Resource families
Transaction code	resfam.*
Function authorization	mdrfam

This document describes the application "Resource Families" on the client.

Purpose

If you look at the assignment of resources to resource types, you soon recognize that in a manufacturing company various resources of the same type exist that are possibly handled quite differently. This means that in general the classification by resource types is not sufficient to organize resources in a useful way.

If you define "resource families" (groups), you can introduce sub-classes of resource types. The diagram below illustrates how the resource type "Tool" is sub-divided into the two resource families "Drill" and "Injection mold". Each of the individual resources is assigned to one of the two resource families.



Integration

The resource families offer another structural level subordinate to the resource types. You can use resource types to define the master/detail user fields of resources. You can improve these master/detail user fields through definition in the resource families. In particular for DNC, you can use resource families as the main search criterion and assignment criterion for machines.

Selection parameters

In the selection panel, you can filter by superordinate or assigned resources. The application provides the following selection criteria:

Resource type

Type of resource.

Resource family

The resource family to which the resource is assigned.

Field descriptions

Resource type

Resource type to which the resource families refers.

Resource family

Unique, descriptive name of the resource family.

You can select this value in the various functions. Only the resource type allows you to identify a resource or its resource ID uniquely. That is why, evaluations also show the resource type of the resource.

Description

This field includes the description of the resource family; serves as a comment.

Responsibility area

Definition of the responsibility area. If you specify the responsibility area for a resource family, you also specify the responsibility area for the assigned resources. The responsibility area controls the visibility and editing options for these resources.

Field description for tab *General*

User field key

Reference to a valid user field key. The user field key entered here overwrites the entries in the resource type.

Note regarding DNC filtering using a DNC family and its search fields (when using HYDRA only):

The definition of suitable user field combinations is important if you want to use the flexible filter and search functions in the DNC module. You can define such user field combinations as part of the configuration. The user is responsible for the assignment and utilization of these user field keys. Use the defined search fields in the terminal to filter the DNC records in addition to the DNC family of the machine. You can also use these fields as search criteria in the MOC.

Starting with release DNC 7.2, the following preconfigured user field keys will be delivered:

User field key	Description of the search fields
DNC_K	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited
DNC_K_V	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited 4. Version, mandatory field, cannot be edited
DNC_K_W	Plastic (tool reference only): 1. Tool, mandatory field, cannot be edited
DNC_K_WV	Plastic (tool reference and version): 1. Tool, mandatory field, cannot be edited 2. Version, mandatory field, cannot be edited
DNC_NC	NC programs: 1. Article, mandatory field, cannot be edited
DNC_NC_V	NC programs: 1. Article, mandatory field, cannot be edited 2. Version, mandatory field, cannot be edited
DNC_NC_M	NC programs: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited
DNC_NCMV	NC programs: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Version, mandatory field, cannot be edited
DNC_FREI	1. Search field 1, Text20, mandatory field, can be edited 2. Search field 2, Text20, optional field, can be edited 3. Search field 3, Text20, optional field, can be edited 4. Search field 4, Text20, optional field, can be edited

Notes on the DNC administration

DNC records are used exclusively with machines. In order to avoid false entries or false allocations, every machine is assigned to a definite DNC resource family. This is stored in the machine resource data (the *Resource family DNC* field). In this way, you can make sure that only programs belonging to a particular resource family and, indirectly, to a particular resource type can be loaded to a machine.

Furthermore, for the management of DNC records certain criteria are necessary, which, among other things, simplify selection and evaluation, thereby simplifying location and editing and enabling inspections. As widely different machine types can be dealt with by DNC administration (including, for example, injection mold machines, printers and NC machines), a rigid determination of these criteria is not advisable. For this reason, the resource family exists. You can use the user fields to assign attributes to the resource families. These attributes describe and specify the variable parameters.

Therefore, you can use the attributes for identification purposes and you can assign validation functions and allocations. In doing so, you establish a connection between the DNC programs on the one hand and the machines and operations on the other (see section entitled "User fields").

There are variables, such as the temperature and humidity, which influence the behavior of the machines and can therefore have an influence on production. You can also record these "environmental factors". For this purpose, you just have to define further attributes in the user fields.

9 Inspection Stations

Summary

Menu	Master data → Quality management → Inspection stations
Transaction code	ista
Function authorization	ista

The inspection station catalog has been designed to categorize characteristics.

Utilization

The "inspection station number" field is the key field, i.e. while saving a new inspection station, the system checks whether there is already an inspection station with this key information.

The input of measures is easy to handle. Only an inspection station number and a corresponding designation have to be assigned.

By assigning an inspection station to a characteristic, it can be achieved that inspection orders are generated that only include the characteristics of an inspection station. This depends on a configuration option within the inspection plan header. In case inspection orders are generated for each inspection station, this information may, in turn, be used as filter for the inspection orders provided at the AIP inspection terminal. For this purpose, the terminal has to be configured accordingly.

Provided that the inspection plan header defines that inspection orders are not to be generated for each inspection station, the inspection station assigned to a characteristic is only a supplementary information.

By differentiating between active and inactive inspection stations, it can be defined whether or not they are still to be available in the selection lists for inspection stations in the later data acquisition process. Inactive inspection stations can be reactivated at any time.

Integration

Inspection stations are used in all applications dealing with characteristics. Moreover, the terminal configuration allows for inspection stations to be assigned.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

The "inactive" filter field allows for the data set to be restricted to active or inactive inspection stations.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies inspection stations that are no longer to be used in the active data acquisition process.

Toolbar

There are no other special function buttons in addition to the standard functions.

10 Customers

Summary

Menu	Master data → Quality management → Customers
Transaction code	cto
Function authorization	cto

The customer catalog has been designed to edit/keep customers. Provided that there is an interface to a higher-level system (e.g. ERP system), customers can be created automatically via interface. As soon as a new customer is created or changed in the ERP system, for example, the customer data record is automatically created or changed in the customer catalog including the defined information.

Utilization

The customer number uniquely identifies customers in all QM applications that refer to the customer catalog. The customer catalog is used as basis in particular for inspection requirements of production and and the goods issue as well as for the complaint management.

The “customer number” field is the key field, i.e. if a new customer is saved it is verified whether a customer with this key information exists already.

By distinguishing between active and inactive customers, it may be defined whether or not they are available in certain selection lists. Thus, no complaint can be created for an inactive customer, for example. However, inactive customers may be evaluated at any time. Moreover, inactive customers can be reactivated at any time.

Extensive address and contact data can be defined in addition to the customer number and the designation.

If a customer is designated as “party in charge” they will be included in the selection list for the parties in charge. Such selection lists are integrated in different detail applications. The list of the parties in charge is accessed mainly in the complaint management function and when measures are generated.

Integration

Customer data is a global catalog that is used in many QM applications. The below list shows the applications referring to the customer catalog.

- External people
- Departments

- Production inspection planning
- Production inspection requirements
- Complaint management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2, and 3 are available in addition to the customer number and customer name.

The active or inactive customers can be restricted using the filter field "inactive".

Field Descriptions

The available fields are self-explanatory and are not described separately, except for the address fields.

The content of the individual address fields is not specified by default and, as a result, may be defined by the user. Normally, address field 1 should include further details on the company, e.g. "site X". As there is no field defined for the street, address field 2 or 3 (to be preferred) is to be used for the street including street number.

Editing functions

The below dialog opens to edit a data record.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there is a 'Main page' tab. The content is organized into two sections: 'Company' and 'Administration'. The 'Administration' section is currently active. It contains a form with the following fields and values:

Field	Value
Type	Customer
Company number	893239
Company name	Office GmbH & Co. KG
Address 1	Lor 11, #01-235
Address 2	
Address 3	
Inactive	☐
Party in charge	☒
P.O. box	
ZIP code	57665
City	Example City 1
Federal state	Johor
Country	Malaysia
Phone	+65 6475 9678
Fax	+66 2 3456 7123
Cell phone	
Pager	
E-mail	office@sample1.com.sg
Web addr.	www.sample4711.com.sg

Toolbar

There are no other special function buttons in addition to the standard functions.

11 Manufacturer

Summary

Menu	Master data → Quality management → Manufacturer
Transaction code	mft
Function authorization	mft

The catalog of manufacturers has been designed to edit and update the list of manufacturers. Provided that there is an interface to a higher level system (e.g. ERP system) the manufacturers can be created automatically by interface. As soon as a new manufacturer is created or an existing manufacturer is changed in the ERP system, the corresponding manufacturer record is automatically created or changed with the defined information in the catalog of manufacturers.

Utilization

The manufacturer number uniquely identifies the manufacturers in all QM applications that refer to the catalog of manufacturers. In particular, the inspection requests of the goods receipt refer to the catalog of manufacturers.

The "manufacturer number" field is the key field, i.e. while saving a new manufacturer, the system checks whether there is already a manufacturer with this key information.

By differentiating between active and inactive manufacturers, it can be defined whether or not they are still to be available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive manufacturer. However, it is possible to evaluate inactive manufacturers at any time. Moreover, inactive manufacturers may be reactivated at any time.

Extensive address and contact data may be defined in addition to the manufacturer number and designation.

If the manufacturer is designated as being a "party in charge" by clicking the corresponding field, the manufacturer is included in the selection list for the parties responsible. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Manufacturer data is a global catalog that is used in many QM applications. Please find below some applications that refer to the manufacturer catalog.

- External people

- Departments
- Goods receipt inspection planning
- Goods receipt inspection requests
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the manufacturer number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive manufacturers.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there's a 'Main page' header. Below it, there are two tabs: 'Company' and 'Administration', with 'Administration' currently selected. The form contains the following fields and controls:

- Type:** A dropdown menu set to 'Manufacturer'.
- Company number:** A text box containing '827363'.
- Company name:** A text box containing 'Spanner International Pte Ltd'.
- Address 1:** A text box containing '5 Ayer Rajah Crescent'.
- Address 2:** An empty text box.
- Address 3:** An empty text box.
- Inactive:** An unchecked checkbox.
- Party in charge:** A checked checkbox.
- P.O. box:** An empty text box.
- ZIP code:** A text box containing '62783'.
- City:** A text box containing 'Example City 2'.
- Federal state:** A text box containing 'Sarawak'.
- Country:** A text box containing 'Malaysia'.
- Phone:** A text box containing '+65 6341 8679'.
- Fax:** A text box containing '+66 2 6574 9900'.
- Cell phone:** An empty text box.
- Pager:** An empty text box.
- E-mail:** A text box containing 'info@sample45.com.sg'.
- Web addr.:** A text box containing 'www.sample4712.com.sg'.

Toolbar

There are no other special function buttons in addition to the standard functions.

12 Suppliers

Summary

Menu	Master data → Quality management → Suppliers
Transaction code	sup
Function authorization	sup

The supplier catalog has been designed to edit and update the list of suppliers. Provided that there is an interface to a higher-level system (e.g. ERP system), suppliers may be created automatically by the interface. As soon as a new supplier is created or an existing supplier is changed in the ERP system, the data record for this supplier is automatically created or changed including the specified information within the HYDRA-CAQ supplier catalog.

Utilization

The supplier number uniquely identifies suppliers in all QM applications that access the supplier catalog. The supplier catalog is used, in particular, for inspection requests of goods receipt and the complaint management.

The "supplier number" field is the key field, i.e. while saving a new supplier, the system checks whether or not there is already a supplier with this key information.

By distinguishing between active and inactive suppliers, it may be defined whether or not the suppliers are available in certain selection lists. Consequently, it is, for example, impossible to create a goods receipt inspection request for an inactive supplier. However, inactive suppliers may be evaluated at any time. Moreover, inactive suppliers may also be reactivated at any time.

In addition to the supplier number and designation, it is possible to specify comprehensive address and contact details.

Being identified as "party in charge", this supplier is included in the selection list for responsible parties. Such selection lists are integrated in different detail applications. The list of the parties responsible is mainly used within complaint management and for the creation of measures.

Integration

Supplier data is a global catalog that is used in many QM applications. Please find below some applications that refer to the supplier catalog.

- External people
- Departments

- Goods receipt inspection planning
- Goods receipt inspection requests
- Complaints management
- Failure mode analysis

Prerequisite

There are no special requirements.

Selection criteria

The address fields 1, 2 and 3 are available in addition to the supplier number and the designation.

The "inactive" filter field allows for the data set to be restricted to active or inactive suppliers.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

The content of the individual address fields is not specified explicitly and, as a result, may be defined by the user. Normally, the address field 1 should include an addition to the company's name, e.g. "site X". As no field has explicitly been defined for the street, the address field 2 or 3 (to be preferred) is to be used for entering the street and street number.

Editing functions

The following dialog opens to edit a data record.

The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there's a 'Main page' header. Below it, there are two tabs: 'Company' (selected) and 'Administration'. The 'Company' tab contains a form with the following fields and values:

Type	Supplier
Company number	928762
Company name	SANYO Co. Ltd.
Address 1	Siglap Drive, #01-367
Address 2	
Address 3	
☐ Inactive	
☒ Party in charge	
P.O. box	
ZIP code	74663
City	Example City 5
Federal state	Sarawak
Country	Malaysia
Phone	+62 21 4659 6789
Fax	+60 7 6576 3311
Cell phone	
Pager	
E-mail	info@allied.com
Web addr.	www.theka.com.sg
Audit date	
Audit setting	
Audit value	

Toolbar

There are no other special function buttons in addition to the standard functions.

13 Failure

Summary

Menu	Master data	Quality management	Failure
	Master data	Quality management	Failure location
	Master data	Quality management	Failure cause
	Master data → Quality management → Originator		
Transaction code	ftyp for failure type		
	floc for failure location		
	fcau for failure cause		
	ori for originator		
Function authorization	ftyp		

There is a standardized catalog for failure types, cause and originator. The system differentiates entries using the failure types. There is an entry for each failure type in the menu. When requesting the application via the menu, the system opens the application and filters it to the individual failure type. The application can also be opened using a pre filter via the relevant transaction code.

Generally, the system uses the catalogs to clearly outline the failure.

Purpose

The "failure analysis number" field is the key field, i.e. when saving a new failure type, the system checks if a failure type with this key information exists.

The input of failure types is quite simple and only requires the assignment of a failure analysis number and name.

The system defines a failure type group beforehand and assigns the failure to the relevant group. This option should not be missed out as it provides improved reports/evaluations. The system assigns groups by opening the group tree using the magnifying button. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. In the field "Groups" of the maintenance dialog of the failure cause the assigned group with hierarchical group structure appears.

The displayed list presents the group hierarchy with the columns "group 1 to group 5".

The system executes the maintenance of groups in the relevant failure group maintenance as is outlined in the manual „[MOC_Groups.pdf](#)“.

Under certain circumstances, we recommend the use of a self-explanatory structure of the failure number as a failure key.

By differentiating between active and inactive failure types, the system defines if the failure type is still available in failure selection lists in the later data acquisition process. Evaluations using inactive failure are still available. The system can also activate inactive failures.

Integration

Defect catalogs are used sometimes in measurement recording and complaint management. The evaluation of measured values can identify and later on evaluate the individual failure during quality deviations. Only in doing so, it is possible to identify the main failures, take appropriate action (measures) and prevent the defects from recurring.

In addition, this is the basis for failure mode analysis.

The system can also integrate failures in the analysis selection catalogs. An analysis selection catalog includes a subset of all failures and restricts the list of failures to be selected during the collection process to those of the analysis selection catalog assigned to the characteristic.

Requirements

Functional requirements from other applications must not be met in order to use this function.

Selection criteria

Selection criteria are self-explanatory and not described separately. The tab "Group" using the symbol



and the selection of a group (in the tree structure) can filter the failure groups. The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows the data set to be restricted to active or inactive failures.

Field descriptions

The available fields are self-explanatory and not explained separately.

The "inactive" check box identifies failures that are no longer used in the active data acquisition process.

In a tree structure, the group field shows the assigned group or allows groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions/features.

14 Measures

Summary

Menu	Master data → Quality management → Measures
Transaction code	meas
Function authorization	meas

The catalog of measures has been designed to define the action to be taken, for example, to prevent the detected failures from recurring or to allow for them to be identified in time.

Utilization

The "measure number" field is the key field, i.e. while saving a new measure, the system checks whether there is already a measure with this key information.

The input of measures is easy to handle. Only a measure number and a corresponding designation have to be assigned.

Measure groups may optionally be defined beforehand and the corresponding group can be assigned to the measure. This option should not be missed out as it provides improved reports/evaluations or structures. Groups can be assigned by opening the group tree using the magnifier function. The requested group may be selected and taken over in the group tree by way of the hierarchical tree entries. The assigned group including the hierarchical group structure can then be found in the "groups" field of the editing dialog for the measures.

If measures are presented in list form the group hierarchy is represented by the columns "group 1" to "group 5".

Groups are edited in the "measure groups" application, which is described in the manual entitled [MOC_Groups.pdf](#).

By distinguishing between active and inactive measures, it can be defined whether or not they are still to be available in selection lists for the measures in the later data acquisition process. Inactive measures can be reactivated at any time.

Integration

The measures catalog is used, among other things, within measurement recording and in complaint management. If deviations are detected within measurement recording actions can be triggered immediately on the basis of the catalog of measures. At first the assigned measures have the status "open". In complaint management a party responsible (e.g. a person) as well as a deadline may be defined. In addition to this, an actual date, the completion in % and effectiveness in % may be specified as well. Normally, these fields are only filled out, once a measure has been completed (measure status switches to "done").

Subject to whether the corresponding license for the escalation management module has been purchased, the measures can also be sent to the defined person by e-mail.

Prerequisite

Functional requirements from other applications do not have to be met to be able to use this function.

Selection criteria

Selection criteria are self-explanatory and are not described separately. Measures of a group can be filtered in the "groups" tab using the icon  and selecting a measure group (in tree structure). The group tree list provides a function to cancel and accept the entries made.

The "inactive" filter field allows for the data set to be restricted to active or inactive measures.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The "inactive" check box identifies measures that are no longer to be used in the active data acquisition process.

In a tree structure the group field shows the assigned group or allows for groups to be assigned in form of the tree structure.

Toolbar

There are no other special function buttons in addition to the standard functions.

15 Analysis Selection

Summary

Menu	Master data → Quality management → Analysis selection
Transaction code	asc
Function authorization	asc

Analysis selection catalogs allow for the set of failure types, failure locations, failure causes, originators, and measures that are available in measurement recording to be restricted specifically.

The overall application is divided into two hierarchical levels. A master-detail grid is used for presentation. The data records of the analysis selection are created on the first level. One level below, the corresponding failure types, locations, causes, originators and measures are assigned to these data records of the analysis selection. There is a separate tab for each assignment type on this second level.

Separate function keys to create, edit or delete data records are available on each level of the master-detail grid.

Utilization

The analysis selection number identifies the data records of the analysis selection uniquely in all QM applications in which they may be selected and assigned.

A data record of an analysis selection catalog only consists of a number and designation as well as of the flag to disable it ("inactive" field). The most crucial factor here is the possibility to assign failure types, locations, causes, originators and measures to a data record of the analysis selection.

If a characteristic is assigned an analysis selection catalog only the failure types, failure locations, failure causes, originators and measures listed in this catalog will be available for this characteristic when measured values are recorded. Consequently, the failure list, for example, can be designed in relation to characteristics.

Analysis selection catalogs are mainly used for the assignment to an inspection chart characteristic. An assignment is almost mandatory for inspection chart characteristics, as in any other case, the user may choose from the whole set of failure types of the entire master data catalog when recording measured values for this characteristic. This would make inspections confusing and too complex.

Integration

Analysis selection catalogs are used in all applications dealing with characteristics. By assigning analysis selection catalogs in these applications, the selection list for failures, originators and measures is restricted for measurement recording.

Prerequisite

The master data for failure types, failure locations, failure causes, originators and measures need to be maintained before this function can be used in a useful manner.

Selection criteria

Selection criteria are self-explanatory and are not described separately.

Field descriptions

The available fields are self-explanatory and are not explained separately.

The check box "inactive" identifies data records of the analysis selection that are no longer to be used for the definition of characteristics (of inspection plans/inspection orders).

16 Norms

5.1 Summary

Menu	Quality management -> Master data -> Norms Master data → Quality management → Norms
Transaction code	Norm
Function authorization	Norm

The application "Norms" provides an overview of all existing norms inclusive of reference values i.e. fit dimensions. This application processes entries for norms and setting up new norms.

Purpose

If a new norm is set up, an entry in the CAQ status table is created for this norm including long and short name. The norm is now available in the selection list for characteristics.

Integration



If all entries of a norm are deleted, then the system removes the entry of the norm in the CAQ status table. Then this norm is not available in the selection list.

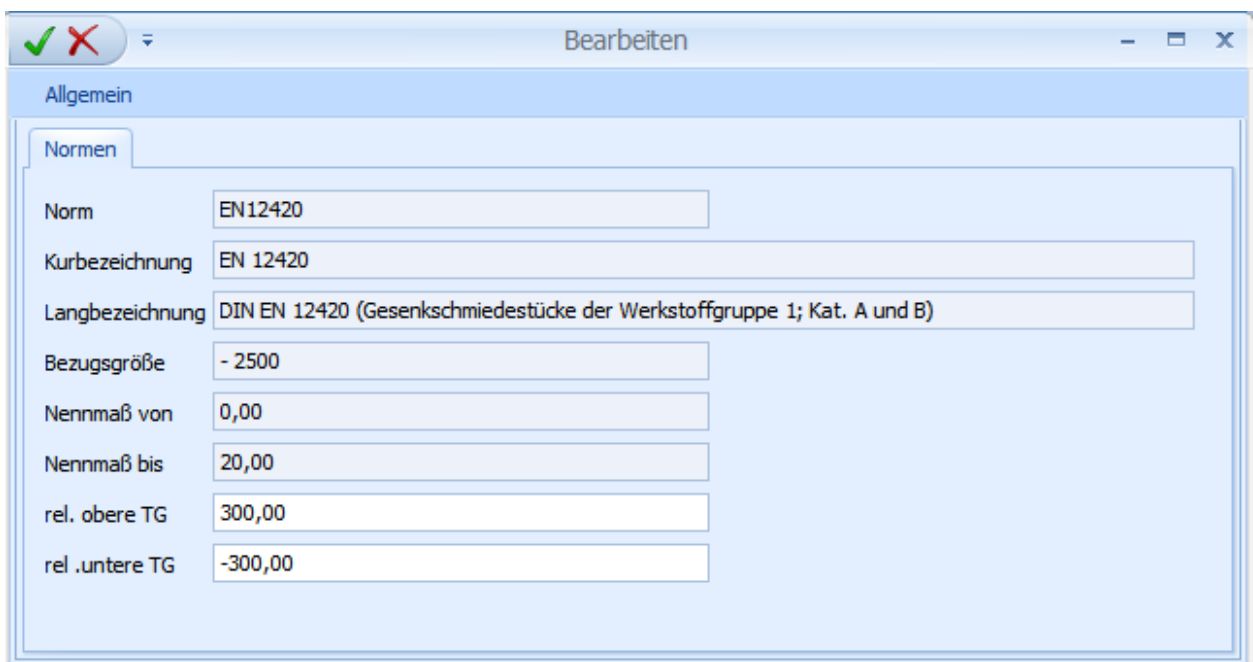


The fields short and long norm are only considered when creating the first entry of a norm. They cannot be changed retrospectively.

Requirements

Editing functions

The following dialog opens to edit a data record:



The screenshot shows a Windows-style dialog box titled 'Bearbeiten'. It has a tab labeled 'Allgemein' and a sub-tab labeled 'Normen'. The dialog contains several input fields for editing a norm record:

Norm	EN12420
Kurzbezeichnung	EN 12420
Langbezeichnung	DIN EN 12420 (Gesens Schmiedestücke der Werkstoffgruppe 1; Kat. A und B)
Bezugsgröße	- 2500
Nennmaß von	0,00
Nennmaß bis	20,00
rel. obere TG	300,00
rel. untere TG	-300,00

17 Units

Overview

HYDRA menu	System administration → System settings → Units
FEDRA menu	System administration → System settings → Units
Transaction code	unit
Function authorization	mdunit

Purpose

The units are stored in an administration table. This table is used during validation checking to compare the quantity units entered (e.g. at an operation) or to check the information transferred at the interface. Application of units:

Integration

The units are stored in an administration table. This table is used during validation checking to compare the quantity units entered (e.g. at an [operation](#)) or to check the information transferred at the interface. The units are used in different areas:

- Quantity units in the order backlog
- Quantity units in order entry and posting
- Quantity units used to enter rolls

Field descriptions

Unit

Unit of quantity, such as kg

Unit ISO

ISO quantity unit, if different

Type

This field can be used to group different quantity units, such as mm, cm, m, km, etc. belonging to the "length" type.

This value can be selected and transferred from the list. Alternately, any other arbitrary value can be entered here manually.

Designation

Designation or description of the unit (e.g. kilogram)

SI unit

Unit as stipulated by the "SYSTEME INTERNATIONAL" agreements.

Values Y/N; there can be a maximum of one SI unit within one type. However, there can also be types that do not have an SI unit.

Usage: One unit of measure is designated as SI unit in each dimension. Among other things, this is the reference point used when converting from one unit to another.

Examples: The SI unit used for length is meter; the SI unit used for time is second, etc.

18 Catalogs

18.1 Overview

Menu	Master data → Quality management → Catalog
Transaction code	cat
Function authorization	cat

18.2 Purpose

Use this function to create, insert, edit, copy and delete catalog entries. If you use imported catalog entries (QMS), do not change the application's contents because this application contains catalog entries, which are exclusively populated by SAP-QM using the QM-IDI interface. The application also includes the usage decisions of inspection points, which are also used in the in-production inspection (no SAP-QM or QM subsystem). Via customization, you can change the inspection point decisions that are not passed via SAP-QM. Contact the MPDV Consulting. As part of the customization, you can change field names, deactivate specific entries or create new inspection point decisions.

SAP-QM provides the following catalogs, for example:

- Usage decisions for the inspection point and the inspection requirement (inspection batch),
- Failure types,
- Failure locations

The application provides an overview of the selection lists that can be called during the inspection process. The higher-level planning system (SAP-QM) then specifies where and which catalog is available.

18.3 Integration

In the inspection processes, the catalog contents are provided as selection lists. You can, for example, select an inspection point decision or an evaluation of a characteristic from a selected catalog. As a subsystem of SAP-QM, this master data catalog includes all data records of the required QMS catalogs. The data records are passed via QM-IDI.

If the function extension of the in-production inspection is available (license FEP-AQF), you can specify an inspection point decision as *Setup inspection*. Measured values and attributive inspection results of a *Setup inspection* are automatically set to invalid when the inspection point is completed. For further details, refer to option 1223 in the procedure document [Configuration QM Options](#).

18.4 Selection criteria

The application provides the following selection criteria:

Catalog type

Number of the catalog type

Selected set

Subgroup of the catalog type

Plant

The production site where the catalog is used.

Code

ID number (alphanumeric)

Code group

Group identification of specific codes

Selection

Checkbox

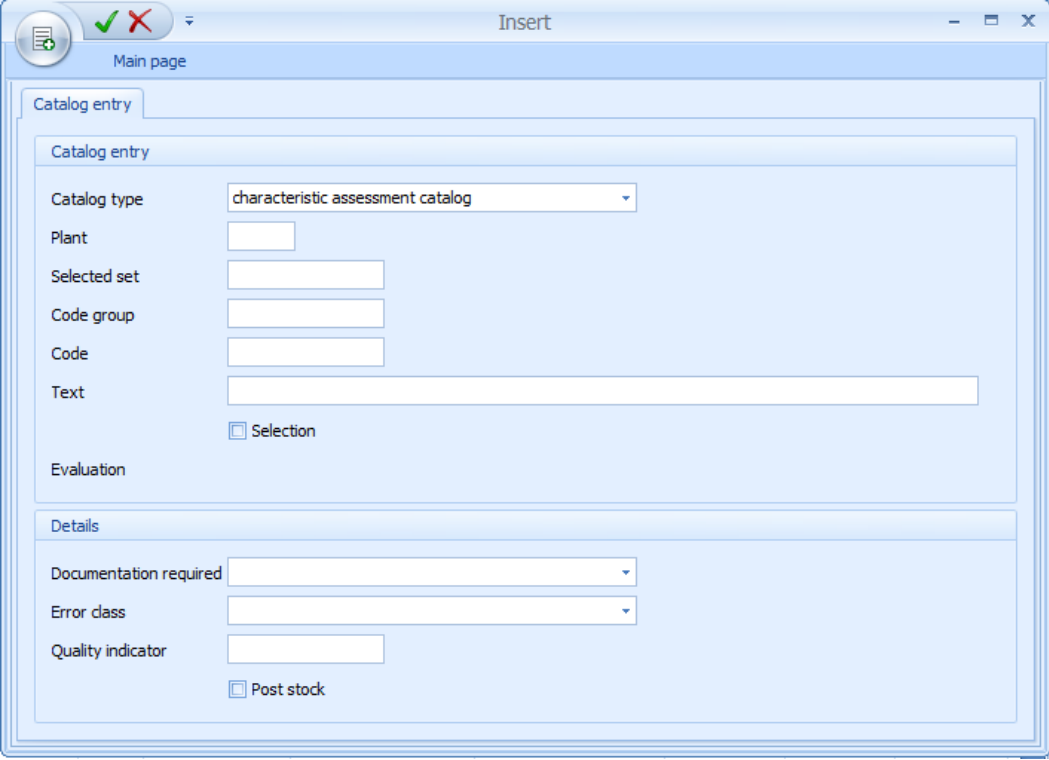
All filter fields provide a match code search, except for the checkbox *Selection*. Use the *Selection* checkbox to specify if all data records are displayed or only the data records with activated or not activated *Selection* field.

18.5 Field descriptions

The available fields are self-explanatory and, as a result, not explained separately.

18.6 Editing functions

The below screenshot shows an example of an editing dialog. Design and alignment of fields can deviate from the example shown.



Although you can enter up to ten characters in the fields *Code group* and *Code*, you must limit the entry in field *Code group* to eight characters and in field *Code* to four characters.

Longer character strings cannot be used in the AIP recording of inspection data. Example: This is important for the inspection point decisions or characteristics that are evaluated using an evaluation catalog.

Toolbar

The application does not provide any special function buttons in addition to the standard functions/features.

19 Inspection planning

Overview

Menu	Quality management → In-production inspection → Inspection planning Quality management → Goods receipt → Inspection planning Quality management → Goods issue → Inspection planning Quality management → Initial sample inspection → Inspection planning Quality management → Gage management → Inspection planning
Transaction code	iplp ¹
Function authorization	iplp

Available user fields		
Where	Object type/user field key	Source (type)
Inspection plan: Table and detail view	Production: CPPL/FEP Goods receipt: CPPL/WEP Goods issue: CPPL/WAP Initial sample: CPPL/EMU Gage management: CPPL/PMV	QM
Inspection plan characteristics: Table and detail view	Production: CPPLMM/FEP Goods receipt: CPPLMM/WEP Goods issue: CPPLMM/WAP Initial sample: CPPLMM/EMU Gage management: CPPLMM/PMV	QM
How to configure user fields? Which user field types are available?		

Purpose

A careful inspection planning is usually the most important function, as inspection plans are the basis for the generation of inspection orders and the resulting inspections. This is true for inspections in the goods receipt process, during production and in the goods issue process. It also applies for the inspection of initial samples and the calibration of test and measuring equipment. As the requirements of an inspection planning are almost identical in all areas, the HYDRA inspection planning functions are almost identical for all these areas.

In general, master data catalogs help the user to generate inspection plans. Therefore, careful maintenance of master data is very important.



¹If you start the application *Inspection Planning* using the transaction code, you cannot create a new inspection plan.

Integration

You require the inspection plan to generate inspection requirements, inspection steps and the resulting inspections/calibrations.

The generated inspection requirements include the used inspection plan. Also some evaluations use the inspection planning. You can thus expand the basic data of a control chart of an inspection order characteristic by the data of the same inspection plan number. The same is possible for the global control chart evaluation. Here, you can use specific filter fields of the inspection plan. The global control chart evaluation does not integrate calibration data and thus you cannot filter by calibration inspection plans.

Another essential relation is established to the production control plan. You cannot create a production control plan without having generated inspection plans beforehand.

Requirements

To create inspection plans, you must maintain/edit the relevant master data. The identification of the master data, that you must maintain, depends on the respective application. However, it is a fundamental to edit the article and the characteristics beforehand. If you create inspection plans for article groups, you must first create the article groups.

Selection criteria

The following list shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt and goods issue. By default, the following areas are available:

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

Active

Enable this option to narrow down the list of inspection plans to view only active inspection plans. If this option is disabled, the list only includes inspection plans with a status "in process" or "released". The third state of this option (grayed out) shows all inspection plans. This is the initial state.

Article number

Filters the article number in case of inspection plans based on articles.

Article designation

Filters the article designation/name in case of inspection plans based on articles.

Article group

Click the icon  to call the article group tree, if you want to filter the inspection plan for an article group. There is a function to accept and cancel the activity.

Operation

Operation number

Customer number

Enter the number directly or call the customer catalog and select an entry to take it over.

Supplier number

Enter the number directly or call the supplier catalog and select an entry to take it over.

Manufacturer number

Enter the number directly or call the customer catalog and select an entry to take it over.

Field descriptions***Inspection plan header*****Area, inspection plan number, inspection plan index**

The "Area", "Inspection plan number" and "Inspection plan index" uniquely identify the existing inspection plans. Select the area. Enter alphanumeric characters for the inspection plan number and inspection plan index. All three fields are mandatory fields.

The input in these three fields must be unique, i.e. no other inspection plan may exist that already includes this information. If you assign a structured inspection plan number, you can include specific information. This information might be helpful later on for sorting. As an alternative, you can also use the article number or the number of the test equipment group as inspection plan number. The inspection plan index corresponds to the inspection plan version. If you must modify an existing inspection plan version, but it is not possible to change the inspection plan as it has already been used for the generation of an inspection order, you must copy the original inspection plan and change the inspection plan index (e.g. increment by one). You should keep the inspection plan number with regard to subsequent evaluations, if possible (e.g. display of order-independent control charts where the inspection plan versions are different and the inspection plan number is identical).



If you generate calibration inspection plans, you can only enter 24 characters into the field *Inspection plan*. Activity calendar: Depending on the system configuration, you enter the inspection plan number in the field *Project number*. There, the field *Project number* is restricted to 24 characters.

Inspection plan type

There are two types of inspection plans: "Inspection plan of articles" and "Inspection plan of groups". If you select "Inspection plan of articles", you assign an article from the article master data to generate an inspection plan for this article. If you want to generate an inspection plan for a group of articles, select the type "Inspection plan of groups". You can then select an article group. The selected article group is displayed in a separate field within the tree structure.

Article

Enter the article number. If you know the article number, you can directly enter it. If not, you can open the article catalog and identify and transfer the requested article using the filter and sort criteria. Select an article to take over the drawing issue number, the article designation, the customer article number and the drawing number from the master data record. The respective data is then displayed in the corresponding fields.



If the field *Article number* or *Article group* is shown in the test equipment management, you must not make an entry here.

Drawing issue number

The drawing issue number can be entered directly, just as it is the case for the article number.



If you directly enter the article number and the drawing issue number, the master data record is identified with this information when you save the data. The fields article designation, customer article number and drawing number are then populated and displayed.



The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.



If the field "Drawing issue number" is shown in test equipment management, you may not enter any values here.

Operation assignment

You can select between two assignment types: "One insp. plan for each OP" and "One insp. plan for all OPs".

Assign the operation type to define if the inspection plan you want to create includes the characteristics of different operations ("one inspection plan for all OPs") or if the inspection plan only includes the characteristics of one operation. If you select the setting "One inspection plan for each OP", the fields "Operation" and "Operation designation" are shown. In any other case, operation details are assigned in the area of inspection plan characteristics. If possible, you should select the option "One inspection plan for all OPs". You can then print all inspection characteristics of an article, although it includes several operations.



In case there is no assignment to a production order, you should use the option "One inspection plan for all OPs". The advantage is that you must only enter the "fictitious" operation once in the inspection plan header, and not for each inspection plan characteristic. This inspection plan method requires a suitable system configuration that must be carried out by MPDV consulting.

Operation, operation designation

If you use operations, you must only complete one of the two fields "operation" or "operation designation". If you want to automatically generate an inspection requirement/an inspection order when you log on an operation/production order, you must only enter the operation number in the "Operation" field.

Note: These fields are used as search criteria when you generate inspection orders later on. That is: If you only enter the designation of the operation in the inspection plan, you must include this and only this information when you generate an inspection order later on.



Also if there is no assignment to a production order (i.e. goods receipt and test equipment management), you must enter a "fictitious" operation (e.g. 9999). This is required for the generation of an inspection step later on.

IO (inspection order) + Inspection station

If you enable the option "An IO for each inspection station", the assignment of inspection stations is used when it comes to inspection plan characteristics. In this case, the system generates a separate inspection step for each inspection station if the CAQ system option 1157 is configured accordingly. The separate inspection steps are also created with identical operations. If you want to use the specific AIP inspection modes for the goods receipt, the inspection in the laboratory/measuring room and for calibration, you must select the option "An IO for each inspection station". Only with this configuration, the system includes the information on the planned machine/machine group later on when the inspection step is generated. It is not necessary to assign an inspection station in this case.

Released/Active

Shows whether the inspection plan is "released" or additionally has the status "active". If the inspection plan is released or active, the respective options are checked. An inspection plan is only released and activated, i.e. its status is changed, if you use the respective toolbar functions. You must release the inspection plan before it can be activated.



When you generate inspection orders at a later point in time, the system only uses active inspection plans.

Released by / on

Shows the HYDRA user who has released the inspection plan. The release date is displayed additionally.

Cavity assignment

You require the authorization "ipl_cav" to display this field.

You can select "None" or "Sample" to assign cavities.



If this field is not available, you require a new program version of this application.

An inspection based on cavities is only possible, if the inspection is based on inspection points and characteristics.



With the function extension for the production inspection, there is the additional option of a piece-related collection based on cavities.

If the inspection is based on samples and/or pieces, you cannot assign cavities.

Valid from/till

The field entries are for information purposes only.

Type of inspection

Characteristic-related or piece-related inspection

If the inspection is based on characteristics and the sample size is for example 5, the characteristic is checked completely first. If the inspection is based on pieces, the characteristic is changed every time a measured value is collected, as all characteristics of one piece are inspected.



You can only use the piece-related inspection, if you collect data with reference to inspection points. If you want to perform a piece-related inspection in the goods receipt process, you must change the sample-related inspection into an inspection based on inspection points.



An inspection based on cavities is only possible in case of a characteristic-related inspection. If the inspection is piece-related, you cannot assign cavities.

Action

"Creation of inspection step" or "Create and release inspection step"

Only inspection steps that have been released can be checked.

Customer / Supplier / Manufacturer

If you select a customer/supplier/manufacturer, this inspection plan only applies for the selected company. Consequently, the customer/manufacturer/supplier is a key field of an active inspection plan. If one of these fields is populated, the inspection requirement must also include a company entry. Only then, you can use this inspection plan to generate the inspection requirement.

Dynamic modification type

Characteristic-related, batch-related or none

You require the authorization "iriscp.dynamic" to display these fields.

Batch-related: Here, the transitional definition, which is used for the dynamic modification, is specified in the inspection plan header. The initial inspection severity is also specified in the inspection plan header. You additionally require the reference to the dynamic modification norm in the inspection plan characteristics. However, the selection is restricted. You can only select dynamic modification norms that reference the same inspection severity definition as the assigned transitional definition.

Characteristic-related: The inspection plan characteristics include the references to the transitional definition, the initial inspection severity and the dynamic modification norm. The inspection severity definition of the assigned transitional definition and the assigned dynamic modifications norm must match.

None: The "dynamic modification" function is not enabled for this inspection plan.

Transitional definition

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics has been selected. It provides the possible inspection severities and controls switching between the inspection severities.

Initial inspection severity

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics has been selected. It defines the inspection severity that is used to inspect the first goods received according to the basics of the dynamic modification history.

Initial sample form

Use this option to select the form type for the initial sample inspection. Currently, only the form type "VDA volume 2, 4th edition" is supported. Depending on the form type, a respective list of categories to generate an initial inspection is shown in the inspection plan characteristics.

Gage method

You can edit the field "Process acc. to" in case of a calibration inspection plan. You can select between "Standard" and "QS9000". If you select "QS9000", the additionally displayed fields (Reference, Acceptance and Amendment) are mandatory fields.

Editing functions

The key fields "area", "inspection plan" and "inspection plan index" cannot be changed in the editing mode.

Toolbar



Copy

To copy an inspection plan, the following dialog opens.

You can enter the target area type and the target area here. You usually select the values that are identical to the ones of the source inspection plan. Then enter the new inspection plan number and inspection plan index. If you generate a new version of an existing inspection plan, you normally use the identical inspection plan number and only change the inspection plan index.

If the option "Reorganization OP sequence" is activated, the characteristics are created in the new inspection plan with OP sequence numbers in increments of 10.



Activate

Function authorization: `ipIp.activate`

Changes the inspection plan status to "Active".



Deactivate

Function authorization: `ipIp.deactivate`

Changes the inspection plan status from "Active" back to "Released".



Release

Function authorization: iplp.release

Changes the inspection plan status from "In process" to "Released".



In process

Function authorization: iplp.inprocess

Changes the inspection plan status from "Released" to "In process".



Document management

Click here to call the [Document management](#).



Import inspection plan

Function authorization: iplpimport

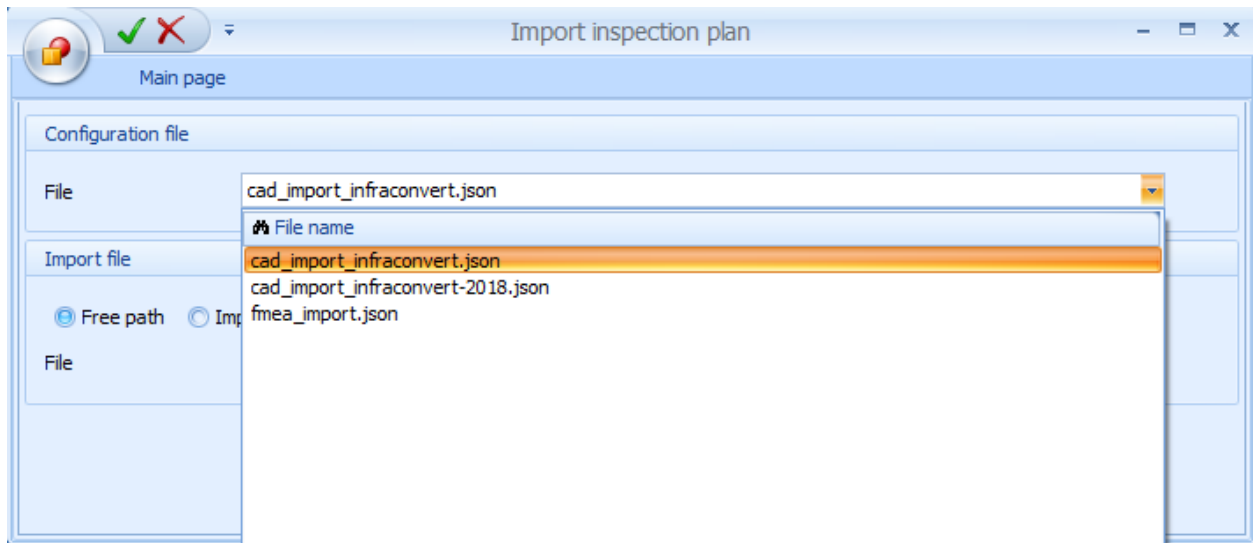
You can use this function to import inspection plan characteristics for the inspection plan selected.

The system uses a previously created data file for the import.

"Import inspection plan" detail application

Function authorization	Iplpimport
License	FEP-PCF or WEP-PCF and EIS-CFM

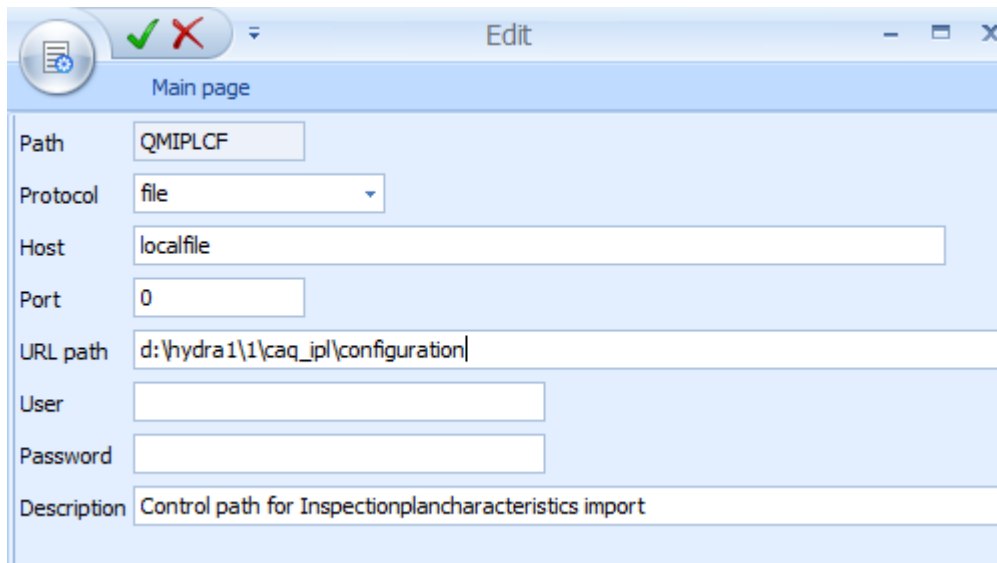
You can use the detail application "Import inspection plan" to add inspection plan characteristics to an inspection plan selected. You use this function if you have previously exported characteristic data into a data file using a CAD drawing or the HYDRA-FMEA application. In both cases, you first select the configuration file that you want to use for the import.



In the configuration file, you define the characteristic data that is transferred from the data file and you define the HYDRA characteristic fields where the data is transferred to. To import data from HYDRA-FMEA, you can use the configuration file "fmea_import.json". To import data from a CAD drawing, the system provides the configuration files "cad_import_infraconvert.json" and "cad_import_infraconvert-2018.json" by default. These two configuration files are especially designed for the export of CAD characteristic data using the tool "infra CONVERT" or "infra CONVERT 2018" created by the company Elias GmbH.

If required, you can create custom configuration files. For details, refer to the procedure document "Configuration_InspectionPlanImport".

If the system does not display any configuration files, you must check the respective path configuration. In the HYDRA path configuration, the path "QMIPLCF" must be available. If the path "QMIPLCF" is not available, you must create the path. The below screenshot shows a possible path configuration for the configuration data.



Path: QMIPLCF

Protocol: file

Host: localfile

Port: 0

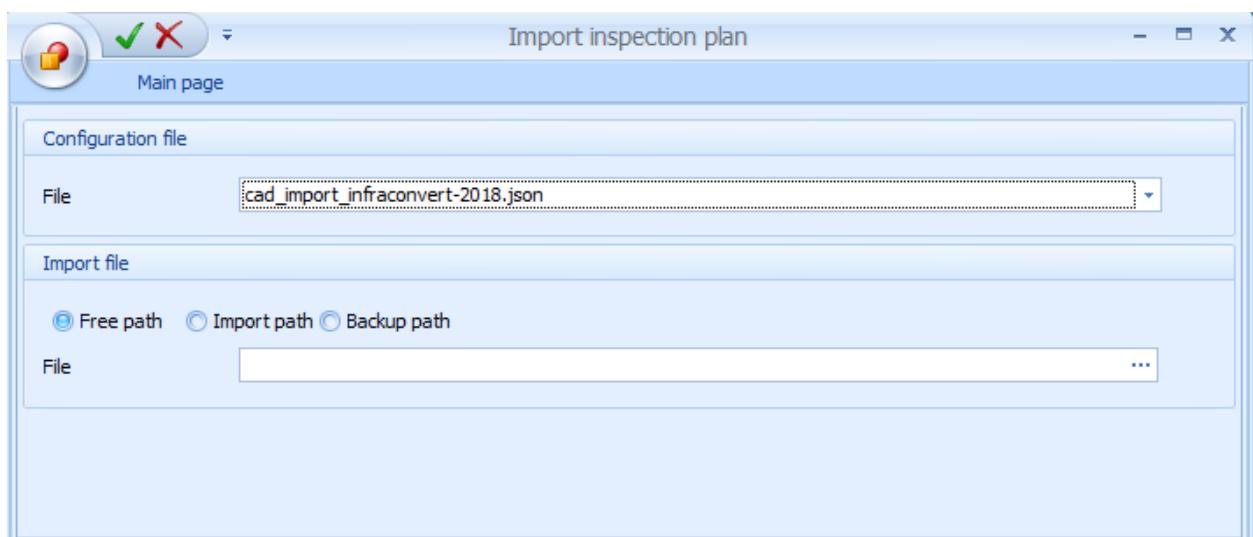
URL path: d:\hydra1\1\caq_ip\configuration

User:

Password:

Description: Control path for Inspectionplancharacteristics import

When you have selected the configuration file, then you select the import file. For the import of data from a CAD drawing, enable the selection option "Free path". Using this selection option, you can select any import file.



Configuration file

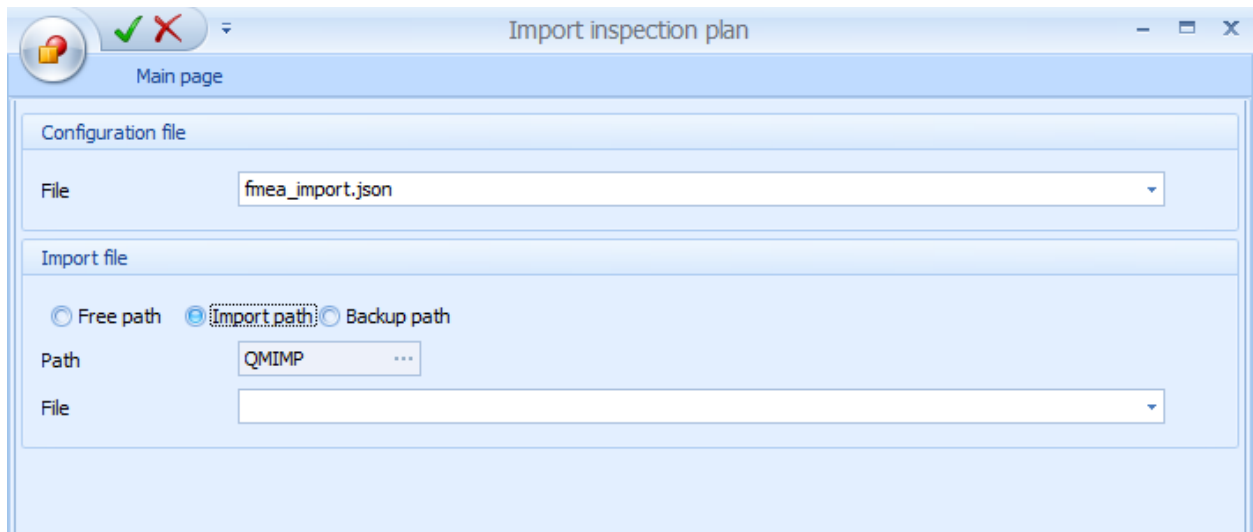
File: cad_import_infracorevert-2018.json

Import file

☒ Free path ☐ Import path ☐ Backup path

File:

For the import of HYDRA-FMEA characteristic data, enable the selection option "Import path".



Import inspection plan

Main page

Configuration file

File: fmea_import.json

Import file

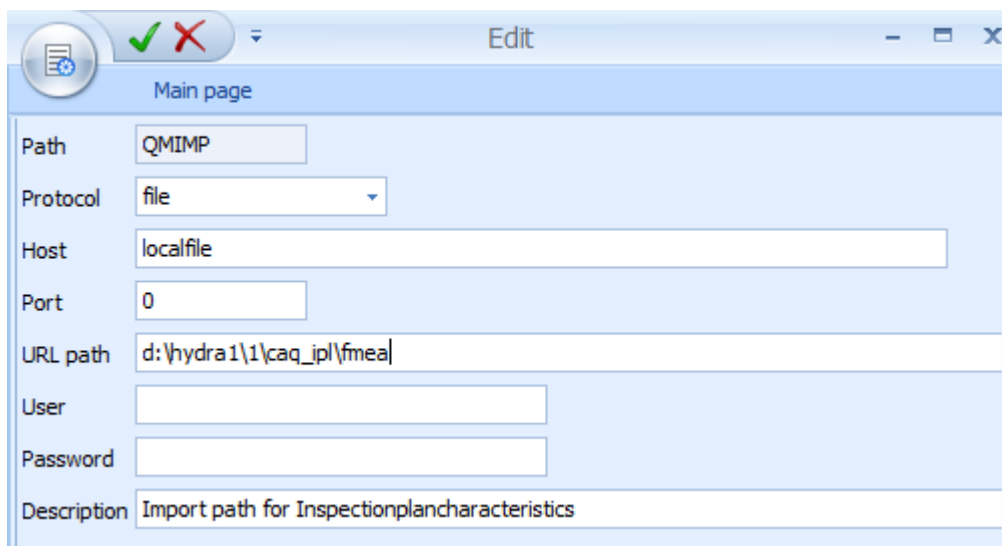
☐ Free path ☒ Import path ☐ Backup path

Path: QMIMP

File:

During the previous export of HYDRA-FMEA characteristic data, the exported characteristic data is automatically stored in the import path. In the HYDRA path configuration, the URL path is configured for the path "QMIMP". The path "QMIMP" must be available before using the import function for FMEA characteristic data. If the import path is not available, you must manually create the path. If the import path is available, you must check the path configuration before the first import. If required, correct the specified URL path. It might be required to correct the system, for example.

The below screenshot shows a possible path configuration for the import of HYDRA-FMEA characteristic data.



Edit

Main page

Path: QMIMP

Protocol: file

Host: localfile

Port: 0

URL path: d:\hydra1\1\caq_ip\fmea

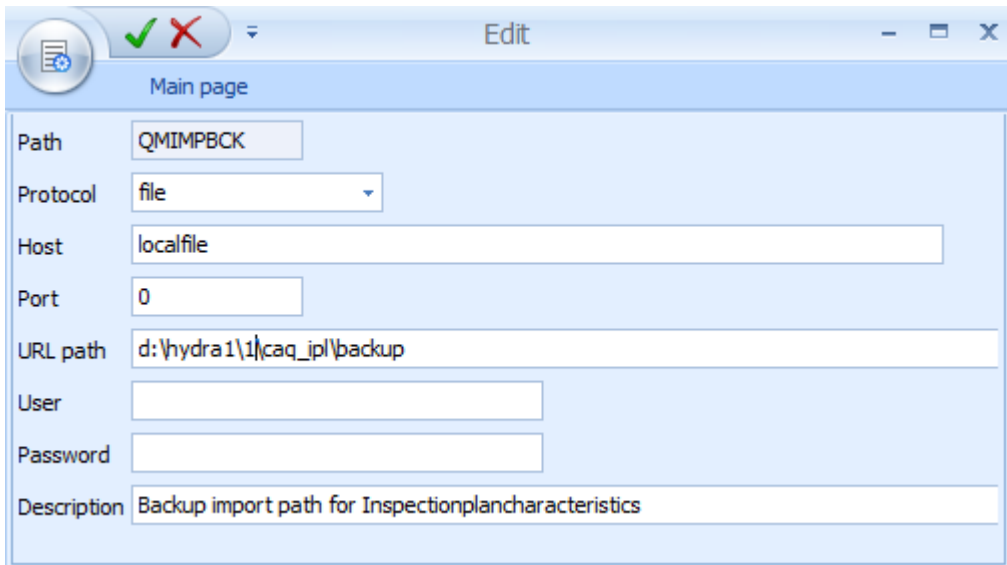
User:

Password:

Description: Import path for Inspectionplancharacteristics

You must not only specify the HYDRA path, but you must also ensure that the specified URL path is available on the HYDRA server. If the specified path is not available, you must manually create the path on the HYDRA server. You must also check the specified URL path for all HYDRA path configurations mentioned below. If required, you must change the path configuration.

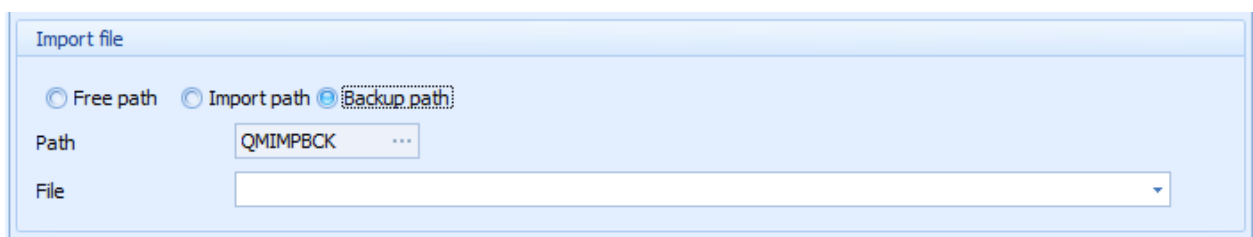
Before the first import, you must also create the backup path "QMIMPBCK" in the HYDRA path configuration because the data file is copied into the backup path after the successful import of HYDRA-FMEA characteristic data. The below screenshot shows a possible path configuration for the backup path.



The screenshot shows a window titled "Edit" with a "Main page" tab. It contains several input fields for configuring a path:

- Path:** QMIMPBCK
- Protocol:** file (dropdown)
- Host:** localfile
- Port:** 0
- URL path:** d:\hydra1\1\caq_ip\backup
- User:** (empty)
- Password:** (empty)
- Description:** Backup import path for Inspectionplancharacteristics

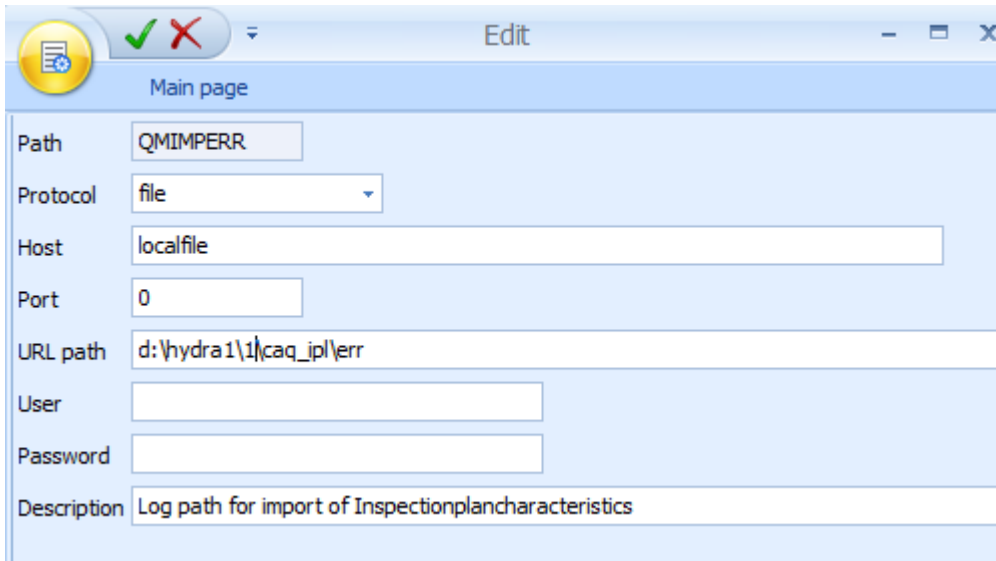
If you want to re-import HYDRA-FMEA characteristic data after a successful import, you must enable the selection option "Backup path".



The screenshot shows a dialog box titled "Import file" with three radio buttons: "Free path", "Import path", and "Backup path". The "Backup path" option is selected. Below the radio buttons, there are two input fields:

- Path:** QMIMPBCK (with a browse button "...")
- File:** (empty)

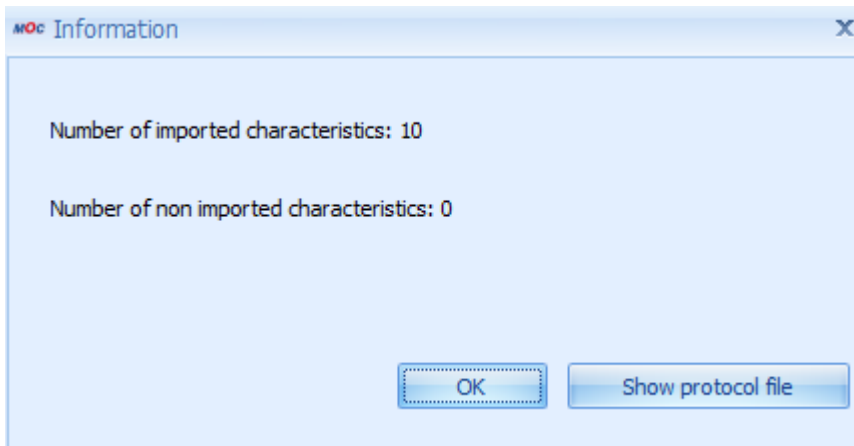
During the import process, a log file is created. To store this log file, you must create the path "QMIMPERR" in the HYDRA path configuration. The below screenshot shows a possible path configuration for the log path. If a valid log path is not available, you cannot import characteristic data.



For the import of HYDRA-FMEA characteristic data, you must finally select the file that you want to import.

The import of characteristic data is always completed using the button .

When the characteristic data have been imported, a message window shows the number of successfully imported characteristics and the number of not imported characteristics.



Use the button "Show log file" to open a log file. This log file specifies the characteristics that have been successfully imported and the characteristics that have not been imported.

"Print" – Detail application

Function authorization	iplp.print
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The print dialog of the inspection plan header opens a list of available reports. These are Word forms. The web services that are available in the respective context determine the potential content of these forms. The form entries, i.e. the content of the list of forms of the respective print dialog, are defined within the master data of quality management. Here, the basics for new forms and the respective form properties are also specified. You require the respective license to be able to change the forms with respect to content and design.

Print – Toolbar

There are no other special function buttons in addition to the standard functions/features.

"Inspection plan characteristics" – Detail application

The detail application *Inspection plan characteristics* is nearly identical to the application of the characteristics master data. Therefore, only additional features or modifications are described here.



Go to

For further information on the definition of characteristics, please refer to the functional description in the document [MOC_CharacteristicsQM](#).

On the level of inspection plan characteristics, the inspection plan header defined beforehand assigns the respective inspection plan characteristics. To assign inspection plan characteristics, create a new data record, open the characteristics catalog and accept the selected characteristic. By accepting the characteristic, all master data entries are copied to the inspection plan characteristic. You can still change and/or complete each (copied) piece of information. The description of the characteristic is often completed to define it in more detail.

You can also create a characteristic that does not exist in the catalog of characteristics. But this is only recommended in exceptional cases, as all analyses (e.g. failure mode analysis) are based on characteristics included in the catalog of characteristics. Consequently, the characteristics catalog should be maintained.

You can define different properties and settings, before you complete the specific characteristic data.

Field descriptions

Inspection plan characteristics

OP sequence

The operation sequence number (OP sequence/AFO) specifies the inspection sequence. Entries must be unique. In the ideal case, operation sequence numbering should be assigned in steps of 10. You can then add a new characteristic between two existing characteristics at a later point in time.

Characteristic number

Number of the characteristic selected from master data.

Characteristic type

This option specifies whether the collection of measured values (variable) or the identification of the number of detected failures (attributive) is used for the inspection. In case of an attributive inspection, the decision is often only based on the "pass" or "fail" results. Further characteristic types are the inspection chart and the information characteristic. The information characteristic is only used to display a document during the inspection process. Subject to the input type, the lower area of the dialog provides the respective sampling schemes.

Inspection result base

This setting defines whether all samples or only the sample recorded last is used to identify the inspection result (pass/fail).

Mandatory inspection

If this option is activated, you must enter at least one measured value for this characteristic, before you can complete an inspection order including this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

Formula

Further details can be taken from the manual dealing with the CAQ master data characteristics.

Operation / Operation designation

this field are only available, if the setting "One inspection plan for all OPs" is activated in the inspection plan header.

If you want to include a reference to a productive operation in the characteristic, you need only store the operation number in the field "Operation".

If you define characteristics of QM operations, we recommend to add the operation. For characteristics of the same operation, this entry must always be identical.



If you define QM operations, you must be careful that the operation created in the production order later on is not the last operation. For example: If the last productive operation is the operation "0030" and you want to define characteristics with reference to a QM operation for this operation, you should use "0029" for this QM operation.

Initial sample creation

Selection list of categories for the creation of an initial sample inspection. The content depends on the form type selected in the inspection plan header. In the initial sample report, the inspection plan characteristics are grouped by the selected category for the creation and for each group a separate page is printed.

Copy characteristic

You can define here which characteristic is copied from an initial sample inspection plan into a production plan, for example. Only those characteristics are copied that have been specified as relevant in this field.

Details

This option specifies if - for this characteristic - the contents of the "details" tab are defined within the inspection plan or if they are identified using the available master data characteristic when an inspection requirement is later generated based on this inspection plan. If the "from characteristic catalog" option is selected, the "details" tab is hidden. The option "from characteristics catalog" is useful, if the same characteristic specifications must be defined for several articles (also for several article groups). In this case, specifications can be defined centrally within the master data. This reduces the input and update efforts considerably.

Specifications

This option specifies if - for this characteristic - the contents of the "specifications" tab are defined within the inspection plan or if they are identified using the available master data characteristic or an entry from the specification list when an inspection requirement is later generated based on this inspection plan. If you enable the option "from list" or "from characteristic catalog", the tabs "specifications", "chart 1", "default values chart 1", "chart 2" and "default values chart 2" are hidden. The option "from list" is only useful with an inspection plan for article groups. You use the specification list if the specifications for a specific characteristic are different for the individual articles.

No cavity

You require the authorization "ipl_cav" to display this field.

You can use this option to label a characteristic as not relevant to cavities, although the respective inspection requirement specifies that the relevant characteristics are (actually) to be recorded in relation to cavities.

You must always enable this checkbox for attributive characteristics and inspection charts.

Dynamically modified

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. You can use this option to specify that a characteristic should not be modified dynamically, although the dynamic modification option is selected in the inspection plan header.

Transitional definition

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. The selected transitional definition provides the possible inspection severities and controls switching between these different inspection severities. For the batch-related dynamic modification, the transitional definition is assigned in the inspection plan header and in this case the transitional definition is not available on the level of inspection plan characteristics.

Initial inspection severity

You require the authorization "iriscp.dynamic" to display this field.

It defines the inspection severity that is used to inspect the first goods received according to the basics of the dynamic modification history. Only the inspection severities of the previously selected transitional definition are available. For the batch-related dynamic modification, the transitional definition is assigned in the inspection plan header and in this case the transitional definition is not available on the level of inspection plan characteristics.

Standard

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. The system uses the master data to assign the dynamic modification norm.

Inspection level

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. An inspection level can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

AQL

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. You can only select an AQL value (Accepted Quality Limit) if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859..

Method

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. A method can only be selected if the assigned norm corresponds to DIN ISO 3951.

Sample group

You require the authorization "ipl_ipsampling" to display this field.

This field specifies which sample group this characteristic belongs to. The sample group specifies which characteristics are part of a sample/sampling characteristic.

Sampling

You require the authorization "ipl_ipsampling" to display this field.

If a characteristic has been assigned to a sample group, this option specifies if you use this characteristic to generate a sample. In this case, you call it a sampling characteristic. Activate this option to generate sampling characteristics. You must assign a sample group to the sampling characteristic.

If you want to inspect a sample using this characteristic, this option is not active.



If these fields are not available, you require a new program version of this application.



You must configure in the inspection plan characteristics, which are assigned to a sample group, that the details are provided by the inspection plan and not by the characteristics catalog, because you can only assign sample groups in the inspection plan.



All sample characteristics of one sample group must be included in the same inspection step. In the inspection plan, the combination of inspection station (machine, machine group) and operation must be identical for the sample characteristics.



If you want to combine the inspections for the producing machine (including sampling characteristic) and the inspections of the samples in one operation, you must consider the following issues when you plan the characteristics for this operation:

- The characteristics, which are used for the inspections of the producing machine, and the sampling characteristic must include a specific inspection station (e.g. MACHINE).
- The characteristics, which are used for the inspections of the samples, must be

assigned to a different inspection station (e.g. LAB).

- In the respective inspection plan, you must set the configuration "IO + inspection station" to "An IO for each inspection station".
- The terminal the producing machine has been assigned to must be configured specifically. For the above-mentioned example, you must assign the inspection station "MACHINE" in tab "QM functions".
- In the configuration of the terminal where the samples are inspected, you must assign the inspection station "LAB" in tab "QM functions" for the above-mentioned example.

Measurement system analysis

If the inspection plan is a QS9000 inspection plan, an additional tab *Measurement system analysis* is displayed.

Besides the sample size that you edit in tab *Specifications*, you must edit the key data of the measurement system analysis here.



Go to

The other fields of the respective tabs correspond to those of the characteristic master data and are outlined in the documentation entitled [MOC CharacteristicsQM](#).

Inspection plan characteristics – Toolbar



Create new data record

Select a characteristic from the master data. All characteristic details of the master data characteristic are transferred to the inspection plan characteristic.



When you select a characteristic from the master data, also the documents assigned to the master data characteristic are included in the inspection plan characteristic.



Copy

To copy inspection plan characteristics, the data of the selected characteristic are opened in insert mode. All fields can be modified. To save the data, you must assign an OP sequence number that does not yet exist in this inspection plan.



The documents of the inspection plan characteristic, for which the “copy” button has been clicked, are transferred.



Copy characteristics

Select a target inspection plan. The selected characteristics are then copied into the specified inspection plan. If the target inspection plan already includes characteristics, the new characteristics are created with OP sequence numbers in steps of 10.



The documents of the different characteristics are also transferred.



Document management

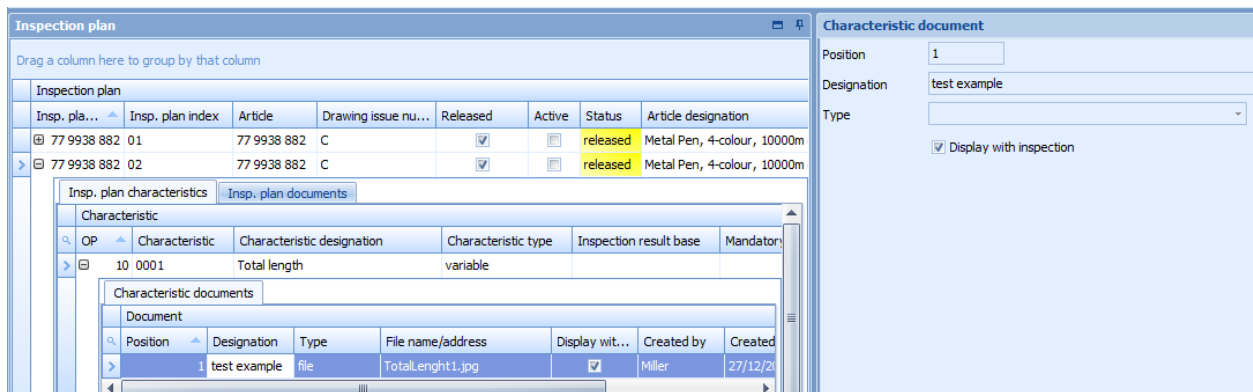
Click here to call the [Document management](#).

"Inspection plan documents" and "Documents of inspection plan characteristics" – Detail applications

Inspection plan	Insp. plan index	Article	Drawing issue nu...	Released	Active	Status	Article designation
77 9938 882	01	77 9938 882	C	☒	☐	released	Metal Pen, 4-colour, 10000m
77 9938 882	02	77 9938 882	C	☒	☐	released	Metal Pen, 4-colour, 10000m

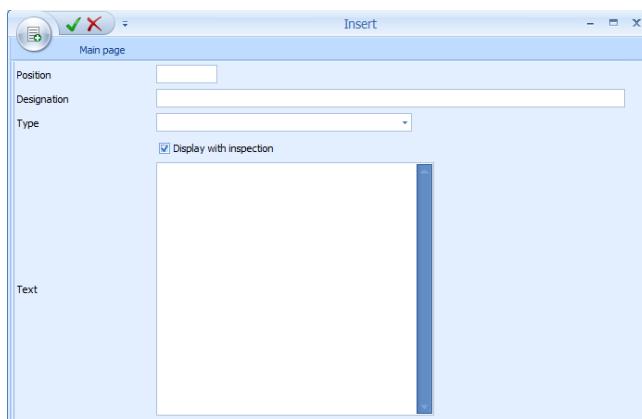
Position	Designation	Type	File name/address
1	reference model	file	metal_pen.jpg

The above screenshot shows how an inspection plan document is assigned.



The above screenshot shows how a document of an inspection plan characteristic is assigned.

You can assign any number of documents to each inspection plan and each inspection plan characteristic, if the tab "Inspection plan documents" or the tab "Characteristic documents" has been activated in the master detail grid. If you activate these tabs, the toolbar provides the corresponding buttons to edit documents.



You can use all formats registered by Windows for the documents you want to assign. You can therefore assign simple documents (e.g. written in Word), drawings of any format and videos. You only have to make sure to install a program that is able to display the used format. The program link stored in Windows opens the documents.

The file types "File", "URL", and "Text" are available. The file name including the path may be entered manually for the "file" type. The file type "URL" allows to access the internet or intranet. Use the file type "text" to directly enter text.

You can assign a designation/name to each document that is added. You can also define the list order of the documents. Use the field "position" to define the order (numeric input). Position numbers must be distinct in this list. Enable the checkbox *Display* to define that the document is displayed during inspection process.

"Inspection plan documents" and "Documents for inspection plan characteristics" – Toolbar

In addition to the standard functions, the button to show documents is available.



Show documents

If a document link is stored, this button opens and shows the linked document. However, a program, which can show the linked file type, must be installed in the PC.

20 Inspection requirements

Overview

Menu	Quality management → In-production inspection → Inspection requirement Quality management → Goods receipt → Inspection requirement Quality management → Goods issue → Inspection requirement Quality management → Initial sample → Inspection requirement Quality management → Gage management → Inspection requirement
Transaction code	irp ¹
Function authorization	irp

Available user fields		
Where	Object type/user field key	Source (type)
Inspection requirement: Table and detail view	Production: CPAN/FEP Goods receipt: CPAN/WEP Goods issue: CPAN/WAP Initial sample: CPAN/EMU Gage management: CPAN/PMV	QM
Inspection step characteristics: Table and detail view	Production: CPAUMM/FEP Goods receipt: CPAUMM/WEP Goods issue: CPAUMM/WAP Initial sample: CPAUMM/EMU Gage management: CPAUMM/PMV	QM
Inspection points: Table and detail view	CPANUMP/PPUNKT	QM
How to configure user fields? Which user field types are available?		

Purpose

The inspection requirement in the inspection process corresponds to the production order in BDE (shop floor data collection). The inspection requirement includes one or several inspection steps (e.g. one inspection step per operation). When an inspection requirement is generated, the system uses the respective inspection plan to create the inspection steps. The inspection requirement includes the general information, such as order or article number. One level below the inspection requirement, there is the inspection step. The inspection step then specifies the characteristics to be checked. To collect measured values, you therefore select the inspection step or the operation of the production order (that includes the inspection step) and log it on to the terminal.

As the functional requirements are almost identical in all areas (e.g. goods receipt, production, goods issue, gage management), the respective applications all have a similar structure.



¹If you use transaction codes to call the application *Inspection requirement*, you cannot create new requirements.

If you create new inspection requirements, you are generally supported by master data catalogs. Therefore, careful maintenance of master data is very important.

Integration

You generate inspection requirements/inspection steps using inspection plans, which you have created previously. These plans are the reference the inspection is based on. When being generated, the contents of inspection plans are copied into the inspection requirements/inspection steps. The inspection plan is the reference used to create inspection requirement or inspection step. If you make changes to the inspection plan (limited options), these changes do not affect inspection requirements/inspection steps that have already been generated.

The inspection steps, in turn, are the "objects" that can be checked/inspected. In production, the inspection steps are connected to the respective operation of the production order via operation number. In gage management, the inspection step actually is a calibration order.

Almost all reports and evaluations reference the data of the inspection requirements/inspection steps and the included inspection step characteristics. To restrict the data in reports, you can filter by the article number, order number, operation, and inspection step characteristic, for example.



In gage management, the inspection requirement is automatically generated from the activity calendar. This generation is triggered by a user action. In exceptional cases, inspection requirements can also be generated directly in the "inspection requirement" application. Here, you must manually enter the inspection plan number.

Requirements

Before you can create inspection requirements/inspection steps, you must first create inspection plans for the required area/context. You also require some master data. It depends on the planned use which master data you must maintain.

Selection criteria

The following list shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt or goods issue. By default, the following areas are available:

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

Inspection requirement no.

The inspection requirement number uniquely identifies the inspection requirement. HYDRA automatically generates the inspection requirement number, which is unique.

Status

Open a status list to filter by the different inspection requirement statuses. This way, you can limit the list of inspection requirements to the completed inspection requirements, for example.

Skip lot (available for the goods receipt and goods issue only)

Due to dynamic modifications, an inspection requirement can be classified as "skip lot" and, as a result, no inspection is performed.

Order

Number of the respective production order or calibration order.

Article number

You can directly enter the number or open the article catalog where you can select an entry and take over the number.

Drawing issue number

The article is uniquely identified using the combination of "article number" and "drawing issue number".

Customer number

You can directly enter the number or open the customer catalog where you can select an entry and take over the number.

Supplier number

You can directly enter the number or open the supplier catalog where you can select an entry and take over the number.

Manufacturer number

You can directly enter the number or open the manufacturer catalog where you can select an entry and take over the number.

Actual date from / to

You can enter a time limit. In production, this field is usually empty. In goods receipt, the "actual date" specifies the date when the goods are delivered.

Planned date from / to

You can enter a time limit. In production, this field is usually empty. In goods receipt, the "planned date" specifies the date when the goods are planned to be delivered (initially).

Manufacturing date from / to

You can enter a time limit.

Consider long-term data

If you enable this option, you can show archived inspection requirements of the medium-term data area.



You cannot show inspection certificates, etc. in Word format for the archived inspection requirements displayed.

Field descriptions of the *Inspection requirement*

In the following, find a description of the most important fields.

Inspection requirement**Area**

The inspection requirement is created for the specified area. The application provides a selection list of the configured areas of in-production inspection, goods receipt, goods issue, initial sampling or gage management. By default, the following areas are available:

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

This field is a key field. Using this field, the active inspection plan or the specified inspection plan including index is identified that is used to create the inspection requirement, the inspection steps and inspection step characteristics.

Inspection requirement no.

The inspection requirement number is automatically generated upon saving. The number is numeric and uniquely identifies an inspection requirement. It cannot be changed.

Operation, operation designation

If you use operations, you must only fill in one of the two fields "operation" or "operation designation". If you want to automatically generate an inspection requirement/an inspection order when you log on a production order, you must only fill in the field *Operation number*. Usually the operation number is the only reference required because this is the most important number in the common work plan structures.

This field can be a key field. Depending on the system configuration, this field is used to search for the active inspection plan. The generated inspection requirement, inspection steps and inspection step characteristics are then based on this plan. If the system is configured accordingly, the system generates the inspection requirement and all inspection steps at the same time for the operations that are included in the respective inspection plan.



Also in case of areas that do not have production orders (e.g. goods receipt and gage management), it might be required to specify a "fictitious" operation (e.g. 9999) depending on the system configuration. This "fictitious" operation has to be included in the respective inspection plan.

Inspection plan number / inspection plan index (specified)

You can use these fields to specify the inspection plan that is used for the generation of the inspection requirement if you do not specify key fields to search for the active inspection plan.

Date

When a new data record is created, this field includes the system date, which may be changed afterwards. Once it has been saved, this field may no longer be changed.

Article

Enter the article number. If you know the article number, you can directly enter it. If not, you can open the article catalog and identify and transfer the requested article using the filter and sort criteria. Select an article to take over the drawing issue number, the article designation, the customer article number and the drawing number from the master data record. The respective data is then displayed in the respective fields.

This field is a key field. You use this field to search for the active inspection plan that is used to generate inspection requirement, inspection steps and inspection step characteristics if no inspection plan including index has been specified before.



In gage management, this field is empty or should be left empty.

Drawing issue number

Like the article number, you can directly enter the drawing issue number.

This field is a key field. You use this field to search for the active inspection plan that is used to generate inspection requirement, inspection steps and inspection step characteristics if no inspection plan including index has been specified before.



If you directly enter the article number and the drawing issue number, this information is used upon saving to identify the master data record. Using this master data, the system can then display article name, customer article number and drawing number.



The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.



If the field "Drawing issue number" is shown in gage management, you may not enter any values here.

Companies

Customer / Supplier / Manufacturer

You can directly enter the respective company number or open the master data catalog and select the respective entry. When you have selected and saved the inspection requirement, the system also enters the company name and address data.

These fields are key fields. You use these fields to search for the active inspection plan that is used to generate inspection requirement, inspection steps and inspection step characteristics if no inspection plan including index has been specified before.

Date - quantities**Actual quantity**

The system uses the actual quantity to calculate the inspection scope. The calculation is based on the selected sampling scheme. This is especially true in goods receipt and goods issue if dynamic modification is enabled. In production, this field is usually empty.

Target quantity

In particular for the goods receipt, it is important to specify the target quantity in order to identify the difference between the ordered quantity and the actually delivered quantity.

Scrap

This field is only relevant if you are just using the CAQ functions of HYDRA and you are not using the shop floor data collection (BDE) module.

Rework

This field is only relevant if you are just using the CAQ functions of HYDRA and you are not using the shop floor data collection (BDE) module.

Actually produced quantity (act. prod. qty.)

This field is only relevant if you are just using the CAQ functions of HYDRA and you are not using the shop floor data collection (BDE) module.

Actual date

In particular for the goods receipt, it is important to specify the actual date in order to identify the difference between the planned delivery date and the actual delivery date. In production, this field is usually empty.

Planned date

In particular for the goods receipt, it is important to specify the planned date in order to identify the difference between the planned delivery date and the actual delivery date. In production, this field is usually empty.

Status - details**Status**

The status specifies the inspection requirement status, e.g. whether it is completed, unchecked ("created" status) or whether inspection results already exist ("partial result" status).

Overall result

The overall result defines whether the inspection requirement is classified as "pass" or "fail".

Usage decision

The usage decision classifies the overall result in more detail. You can extend the list of usage decisions via customizing the system. When you complete an inspection requirement, the usage decision is predefined for each overall result. A "pass" result leads to the usage decision "release" and a "fail" result leads to the usage decision "reject". You can change the predefined usage decision if you manually complete the inspection requirement.

Skip lot

Due to dynamic modifications, an inspection requirement can be classified as "skip lot" and, as a result, no inspection is performed.

PPS/ERP reference

If higher-level systems automatically generate inspection requirements via interface, you can use this reference number to upload and uniquely assign results.

PPS/ERP addition

If higher-level systems automatically generate inspection requirements via interface, you can use this unique reference number to upload results and add an additional information. Only the interface populates this field.

Uploaded to ERP/PPS

If this field is enabled (by interface) it can specify that the upload to the higher-level system has been performed and the higher-level system has confirmed this via interface. Only the interface populates this field.

Inspection plan number/inspection plan index (used)

The inspection plan number and inspection plan index specify the inspection plan that is used as basis for the generation of the inspection requirement and inferior data (inspection steps, inspection step characteristics).

Cavity assignment

You require the authorization "ipl_cav" to display these fields.

The options "none" or "sample" are available for cavity assignment. The respective setting is based on the inspection plan.

Dynamic modification

Dynamic modification type

Shows the dynamic modification type of the respective inspection requirement.

Batch-related: The inspection plan used includes a dynamic modification based on batches.

Characteristic-related: The inspection plan used includes a dynamic modification based on characteristics. In this case, the transitional definition, the initial inspection severity and the dynamic modification norm are specified on the level of inspection step characteristics.

None: The inspection plan used does not include any dynamic modification.

Transitional definition

These fields (number and designation) are only available if dynamic modification based on batches or characteristics is defined for the inspection plan used. The transitional definition provides the possible inspection severities and controls switching between these different inspection severities. The content cannot be changed.

Determined inspection severity

This field defines the inspection severity that is/was used for the inspection of the incoming goods according to the basics for the dynamic modification history. This field is only available if dynamic modification based on batches is defined in the inspection plan used. The content cannot be changed.

Revised inspection severity

This field defines the inspection severity that is/was used for the inspection of the incoming goods according to a manual revision of the inspection severity. This field is only available if dynamic modification based on batches is defined in the inspection plan used.

Form

Party in charge type

Provides the shortlist of the different categories of responsible parties, e.g. department, customer, supplier, external persons.

Party in charge

List of the responsible parties that includes only the previously selected type. This list includes all master data that is marked accordingly in the "party in charge" field.

Party in charge name

Only shows the name/designation of the selected responsible party.

Form

Form type of the inspection plan used.

Calibration

Resource type

Shows the resource type of the resource to be calibrated. For gages, this is the type "PRM".

Resource

Shows the resource number (gage number) that is assigned to this inspection requirement.

Maintenance

Shows the activity number that is assigned automatically by creating an activity in the activity calendar. This unique ID number identifies or references the respective activity.

Calibration findings

Shows the result entered or identified when the calibration process was completed, e.g. capable without restrictions.

Calibration status

Shows the status entered or identified when the calibration process was completed, e.g. released.

Inspection requirement - editing functions

The key fields "area" and "inspection requirement number" as well as the fields "skip lot" and "uploaded to ERP/PPS" cannot be changed in the editing mode.

Inspection requirement - toolbar



Copy

Function authorization: irp.copy (available as of SP13)

The selected data record is opened in copy mode. To create a new inspection requirement, you must normally change the order number. The data structures that are below the inspection requirement are not copied (inspection steps, characteristics, inspection points, etc.) Upon saving, the system searches for an inspection plan using the specifications of the inspection requirement and creates the respective inspection steps, etc.



Release

Function authorization: irp.release

The system sets the status for the inspection requirement that was available when it was completed, e.g. to the "partial result" status. This is required if the inspection requirement has already been completed and if it must be reset into a status that allows inspection.

**Complete**

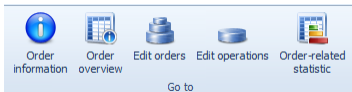
Function authorization: irp.complete

Opens a dialog to assign a usage decision. The system has already preset a usage decision using the available inspection results. The inspection requirement is finally transferred to the "completed" status, once it has been completed. With this status, you can no longer inspect the included inspection steps.

**Cancel**

Function authorization: irp.cancel

Transfers the inspection requirement to the "canceled" status. With this status, you can no longer inspect the included inspection steps.

**Order/operation**

Function authorization: irp_goto

Click the respective button to jump to the application.

"Print" detail application

Function authorization	irp.print
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The print dialog of the inspection requirement header opens a list of available reports. These are Word forms. The web services that are available in the respective context specify the potential content of these forms. The form entries, i.e. the content of the list of forms of the respective print dialog, are defined within the master data of quality management. Here, the basics for new forms and the respective form properties are also specified. You require the respective license to be able to change the forms with respect to content and design.

The report, 4th edition, 2nd volume VDA (German Association of the Automotive Industry), is available as special form for initial sampling. In addition to the cover sheet, the relevant attachments of the inspection characteristic results are printed as well. Each attachment includes the inspection characteristics and the relevant results for an attached category, e.g. data for dimensional checks (attached category: dimension).

Print – Toolbar

There are no other special function buttons in addition to the standard functions/features.

"Complete inspection requirement" detail application

Function authorization	irp.complete
------------------------	--------------

To complete an inspection requirement, click the respective button in the toolbar. An editing dialog opens.

The system displays information on the inspection requirement (i.e. order, batch, article, company data) and also information on the inspection steps included in the inspection requirement. You cannot complete an inspection requirement if any inspection steps are not completed, i.e. the field "without result" includes a value greater than zero. In this dialog box, you can complete the inspection orders that are still open using the button "complete all inspection orders". This is only possible if the conditions of a mandatory inspection that might exist are fulfilled. You can then specify the result and further use of the inspection requirement. If all inspection steps have been completed with "pass", the system proposes "pass" as the result and "release" as usage. If one inspection step has been completed with "fail", the system proposes "fail" as the result and "reject" as usage. Another result and usage may still be selected though.

The list specifying the available usage decisions can be changed as part of an MPDV customizing service.

"Inspection step" detail application

In the master detail grid, if you change from an inspection requirement to the inferior level of "inspection steps", the grid displays all inspection steps that are included in the selected inspection requirement. The system displays the data of the selected inspection step in the detail view. The number or the combination of the inspection steps depends on the configurations made for the existing inspection plan. If an inspection plan includes a configuration that specifies that the operation is defined on the level of inspection plan characteristics and if the system configuration specifies in addition that all included inspection steps are created when the inspection requirement and the first inspection step are created, then the inspection requirement includes an inspection step for each operation included in the inspection plan.

You can also release, complete or cancel an inspection step using the toolbar functions.

You can delete inspection steps, but you cannot create or edit them manually.

You can find the current status of the inspection step in the "rating" category of the "inspection step" tab. The following statuses are available:

Status	Description
No characteristic (KMM)	The inspection step has no characteristics that can be inspected. The inspection step is faulty and no inspection can be performed.
Created (ERS)	The inspection step has been created but not yet released for inspection. No measured values have been recorded yet.
Inspection step released (FRE)	The inspection step has been released for inspection. No measured values have been recorded yet.
Partial result (TER)	The inspection step has been released; some measured values have been recorded.
Inspection in progress (IPR)	Measured values for this inspection step are currently being recorded on another terminal.
Completed (ABG)	The inspection step has been completed.
Canceled (STO)	The inspection step has been canceled.
Skip lot (SKL)	The inspection step is in the "skip-lot" status.

"Inspection step" field descriptions

In the following, the most important fields of the *Inspection step* are described. Self-explanatory fields are not explained.

Inspection step

Inspection step number

Unique number of the inspection step

Status

Status description of the inspection step, e.g. "completed", "partial result", etc.

Result

Qualitative classification of the inspection result based on the respective inspection results of the characteristics pertaining to it.

Workplace

Number of the workplace where this inspection step is performed.

Workplace designation

Name of the workplace where this inspection step is performed.

Tab *Inspection point identification*

You can activate up to nine different fields, which can or must be filled out later when inspection points are generated. By customizing the CAQ system options, it may be defined which fields are to be enabled, how they are labeled (for some fields) and whether or not it is a mandatory field.

If a field is activated and it is a mandatory field, the respective checkbox is always "checked". The respective checkbox is checked and grayed out if it is not a mandatory field but enabled for the resulting data collection. If the field is disabled, the checkbox of the field pertaining to it is not activated at all.

Once activated within the inspection point, the fields 1 to 3 are available for entering values with respect to "equipment", "functional location" and "physical sample". The fields 4 to 9 are user fields relating to inspection points. The fields 4 and 5 are "alphanumeric" fields, the fields 6 and 7 are numeric, the field 8 is a "date" field and field 9 is a "time" field.

Order data from inspection step**Order**

This is e.g. the production order number in production.

Operation

Number of the operation this inspection step has been generated for. The operation number is derived from the inspection plan used.

Operation name

The operation number, which is derived from the inspection plan used, identifies the operation name. Normally, this field remains empty, because usually only the operation number is used in inspection planning. The operation designation is only used in exceptional cases.

Quantities**Rework**

A rework quantity may be entered in this field when the inspection step is completed. This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. This field is also useful in goods receipt and goods issue.

Scrap

A scrap quantity may be entered in this field when the inspection step is completed. This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. This field is also useful in goods receipt and goods issue.

Actually produced quantity (act. prod. qty.)

The actually produced quantity may be entered in this field when the inspection step is completed. This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. This field is also useful in goods receipt and goods issue.

Inspection step - editing functions

You cannot edit an inspection step. You can only add quantity specifications when you complete inspection steps.

Inspection step - toolbar**Release**

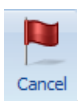
Function authorization: irisp.release

Sets the status "released" for an inspection step with status "created" or "completed". Or the system sets the status that existed when the inspection step was completed, e.g. the "partial result" status.

**Complete**

Function authorization: irisp.complete

Opens a dialog where the (inspection) result is preset. The result is based on the existing inspection results of the characteristics. The inspection step is completed and the status "completed" is set. You can no longer perform inspections for inspection steps with this status.

**Cancel**

Function authorization: irisp.cancel

Sets the "canceled" status for the inspection step. You can no longer perform inspections for inspection steps with this status.

**Order/operation**

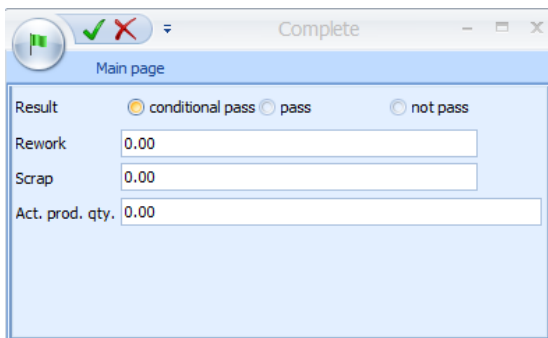
Function authorization: irp_goto

Click the respective button to jump to the application.

"Complete inspection step" detail application

Function authorization	irisp.complete
------------------------	----------------

To complete an inspection step, click the respective button in the toolbar. The following editing dialog opens.



If all inspection step characteristics have the result "pass" or if they are not checked, the system suggests the result "pass". The system suggests the result "fail" if there is at least one inspection step characteristic that has the inspection result "fail". The user can also select another result.

When the inspection step is completed, the user can record a "rework quantity", "scrap quantity" and the "actually produced quantity" (not relevant for gage management). This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. These fields are also useful in goods receipt and goods issue. In particular in the goods receipt area, it is useful to record scrap and rework. These details can be uploaded for several inspection steps to a higher-level system at a later point in time when inspection requirements are completed.

"Inspection points" detail application

By default, the system only uses inspection points in production. It is only useful to have inspection points if you take and check several samples for one order/inspection requirement. For example, this is useful if you perform inspections in production that are based on different events. Example of an event: A specified inspection interval that is based on quantities or time is reached. The events that trigger inspections can be different for the individual inspection plan characteristics. The inspection points of an inspection step can therefore have different inspection characteristics each.

Once an inspection has been performed, you must complete each inspection point. Each inspection point has an inspection result (pass, fail) and a status (completed, open). Inspection points that have not yet been checked have a "pass" inspection result by default.

Note

If you enter a text in the inspection point fields on the AIP terminal that is longer than the field provided in the respective inspection point dialog, the text is cut off when you save the inspection point on the AIP. Therefore, the field contents should not be longer than the field lengths configured for the AIP inspection point dialog. The field lengths of AIP dialogs can be adjusted as part of system customization.

"Inspection points" field descriptions

In the following, the most important fields of the *Inspection step* are described. Self-explanatory fields are not explained.

Inspection point**Inspection point number**

The inspection point number uniquely identifies the inspection point of the inspection step. In addition to the "inspection point" field, the higher-level key fields "inspection step number" and "inspection requirement number" are shown .

Inspection result

The inspection result can be "pass" or "fail". Inspection points that have not yet been checked have the inspection result "pass" by default (standard system configuration).

Status

An inspection point can be completed or open. Only open inspection points can still be checked on the AIP terminal.

Tab *Inspection point identification*

This tab displays the fields, which have been enabled in the higher-level inspection step. You define in the CAQ system options which fields identify the inspection point and which fields are mandatory fields. By default, the following fields are displayed, but none of these fields is mandatory.

Date

You can record a date. If you create a new inspection point, the system date and time is preset. If you manually create the inspection point, you can directly change date and time. Or you can change date and time in the editing mode.

Time

You can record a time. If you create a new inspection point, the system date and time is preset. If you manually create the inspection point, you can directly change date and time. Or you can change date and time in the editing mode.

In addition to the configurable identification fields, there are still the following identification fields that are always available.

Workplace

If the inspection point is generated on an AIP terminal or if the inspection point is automatically generated via inspection event on a specific terminal, this field shows the workplace of the triggering inspection station. If you specify an inspection station, this inspection point is only available at this inspection station.

ERP batch

You can enter a respective ERP batch.

Partial batch

You can enter a respective partial batch.

Details

Generation of inspection points

If the system automatically generates an inspection point because a defined time or quantity interval is reached, the inspection point has the identifier "relating to time" or "relating to quantities". An inspection point that is assigned the identifier "free" has been generated manually or on the basis of other events that make inspections become due.

Inspection points - editing functions

You cannot create new inspection points for completed or canceled inspection steps. And you cannot edit existing inspection points if the respective inspection step is completed or canceled. If you want to create or edit an inspection point, you must first release the inspection step.

Inspection points - toolbar



Release

Function authorization: iripp.release

Sets the "open" status for a completed inspection point.



Complete

Function authorization: iripp.complete

Opens a dialog window where you can complete an inspection point.



Order/operation

Function authorization: irp_goto

Click the respective button to jump to the application.

"Complete inspection point" detail application

Function authorization	iripp.complete
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To complete an inspection point, click the respective button in the toolbar. The following editing dialog opens. The dialog includes an inspection result that has been calculated using the results of the inspected characteristics. This result cannot be changed. The category "Rating" also includes a preset usage decision for the inspection point that is based on the inspection result of the characteristic. You can manually change the usage decision. The system provides a selection list for usage decisions of inspection points that is filtered by the preset production site. Each usage decision classifies the inspection point as "pass" or "fail", irrespective of the inspection result. When you have selected a usage decision, the system displays additional information on the usage decision of the inspection point. This additional information is, for example, the code group, code number, catalog type and the factory/site.

MPDV services for customizing the system may provide further usage decisions for inspection points.

If the identification fields are configured accordingly, you can edit these fields when you complete the inspection point. If an identification field is a mandatory field, you must enter a value before you can complete the inspection point. In addition to this, quantity fields may also be filled out.

"Inspection step/inspection point characteristics" detail application

Function authorization	iriscp.*
------------------------	----------

The detail application of inspection steps/inspection point characteristics is almost identical to the master data application of characteristics. Therefore, only additional features or modifications are described here.



Go to

For further information on the definition of characteristics, refer to the functional description in the document [MOC_CharacteristicsQM](#).

The master detail grid shows the inspection step characteristics one level below the inspection step. If you use inspection points and if you have created inspection points, you can open the list of inspection points one level below the inspection steps. If you open this list, the characteristics of the selected inspection point are displayed in a separate tab.

The characteristics of the inspection step are specified in the inspection plan used. If the inspection plan settings specify that a separate inspection step must be generated for each operation (global setting for all workplaces), then the inspection step only includes the characteristics of a specific operation. If the inspection plan settings additionally specify that a separate inspection step must be generated for each workstation, then an inspection step always includes the characteristics of a fixed combination of operation and workplace.

An inspection point only includes characteristics that become due because of the same event. Also the manual generation of inspection points represents a due date event. For example, if a quantity interval defined for the inspection step characteristics has been reached, an inspection point is automatically generated. This inspection point includes all characteristics that have the same quantity interval. And it also includes the characteristics that have a quantity interval that is the least common multiple of the quantity interval reached.

Example:

Characteristic 1 has a quantity interval of 100. Characteristic 2 has a quantity interval of 200. After the first 100 parts produced, an inspection point is generated, which only includes characteristic 1. After 100 more parts, another inspection point is generated, which includes the characteristics 1 and 2.

If an inspection becomes due because the machine status changes, the generated inspection point only includes characteristics that must be checked because of the same machine status change.

Only manually generated inspection points are not limited with respect to the respective characteristics. These inspection points always include all characteristics of the related inspection step.

Inspection step/inspection point characteristics - field descriptions

In the following, only the fields are described that are not included in the master data of characteristics or in the inspection plan characteristics.

Characteristic

Inspection result (available as of SP13)

The inspection result is identified using the settings for the characteristics in the inspection plan according to the "Inspection result base" and the value of the CAQ option 1038. For details, refer to the description of the CAQ option 1038 in the procedure document "[Configuration QM Options](#)".

Initial sample creation

Shows the field content that is transferred from the characteristic included in the inspection plan used. A separate sheet/page is printed for each category in the printed initial sample report.

Properties

Dynamically modified

This checkbox is enabled, provided that a characteristic is dynamically modified.

Dynamic modification

Determined inspection severity/inspection severity designation

This field shows the inspection severity identified for this characteristic on the basis of the dynamic modification rules specified in the inspection plan.

Revised inspection severity/inspection severity designation

This field only exists if dynamic modification is based on characteristics. A revised inspection severity is displayed here, if the user has changed the process of dynamic modification and if the user has revised the current inspection severity (new initial inspection severity).

Note

A note is displayed in this field, in case an error has occurred with the dynamic modification of this characteristic while inspection steps are generated.

Details

No cavity

You require the authorization "ipl_cav" to display these fields.

You can use this option to label a characteristic as not relevant to cavities, although the respective inspection requirement specifies that the relevant characteristics are (actually) to be recorded in relation to cavities.

You must always enable this checkbox for attributive characteristics and inspection charts.

Sample group

This field specifies which sample group this characteristic belongs to.

Sampling

If a characteristic has been assigned to a sample group, this option specifies if you use this characteristic to generate a sample. In this case, you call it a sampling characteristic. You must enable the checkbox in this case. You must assign a sample group to the sampling characteristic.



If these fields are not available, you require a new program version of this application.

Specifications

Sample size

This field shows the identified sample size if a sampling scheme is defined in the inspection plan characteristic and if this sampling scheme does not specify the sample size because the sample size is calculated using the actual quantity of the inspection requirement. This is the case for the sampling schemes "batch inspection" or "100% inspection".



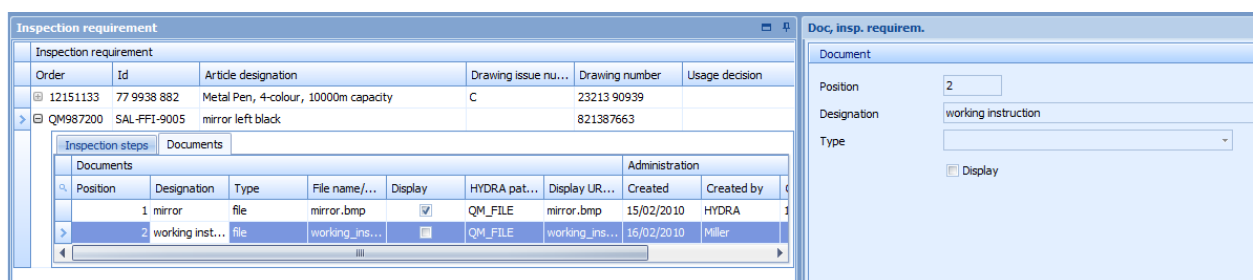
Go to

The other fields included in the tabs are the same as the fields of the characteristic master data and are described in the documentation "[CAQ characteristic master data](#)".

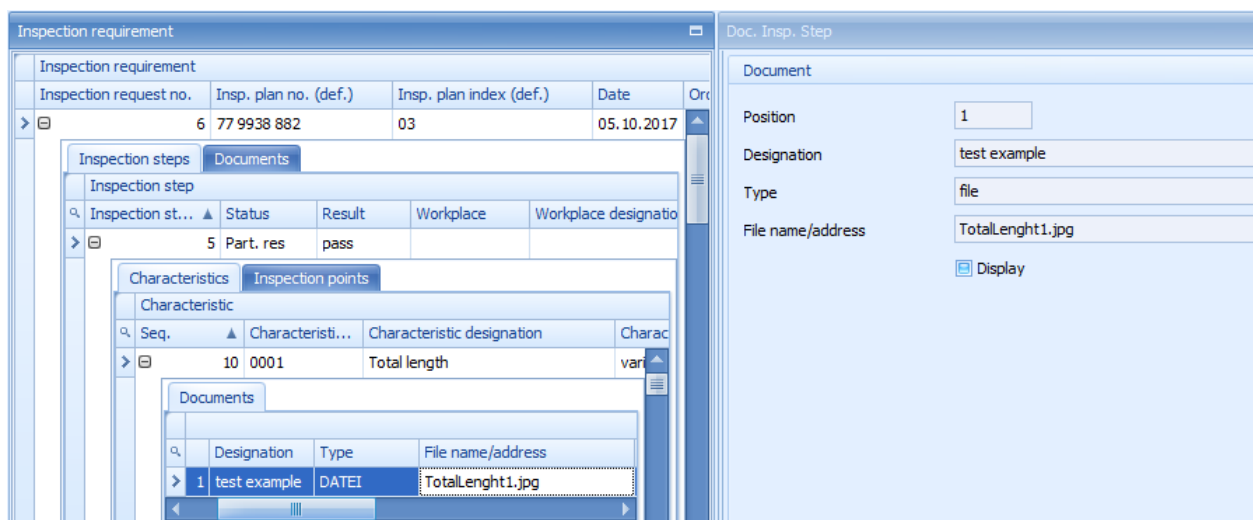
Inspection step/inspection point characteristics - editing functions

Inspection step or inspection point characteristics cannot be edited.

"Inspection requirement documents" and "Characteristic documents" detail applications



The above screenshot shows how an inspection plan document is assigned.



The above screenshot shows how a characteristic document is assigned.

If you have enabled the "documents" tab in the master-detail grid, you can assign any number of documents to each inspection requirement and to each inspection step characteristic. If this tab is enabled in the master-detail grid, the respective editing buttons are enabled in the toolbar to edit the documents.

You can use all formats registered by Windows for the documents you want to assign. You can therefore assign simple documents (e.g. written in Word), drawings of any format and videos. You only have to make sure to install a program that is able to display the used format. The appropriate program linked in Windows opens the documents.

The file types "File", "URL", and "Text" are available. For the type "file", you can manually enter the file name including the path. The file type "URL" enables access to the internet or intranet. With the file type "text", you can directly enter a text.

You can assign a designation/name to each document that is added. You can also define the list order of the documents. Use the field "position" to define the order (numeric input). Position numbers must be unique in this list. Enable the checkbox *Display* to define that the document is displayed during inspection process.

"Inspection plan documents" and "Documents for inspection plan characteristics" – Toolbar

In addition to the standard functions, the button to show documents is available.



Show documents

If a document link is stored, this button opens and shows the linked document. However, a program, which can show the linked file type, must be installed in the PC.

"Failure" detail application

A "Failure" detail application is provided on the level of inspection requirements. This list shows all failures of the type "failure type", "failure location", "failure cause" and "originator", which have been assigned to the characteristics, samples or measured values during the inspection process. The list even shows the failure types that have been generated automatically, e.g. "upper tolerance limit violated".

In addition to the respective failure, the following referenced data is also available.

- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number
- Failure type
- Characteristic number and description
- Workplace number and description
- Weighting (number, e.g. for an inspection chart characteristic)

- Comment,
- Failure date
- Failure time

You can restrict the list in a flexible way using the individual column filter. You can also use the *group by* function. Example: You can list the failures for each characteristic.

Failure list - editing functions

For administration purposes, you can create, edit or delete failure analysis entries for inspection requirements. If a new data record is created, you can enter further key information to specify the assignment. The following information entries can be added:

- Inspection step number,
- OP sequence number (uniquely identifies the characteristic),
- the sample number
- and value number.

To precisely assign this key information, the user requires "administrative" skills. An administrative environment is recommended.

The user requires the following function authorizations:

Function authorization	failure.create => create new failure failure.edit => edit failure failure.delete => delete failure
------------------------	--

Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and a validation process is not performed.

The fields "area" and "inspection requirement number" cannot be modified and you cannot enter values. This is also true when you create new data records. When you create a new data record, this data is taken from the previously selected inspection requirement.

You can enter values for all other fields when you create a data record. In the editing mode, you can only change the fields "weighting" and "comment".

"Measures" detail application

A detail application including a "measure list" is provided on the level of inspection requirements. This list shows all measures and actions which have been assigned to the characteristics, samples or measured values during the inspection process.

In addition to the respective measure number and designation, the following data is also referenced.

- Inspection requirement
- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number
- Measure type
- Party in charge incl. type
- Status
- Comment
- Text
- Effectiveness

You can restrict the list in a flexible way using the individual column filter. Use the "group by" function to have further options of analysis.

"Measures" toolbar

Use the button "measure tracking" to open the measure tracking dialog where you can edit and create measures. Measure tracking uses the internal and unique "serial number" of the measure in the "references" tab for filtering. Filtering is omitted if no measure is selected.



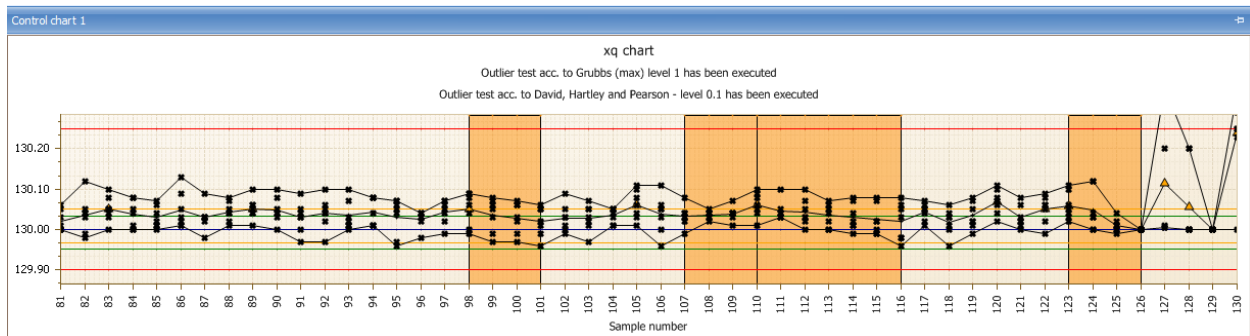
Go to measure tracking

Function authorization	failure.* => opens measure tracking
------------------------	-------------------------------------

"Control chart 1/2" detail application

By default, the application shows the control charts 1 and 2 defined for the inspection step characteristic. If the charts are not defined for the inspection step characteristic, the xq and s charts are used for variable characteristics and the p and u chart for attributive characteristics.

All data of the selected inspection step characteristic is used for the graphic presentation.



Open the dialog to configure "control chart 1" or "control chart 2" to define the contents of this application. If you use this dialog to make changes, the changes are saved per user. If no specific setting has been made, the xq chart ("variable" characteristic type) or the p chart ("inspection chart" or "attributive" characteristic type) is displayed by default. The available control charts depend on the characteristic type. By default, the user can select one of the following control charts.

Variable characteristic

- Xq chart
- S chart
- R chart
- Single value chart
- Median chart

Attributive characteristic

- p chart
- np chart
- c chart
- u chart

SettingsDialog

Chart type:

Number of samples/measured values to be requested: ☐ All ☒ Quantity

☒ Show trend ☒ Show information on outlier tests

☒ Show run ☒ Perform outlier test according to Grubbs max. (1%)

☐ Show middle third ☐ Perform outlier test according to Grubbs max. (5%)

☒ Show single values ☐ Perform outlier test according to Grubbs min. (1%)

☒ Show target value ☐ Perform outlier test according to Grubbs min. (5%)

☒ Show tolerance limits ☒ Perform outlier test according to David-Hearty-Pearson (0.5%)

☒ Show action limits ☐ Perform outlier test according to David-Hearty-Pearson (1%)

☒ Show warning limits ☐ Perform outlier test according to David-Hearty-Pearson (5%)

☒ Show mean value

☐ Consider long-term data

☒ Show title of current label Control chart labeling:

☒ Consider number of decimal places Number of decimal places:

☐ Show characteristic title

☒ Show control chart type

☒ Combine minimum and maximum values ☒ Show legend of y-axis

☒ Show grid ☐ Automatic scaling

OK Cancel

In the following, the most important configuration options are described:

Number of samples/measured values to be requested

Specifies the number of samples or single measured values that are displayed in the control chart.

Show ...

The options that include the term "Show ..." enable or disable the display of the respective data in the control chart.

Consider long-term data

Check this option to integrate archived data of the medium-term data area.

Combine minimum and maximum values

It is recommendable to show the minimum and maximum values that are each connected by a line to improve the presentation of the range of dispersion of single values.

Automatic scaling

Use the "automatic scaling" function to display all values, irrespective of the existing limit values. This function also shows extreme outliers in the control chart. Disadvantage: The other measured values are much smaller in the layout and it is more difficult to identify changing values or a trend.

Show trend / run / middle third

Use the monitoring functions "trend", "run" and "middle third" to better monitor a process. These functions are only available with the control charts xq and median. If you show the trend, you can visualize a process trend that may rise or fall. The system uses several samples to generate a trend. By default, these are seven consecutive rising or falling values. The run shows sections where the process runs above or below the mean value (when displayed, otherwise the target value). The run covers several samples. By default, these are seven consecutive values that are above the mean value. The default number of seven samples/values that is set to identify a trend or run can be changed via system customization. The system identifies a "middle third", if an unusually high or low number of values lies within the middle third of the range between the action limits.

The system automatically makes the respective analyses for "trend", "run" and "middle third". If a trend/run and/or middle third is identified, the control chart shows it in a graphic form. The following events are shown in the control chart via symbols or colors:

- Trend (colored area)
- Run (colored area)
- Middle third (colored area)
- Outlier (symbol: )
- Xq violates action limit (symbol: )

The presentation of outliers is only possible in the xq chart and median chart. Outliers can only be displayed when single values are shown. The function to perform outlier tests and the function to display outliers are separated. You can activate the different outlier tests for the different levels separately. If you have enabled the display of outlier tests, the test result is shown in text form on top of the respective control chart.

The below outlier tests are available for the different inspection levels. The inspection level is specified in parentheses.

- Grubbs max. (1 %)
- Grubbs max. (5 %)
- Grubbs min. (1 %)
- Grubbs min. (5 %)
- David-Hartley-Pearson (0.5 %)
- David-Hartley-Pearson (1 %)
- David-Hartley-Pearson (5 %)

Notes on outlier tests

- The outlier tests do not refer to the total of all samples, i.e. each sample is considered individually. Consequently, the following phenomena may appear:
 - Despite a large range between minimum and maximum value no outlier can be identified. A reason for this may be the equal distribution of the single values within the sample.
 - Despite a low range between minimum and maximum value, an outlier is identified. A reason for this may be an accumulation of many values at one "point" so that an individual value having a certain distance to this agglomeration is identified as outlier within the sample.
- To perform an outlier test, there must be at least three values in the sample. The larger the number of values in a sample, the more uniform is the overall picture of the outliers compared to all samples.
- The Grubbs outlier test is performed with a sample size of $2 < n < 148$ (n = number of values within a sample). The outlier test according to David, Hartley und Pearson is performed with a sample size of $2 < n < 1251$ (n = number of values within a sample).

X-axis labeling

The below information is available to label the x-axis of control charts.

- Sample number
- Order number
- PPS reference number
- Purchase order number
- ERP batch
- Article number
- Machine number
- Cavity number
- Date + time of the first measured value of a sample
- Date + time of the last measured value of the sample
- Date + time of sample completion
- Badge number of the first measured value of the sample
- Badge number of the last measured value of the sample
- Badge number of sample completion



If the following fields are not available, you require a new program version of this application:

- Partial batch
- Workplace

- Production workplace
- Field 1
- Field 2
- Field 3
- Field 4
- Field 5
- Field 6
- Field 7
- Field 8



The fields "field 1" to "field 8" are enabled as part of an MPDV customizing. The field labels change according to the individual requirements. Because these field names are flexible, they are only labeled "field 1" to "field 8". Field 1 includes, for example, the tool if cavity-related data collection is enabled.

Tooltips in the control chart

If the system has identified a value as outlier, the detailed information is shown when you mouse over the symbol (red rhomb). The tooltip includes the test(s) that issued this outlier result.

For mean values, the tool tip shows the value and date and time of the first and last measured value of this sample and the respective inspection requirement number and inspection step number.

For single measured values, the tooltip shows the exact value.

For the colored areas that identify a trend, run or middle third, a respective note including the reason is shown .

Sorting of measured values

The server decides on the sorting of a control chart. The sorting type depends on the settings of control chart filters. If the filters "operation sequence" and "inspection requirement number" or "operation sequence" and "inspection step number" are entered, the characteristic is identified uniquely and sorting is based on the sample number.

Sorting is based on date and time if either the filter field "operation sequence" is left empty or if it is filled and the fields "inspection requirement number" and "inspection step number" are left empty.

The displayed sorting of measured values influences the visual presentation of a trend or run. However, the automatic generation of the failure type "trend" is always based on the measured values being sorted by the sample number, as the numbers of the operation sequence, the inspection requirement and the inspection step are always known at the time data is recorded.

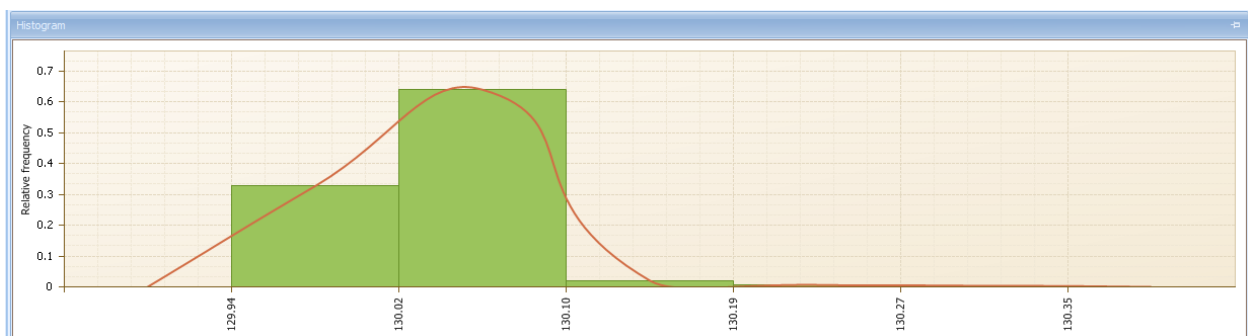
Consequently, the following scenario might be possible.

- The control chart is sorted by date and time and a trend is shown, although the automatic failure "trend" has not been generated.

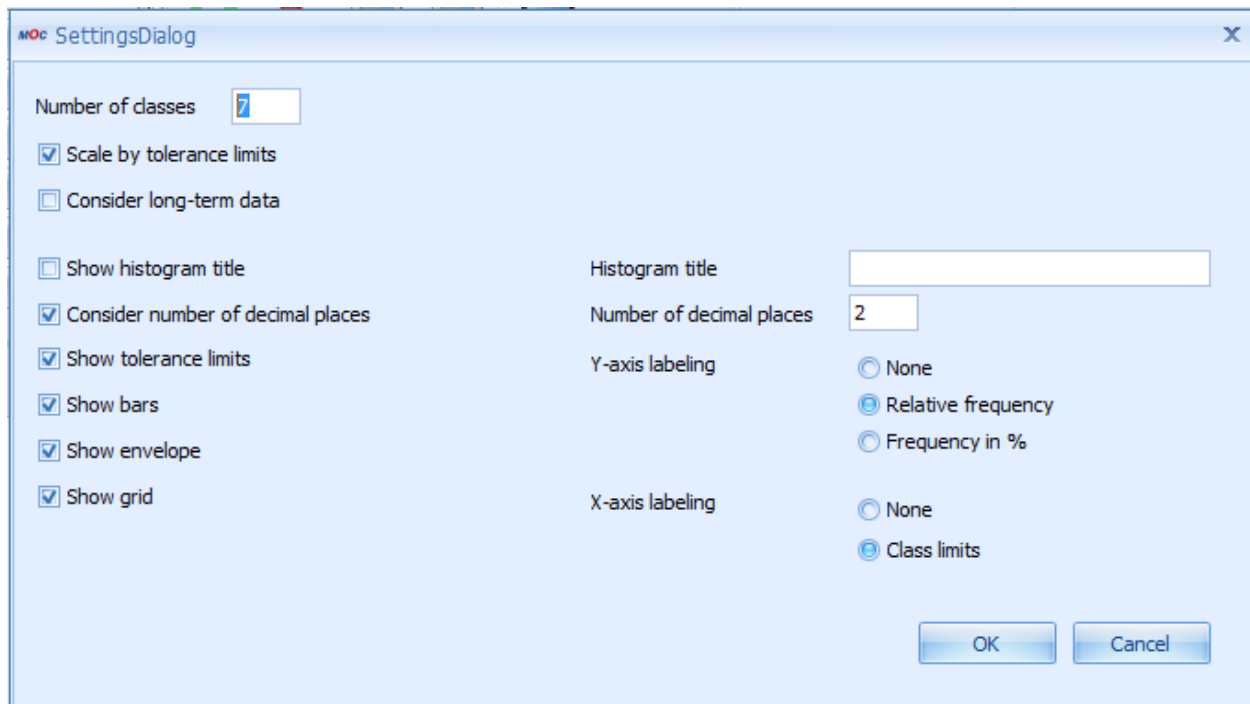
The control chart is sorted by the sample number and no trend is shown, although the automatic failure "trend" has been generated.

"Histogram" detail application

All data of the selected inspection step characteristic is used for the graphic presentation.



You can restrict the number of samples displayed in a control chart. This is not possible with a histogram. The histogram is always based on the total of available samples matching the selection filter criteria. The number of classes and the additionally displayed information influence the histogram presentation. The contents of this application are defined by opening the dialog to configure the "histogram". If you use this dialog to make changes, the changes are saved per user.



In the following, the most important configuration options are described:

Number of classes

Specifies the number of histogram classes. The measured values are classified and displayed in these classes. If the histogram shows the values within the tolerance limits (option "Scale by tolerance limits"), two histogram classes include the values exceeding the tolerance limit (upper and lower).

Scale by tolerance limits

Enabled: The classes are between the tolerance limits - two "outlier classes" being displayed, one to the left and one to the right.

Disabled: The classes include the total range of all measured values (no separate classes for values exceeding the tolerance limits).

Consider long-term data

Includes the archived data of the medium-term data area.

Show histogram title

Enable this option if you want to show a special histogram title.

X-axis labeling

If the option "Class limits" is set, the x-axis shows the respective values of the class limit. You can use the two configuration options for decimal places to specify the detail, i.e. the number of decimal places.

Consider the number of decimal places

With this option, the defined number of decimal places is used for the x-axis labeling.

Control chart 1/2 and histogram - toolbar



Control chart 1 settings

Opens a dialog to configure the settings of control chart 1. The respective details are described in the respective detail application.



Control chart 2 settings

Opens a dialog to configure the settings of control chart 2. The respective details are described in the respective detail application.



Histogram settings

Opens a dialog to configure histogram settings. The details are described in the respective detail application.

"Samples" detail application

The "samples" detail application is on the level of inspection step characteristics. It includes statistical key figures for each sample. In addition to the referenced key fields, such as the inspection requirement number, inspection step number and sample number, the following statistical values are listed if available/calculated:

- $\bar{X}_q$
- $\bar{X}_q$ floating
- R floating
- Standard deviation s
- s floating
- Minimum
- Maximum
- Range
- Median
- P

- U
- Number of defects

The list additionally shows the characteristic specifications (tolerance, action and warning limits) as well as the referenced machine.

"Samples" - toolbar



Complete sample

It might be necessary to complete samples manually, because measured values can be entered for administration purposes or because there might be inspections without inspection points but including the initial creation of samples and/or inspections without having reached the sample size. You can also complete samples that have not been completed by the AIP inspection data collection.



Release sample

You can use this function to re-release samples and to record further measured values for this sample number if the specified sample size has not yet been reached.

"Single value" detail application

The "Single value" detail application is on the level of the inspection step characteristics and includes single measured values for all variable inspection step characteristics. For attributive characteristics the following information is displayed:

- Number of defects
- Number of NCU (non-conforming units) and
- Failure

You can also identify if the measured value or the attributive assessment is valid or invalid.

In addition to the referenced key fields such as inspection requirement number, inspection step number, sample number and value number, the detail application lists the characteristic specifications (tolerance, action and warning limits). For each entry, you also have details on the date and time when the entry was recorded and edited and on the responsible user.

Single value - editing functions

For administration purposes, you can create, edit or delete measured values for a variable inspection step characteristic and/or inspection point characteristic. Or you can create, edit or delete the number of non-conforming units for attributive characteristics. If a new data record is created, you can enter further key information to specify the assignment. The following information entries can be added:

- the sample number
- and value number.

To precisely assign this key information, the user requires "administrative" skills. An administrative environment is recommended. Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and a validation process is not performed.

The user requires the following function authorizations:

Function authorization	value.create => create new measured value value.edit => edit measured value value.delete => delete measured value
------------------------	---

Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and a validation process is not performed.

The fields

- Area,
- Inspection requirement number,
- Inspection step no.,
- Inspection step number (only available for inspection point characteristics),
- OP sequence (uniquely identifies the characteristic),
- Upper tolerance limit,
- Target value,
- Lower tolerance limit and
- Number of decimal places

cannot be modified and you cannot enter values. This is also true when you create new data records. When you create a new data record, this data is taken from the previously selected characteristic and the referenced inspection requirement as well as from the inspection point.

You can enter values for all other fields when you create or edit a data record.

The field "single value of cavity" is only visible if

- the inspection requirement includes a setting specifying that data is generally collected in relation to cavities and
- the relevant characteristic does not include a setting specifying that the inspection must be performed without relation to cavities.

If a new data record is created, the field for the measured value will be restricted to the number of decimal places. The other decimal places are filled up with zeros. For technical reasons, these "additional" decimal places are shown during processing.

The field "measured value" is shown for variable characteristics. Instead of the measured value field, the fields "sample size (checked)", and "number of DU" are shown for attributive characteristics, i.e. this also includes inspection chart characteristics.

Detail application "Statistics"

The detail application "Statistics" is on the level of inspection step characteristics. It includes the usual statistical key figures for the selected characteristic.

For variable characteristics, the following statistical key figures are included:

- Number of samples
- Number of measured values
- $\bar{x}$
- Minimum
- Maximum
- R
- S
- S relative
- Sigma
- Cp and
- Cpk

For attributive characteristics, the following statistical key figures are included:

- Number of non-conforming units
- Number of defects
- p and
- u.